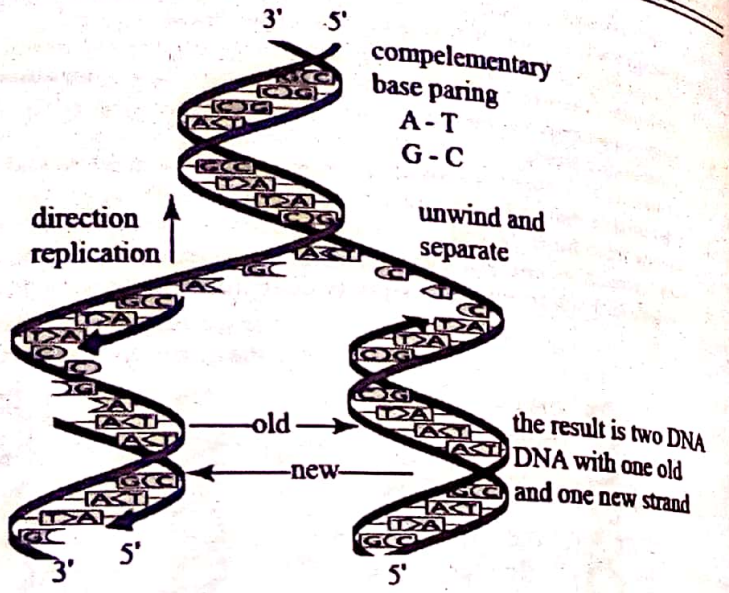

Figure 8.17 steps of DNA replication

Gene and organism cloning

Gene cloning - is the **process** of making **multiple copies** of a gene. It can be done in **two** ways.

a) *In vivo* **cloning** - is a situation in which the **gene** is **introduced** into a living **cell**, and is **copied** many times as the **cell divides**. In such cases, gene products (Example: insulin) may be produced by the cell.

b) *In vitro* **cloning** - is the use of a **machine** called **polymerase chain reaction (PCR)** to duplicate **DNA**. This is **faster** and **cheaper**, and it mimics the **semi-conservative** replication of **DNA** that occurs in cells.

Cloning of organisms - is the production of **identical organisms** by **asexual reproduction**. The resulting **offspring** are **genetically identical** to **each other** and to their **parent**.

Example: Plant propagation by **cutting, tissue culture**, etc.

A **clone** is a **group** of **genetically identical organisms** produced by cloning.

Example: Bananas, sugar cane, crops, flowers, etc.

Genetic engineering

Genetic engineering is the **process** of **modifying** the **genome** of an **organism**, usually by **adding** a **foreign gene**.

Transgenic organism - is a **genetically modified organism**. Most genetic engineering was done on **bacteria** as it is easier to handle them. **Insulin** production controlling **genes** were isolated from **mammalian pancreas** and introduced into **bacteria**. The **insulin** they produced is **extracted, purified** and used to **treat diabetic patients**. **Enzymes** for different purposes, **human growth hormone, bovine somatotrophin** (which increases **milk yield** and **muscle development** in cattle) and **vaccines** can also be produced by such **transgenic bacteria**.

Plants can also be **genetically modified** to **increase** their **yield** and **disease resistance**. **Transgenic plants** can also produce **specific products**.

Example: Golden rice (a transgene) produces **P-carotene** (prevents night blindness). Practically, it is **easier** to clone **plants** than **animals**, because **plant cells retain** their ability to **divide**. Still it is possible to clone animals. **Dolly** is a sheep, the first cloned mammal.

Dolly, an example of *in vivo* cloning

A sheep called **'Finn-Dorset'** was used as a **nucleus provider**. Another sheep called **'Scottish Blackface'** was egg cytoplasm provider. The nucleus from mammary cells of 'Dorset' and the enculeated egg from 'Blackface' were **fused** and stimulated to divide by **electric current**. At **blastocyst** stage, the dividing diploid cell (**embryo**), was **implanted** into a **surrogate 'mother' ewe**. Dolly was born after **7 months**. She was **genetically identical** to **'Finn Dorset'** and the process can be considered as **cloning** (multiplying) **'Finn Dorset'**.

Theoretically and **technically**, it is possible to **clone humans** in similar way, but, it is **ethically illegal** and **forbidden** now! The technique may solve the problem of **sterility** in some individuals. Should it ever be legal or stay **illegal**. Generally, fewer animals have been genetically modified.

Example: Zebra-fish (glow-fish) has had a **gene** that controls **glowing** in darkness from **jelly-fish**, and it glows in **different colors!**

Main steps of genetic engineering

a) Cutting out the **gene** of **interest** from the **donor DNA** by using an enzyme called **restriction endo-nuclease**.

b) **Extracting plasmids** from a **bacterium**, and **opening** it by **restriction endo - nuclease**.

c) **Inserting** and **tying (splicing)** the **gene** of **interest** into the **plasmid** using **DNA-ligase**.

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d) **Introducing the hybrid (recombinant) plasmid into a suitable bacterium.**

e) **Extracting, purifying and utilizing the gene-product (Example: Insulin, enzyme . . .) released by the recombinant bacterium.**

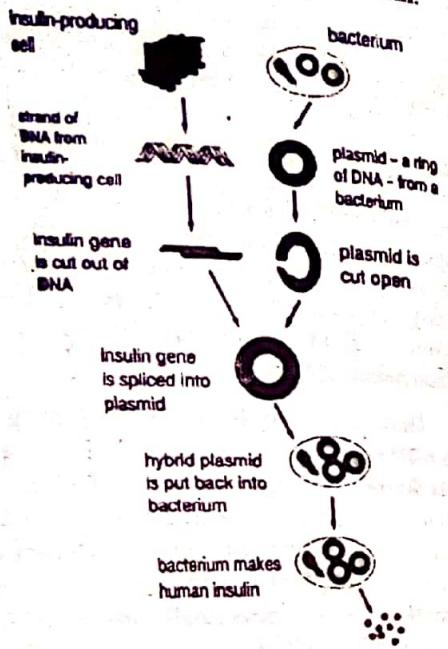

Figure 8.18 steps of genetic engineering

Potential benefits of genetic engineering

a) **Disease prevention** by detection of **hereditary disease susceptibility.**

b) **Treatment of viral disease** by transfer of **antiviral protein coding genes.**

c) **Reduction** in use of **fertilizers & pesticides,** thereby **minimizing pollution,** as the **transgenic plants and animals have high growth rate and less diseases susceptibility.**

d) **Engineered trees may absorb more CO_2 to reduce global warming.**

e) **The transgenic organisms will add to the existing organisms, and may increase genetic diversity of our ecosystems provided that the original organisms are conserved.**

f) **Genetic engineering may open way for faster improvement of organisms. Traditional selection breeding is time consuming.**

Genetic finger-printing

Genetic finger printing - is a technique of **analysing** and **comparing** the **DNA fragments** of **2** or **more** individuals. The use of **genetic finger-printing** to resolve **criminal issues** by making use of **DNA** found at crime scenes is known as **forensic technology (science).**

Generally, the **DNA** in cells can be categorized into **coding** and **non-coding** regions. The **coding regions** represent the **genes.** The non-coding regions are made up of short sequences of repeating bases. The **repeats** may be found

scattered through out the **DNA,** but some are found in clusters. These clusters are called **satellite DNA.** The **number** of **repeats** in each cluster **varies** between **individuals.** Those **satellites** with **small number** of **repeats** are called **mini-satellites. Genetic finger printing** makes use of **mini-satellites.** Any **cell** that has **nucleus** can be used as a source of **DNA** for **genetic finger-printing.**

Example: Blood, skin, semen, hair-root, etc.

If the **amount** of **DNA** in the **sample** found is **very small,** it can be **amplified** by PCR.

Main steps of genetic finger-print preparation

1) **DNA isolation** from cells.
2) **DNA fragmentation** by **restriction enzymes,** and **treatment** by **alkali** to get **single-stranded DNA.**
3) **Separation** of **fragments** by **gel-electrophoresis. Smaller fragments** move **faster.**
4) A **nylon membrane** is placed over the gel to **take up** (transfer) the (invisible) pattern of separated DNA fragments to it. The gel is soaked in a **buffer solution. Blotting paper** is placed over the membrane to absorb the buffer. The **fragments** are **sucked up** by the **blotting paper,** but remain on the membrane **retaining** their **relative positions** as in the gel.
5) Radioactive single-stranded DNA, i.e. complementary **DNA** (C-DNA) is applied to the membrane, and **bind** to the **mini-satellite** regions of the fragments, i.e. hybridization with **radioactive probe.**
6) **Auto-radiography** - the membrane is placed over a piece of **x-ray film,** and the images of the fragments are **developed** to reveal the **hybridized** radioactive regions.

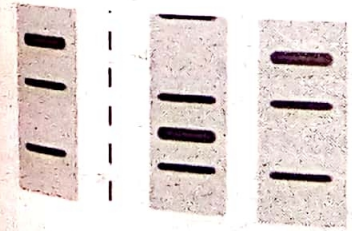

Figure 8.19 DNA fingerprint of a **specimen (semen)** obtained from **cloth** of a **rape victim,** and **blood** of **three suspects.** The genetic fingerprint of **suspect-2** is **similar** to that of the **semen,** and most probably he is the **criminal** (rapist).

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Gel-electrophoresis - is a technique of separating **DNA fragments** or **protein molecules** by applying **electric field** across a **gel**. **Size** and **charge** on the **molecules** affect their **movement** in the gel. **Smaller fragments** (molecules) move **faster** in the gel. As **DNA fragments** are **negatively charged**, they move towards **anode**, but away from **cathode**. Apart from **identical twins**, the **chance** of **two people** having the **same genetic finger-print** is only one in a million (**-1 in 1,000,000**). So, **genetic finger-printing** can give strong **evidence** for the **involvement** or **innocence** of a **suspect** in a **crime**.

Example: If the **DNA finger-printing** of a **victim's blood** and blood from a **defendant's cloth** is **similar**, it is clear that the **defendant** was, atleast, at the crime scene.

Is using gene technology morally/ethically correct?

Morality/ethics - refer to our sense of what is **right/acceptable**, and what is **wrong**. They are not linked to **legality**. But, **morality** is **personal**, and **ethics** refers to the '**code**' adopted by a **group**.

There is **debate** between people (scientists as well as ordinary) on the use of **genetic engineering**. The following **issues** are **raised.**

- **Species** should not be altered.
- There is no enough **knowledge** about the **long term effect** of **transgenes (GMOs)**.
- Introduced **genes** may **jump** from the **genetically modified organisms (GMOs)** to other species.
  Example: **Herbicide resistance genes** may **jump** from **crops** to **weeds** and complicate the matter by making them resistant.
- **Designer babies** may be created by **mixing medical gene therapy** and **cosmetic gene therapy**.
- **Genetic data base (genetic finger-print)** of everyone in a country is needed to compare the **DNA finger-print** found at a **crime scene**, and instantly **identify** the **criminal**. But, if this data is freely **accessed** by many individuals, it may create problem.
  Example: insurance companies may refuse **insurance** to people with an **increased** risk of **genetic diseases**.

Illustrative Examples

31. The sheep 'Dolly' is an example of which biotechnological manipulation of animals by human? **(EUEE, 2005 E.C)**
(A) Transgenic animal (C) Cloned animal
(B) Genetically engineered animal (D) Hybrid animal

Explanation:
- Dolly is the first **cloned mammal** (animal). Her genome was not modified.

Answer: C

32. Choose the one that is **different** from all the others. **(EUEE, 2005 E.C)**
(A) Genetically modified organisms (C) Pathogenic organisms
(B) Genetically engineered organisms (D) Transgenic organisms.

Explanation:
- **Pathogenic organisms** are **disease causing** organisms, and are different from the rest. A, B & D are the same.

Answer: C

33. DNA is wrapped around_____to form a double stranded chromatin fiber. **(EUEE, 2005 E.C)**
(A) Telomere (C) Cenetromere
(B) Histone (D) Ribose

Explanation:
- DNA is wrapped around a **'bead'** of **8 protein molecules** at intervals along its length. These molecules are called **Histones.**

Answer: B

34. Which of the following enzymes are used to polymerize DNA-nucleotides in to DNA strand?
(A) RNA-polymerase (C) DNA-polymerase
(B) DNA-helicase (D) DNA-ligase

Explanation:
- RNA-polymerase polymerizes **RNA-nucleotides** into **RNA**; **DNA-helicase** unizip **DNA double strand**; DNA-polymerase polymerizes **DNA-nucleotides** into **DNA**, and **DNA-ligase** ties DNA fragments together.

Answer: C

35. The machine used to clone DNA outside a cell is:
(A) X-ray machine (C) Golden gun
(B) PCR machine (D) Paper blot

Explanation:
- *PCR machine* is used to duplicate **DNA** outside a cell (i.e. ***in vitro*** clonning).

Answer: B

36. In the genetic finger-print preparation, DNA fragments are separated according to their size by:
(A) Auto-radiography (C) Hybridization
(B) Blotting (D) Gel-electrophoresis

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Explanation:
- The **DNA fragments** are separated by **agarose gel-electrophoresis**.

Answer: D

37. What are sex-linked traits?

Explanation:
- **Sex-linked** traits are **features** of an organism controlled by genes located (found) on the **sex chromosomes**.

38. If a cell having **2n = 60** chromosomes **divides** by **meiosis**, how many chromosomes are expected in the **daughter cells**?

(A) 60 (C) 30
(B) 15 (D) 20

Explanation:
- **Meiosis** is a **reduction division**, therefore, the daughter cells will contain **n = 30** chromosomes.

Answer: C

39. A double stranded DNA molecule has 40 adenine and 60 cytosine bases. What is the total number of base found in the DNA?

(A) 80 (C) 100
(B) 120 (D) 200

Explanation:
- In DNA molecule, **adenine** always combines with **thymine**, and **cytosine** with **guanine**. The number of **adenine** equals that of **thymine** (i.e. A = T = 40). The number of **cytosine** equals to that of **guanine** (i.e. C = G = 60). So, A+T+C+G = 40+40+60+60 = 200 bases.

Answer: D

40. Among the following choices, identify the ones that have identical DNA finger-prints. **(UEE, 2003 E.C)**

(A) Parents & children (C) Fraternal twins
(B) Non-twin sibs (D) Monozygotic twins

Explanation:
- Only **monozygotic (identical) twins** will have **identical DNA finger print** because they are formed the **same egg and sperm**.

Answer: D

41. It is known that the total amount of DNA in a cell contains 300 nucleotides and **adenine** contributes to **80** of these. **How** many nucleotides go to **guanine**? **(UEE, 2003 E.C)**

(A) 35 (B) 70 (C) 80 (D) 140

Explanation:
- The number of **adenine** equals that of **thymine** } A+T = 80+80 = 160
  } G+T = 300-160 = 140 } G = 140/2 = 70

Answer: B

42. If it is known that the total number of the **purine** bases account for **50%** of a DNA molecule and if each of the remaining bases are known to have the same proportion, what proportion is accounted for by **thymine** alone in the same molecule? **(UEE, 2003 E.C)**

(A) 25% (B) 50% (C) 75% (D) 100%

Explanation:
- **Purines** include **adenine** and **guanine**, and they are **double-ringed** **Pyrimidines** include **cytosine** and **thymine**, and these are **single-ringed**. On the other hand, **adenine** combines with **thymine**, and **guanine** with **cytosine**. If purines and pyrimidines contribute 50% each for DNA, individual bases will have a share which is **less than** 50%. So, 25% is the probable share of **adenine**.

Answer: A

43. Among the following couples whose **ABO** blood genotypes are shown, which one can produce children of A, B, AB and O blood types? **(UEE, 2004 E.C)**

(A) OO & AB (C) BO & AO
(B) BO & AA (D) BB & AO

Explanation:
- A cross between **BO & AO** only can produce **all the 4 blood group** phenotypes.

♂ \ ♀	B	O
A	AB	AO
O	BO	OO

} The **chance** of each of the **4 phenotypes** to be produced by this couple is **equal**, and is **1/4** (or **25%**)

Answer: C

44. What is the number of different combinations of **chromosomes** in the pollen grains of the crocus (*crocus balansae*) which has a **diploid** number of six **(2n = 6)**?

Explanation:

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$C = 2^{n-1}$ Where C = number of combinations
n = number of **bivalents**, or **haploid number**.

$C = 2^{3-1} = 2^2 = 4$ different **combinations** of chromosomes are possible during **bivalent** formation in **meiosis**.

45. The molecules of **DNA** in **prokaryotic** cells are **different** from those in **eukaryotes** because
(A) They are much smaller.
(B) They are circular, not linear as in eukaryotes.
(C) They are not associated with hisotnes to form chromosomes.
(D) All are correct.

Explanation:
- **DNA** molecules of **prokaryotes** are **smaller, circular** and **bare** (i.e. not associated with histones).

Answer: D

46. Which of the is **true** about the **ethical** and **moral** considerations of using **gene technology**?
(A) There is **fear** that **doctors** may create **designer babies**.
(B) There is **no enough knowledge** of the **future effect** of **transgenic organisms**.
(C) **Genes** may **jump** from **genetically modified organisms** to other **species** in the environment.
(D) All are correct

Explanation:
- All the statements refer to the **ethics** and **morality** associated to gene **technology**.

Answer: D

47. Which of the following is the **first step** in **DNA finger-printing**?
(A) Autora diography
(B) DNA preparation
(C) Probe hybridization
(D) Gel-electrophoresis

Explanation:
- **DNA preparation** comprises **DNA isolation** form the source cells and **DNA fragmentation** by **restriction endonucleases**.

Answer: B

48. In the first mammalian cloning (production of '**Dolly**'), who was the **genetic mother** of '**Dolly**'?
(A) Finn-Dorset
(B) Surrogate mother ewe
(C) Scottish blackface
(D) 'Dolly' herself

Explanation:
- **Finn-Dorset** was 'Dolly's' **genetic mother** because she provided her mammary cell **nucleus**. Scottish blackface provided **egg** without nucleus. Surrogate ewe carried 'Dolly' in her womb for 7 months.

Answer: A

49. The bead-like structures formed at intervals along the length of DNA when it wraps around protein molecules is called _____
(A) Nucleosomes
(B) Chromatins
(C) Histones
(D) Chromosomes

Explanation:
- **Nucleosomes** are formed when **histone proteins** and **DNA** are associated.

Answer: A

50. Which of the following sugars is the component of **DNA**?
(A) Ribose
(B) Deoxyribose
(C) Ribulose
(D) Deoxyribulose

Explanation:
- **Deoxyribose** is a **5-carbon sugar** found in DNA. It differs from **ribose** which is the **component** of **RNA** in that it has **lost** its **oxygen** at one point in its structure.

Answer: B

8.3 Protein synthesis

Proteins are essential molecules in our body that **determine** our **characteristics**. They act as **hormones, enzymes, structural** materials, and as **cell markers**. Their production needs *flow* of **information** from **DNA** in the **nucleus** to ribosomes in the **cytoplasm**. **Ribosomes** are **factories** where **proteins** are **manufactured**. The information on **DNA** (i.e. **code**) is **copied** by **messenger** RNA (mRNA) as a **codon** and moves out of nucleus into cytoplasm, where it is used to **synthesize** a **protein** molecule (i.e. **translation**). the synthesis of mRNA is known as **transcription**.

Steps of protein synthesis

1) The **code** for a particular **protein** is **copied** by **mRNA** from a **gene** on DNA by a process called **transcription** (RNA-synthesis).
2) **mRNA** crosses **nuclear envelope** and **attaches** to a **ribosome** in the cytoplasm.
3) Free **amino acids** are carried from **cytoplasm** to the **ribosomes** by **transfer RNA (tRNA)** molecules.

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4) **The ribosome reads the information in mRNA and assembles the amino acids into a protein by a process called translation.**

The genetic code (a gene)

The genetic code (a gene) is a **sequence of base triplets** in the **DNA** molecule that carries **information** for a **polypeptide**. There are **20 amino acids** in nature to make **polypeptides (proteins)**. But, there are only **4 bases** in **DNA** (i.e. A, G, C, & T). If a base is considered to represent an amino acid in the genetic code, only 4 amino acids would be coded for by the 4 bases. If a **2-base code** is ???

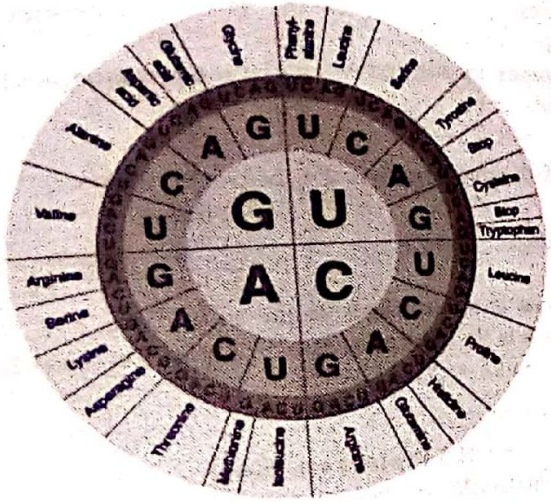


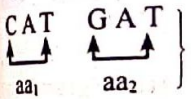
Figure 8.20 The **relationship** between the **triplets** and the 20 amino acids. There are **3 circles** in the figure. **If we restrict our movement** to a quadrant, **select a base** from the **central circle**, move to a **division** in the second circle, pick a **second base**, and finally pick a **third base** from the **third** circle, we will end up with a 'stop' information or an **amino acid** that is coded for by the **triplet** we have **constructed**. **Example**: If the arrows are followed, GTA codes for **valine**, **TAG** for a 'stop' and **CTA** for **leucine**.

considered to stand for an amino acid, **4x4 = 16** possible combinations of the **4** bases can be **obtained**. Still all the **20** amino acids **could not** get their **own code**. Therefore, a **3-base code** would be a **remedy** for the problem. In this case, the **first**, the **second** and the **third** positions can be filled by **one** of the **4 bases**, and consequently, **4x4x4 = 64** possible **base combinations** (i.e. triplet bases) can be **obtained**. Here, since there are **only 20 amino acids**, the 64 triplet codes are unevenly distributed among the amino acids. Three triplets (**TAA, TAG & TGA**) are '**stop**' codes during **protein synthesis**. Only **2** amino acids (**methionine** and **tryptophan**) have **single triplet code** (**ATG** and **TGG**, respectively). **Agrenine, leucine & serine** have **6 triplet codes** each. The remaining amino acids have **triplet codes** ranging between **2 & 6**.

Properties of the genetic code

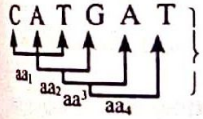
a) **The genetic code is a triplet code** - that consists of a **sequence** of **triplet** (3) bases.

b) **The genetic code is non-overlapping**
Example:



Non-overlapping --- the correct scenario (situation) in nature.

- Only **2 amino acids** are coded for by **these 6 bases.**



Overlapping codes

- **4 amino acids** could have been **coded** for by the **6 bases**, but this is **not** the case in **DNA** in **nature.**

c) **The code is commaless** - this means that there are **no bases** acting as **commas** between **triplet codes**. Therefore, the **genetic code** is **read** from one end to the other in an **uniterupted** manner.

d) **The code has polarity** - the **code** that specifies a **polypeptide** has **fixed start** and **end** points which are the **initiation** and **termination** triplets, respectively. The **code** should be **read** in that **fixed direction**. If it is read in the opposite direction, it would specify a different polypeptide, or become meaningless.
Example: If the triplet **codons** in **mRNA** below are read in **opposite directions**, they would automatically specify **2 different polypeptides**.

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Codons: --- UUG AUC GUC UCG CCA ACA AGG
Leu - Ile - Val - Ser - pro Thr - Arg

Left to right ► Val - Leu - Leu - Ala - Thr - Thr - Gly ◄ Right to left

e) **The codes are degenerate** - this means that there are **excess triplet codes** than what is needed. There are **20 amino acids**, but **64 triplet codes**. **Three** of these are **termination** ('**stop**') codes. The remaining **61 triplets are unevenly** distributed among the **20 amino acids**.

f) **The code is universal** - the genetic code is essentially the **same for all organisms** ranging from **bacteria** to **man**.

Transcription in Eukaryotes

The **information** (**gene**) for **one** type of **polypeptide** is **copied** from one **strand** of **DNA** by synthesizing **messenger RNA** (**mRNA**) in a process called **TRANSCRIPTION**. The **DNA strand** used to synthesize **mRNA** is called the **antisense strand**. This strand is used as a **template** for **RNA synthesis**. The other **complementary** strand is called **sense strand**.

Steps of transcription

1) DNA - dependent **RNA-polymerase** binds to **DNA** near the gene to be **copied** (**transcribed**).
2) **Transcription factors activate** the **enzyme**.
3) The enzyme '**unwinds**' (**opens**) **DNA** double strand and moves forward along the **antisense strand**.
4) **RNA - polymerase** assembles free **RNA-nucleotides** into a chain of **mRNA** using the **antisense strand** as a **template**. The mRNA so formed will be **complementary** to the **antisense strand**, but **identical** to the **sense strand** except for the **replacement** of '**T**' by '**u**' in the RNA.
5) The **completed mRNA** detaches from the **DNA**. The **DNA strands rejoin** and **recoil**.
6) The synthesized **mRNA** comes out of the **nucleus** and **attaches** to the **ribosome** in the **cytoplasm**.
Transcription is completed.

DNA & RNA comparison

		DNA	RNA
1.	Size	Huge	Smaller
2.	Strand	Double	Single
3.	Bases	A, G, C, T	A, G, C, U
4.	Sugar	Deoxyribose	Ribose

Code, codon & anticodon

Code - refers to a **base-triplet** on **DNA** that **codes** for an **amino acid** or a '**stop**' information. **Example:** ATG.
Codon - is a **base-triplet** that represents an amino acid or a termination information on mRNA. **Example:** UAC.
Anticodon - is a base-triplet on **transfer RNA** (**tRNA**). It is **complementary** to one of the **codons** on **mRNA**, and **codes** for an **amino acid**.

Translation

Translation is the process by which the **code** (information) in **mRNA** is converted into a **polypepetide** (**protein**). **Enzymes**, **transfer RNA** (**tRNA**) and **ribosomes** are required. **Transfer RNA** has a '**clover-leaf**' **shape** (configuration). One end of **tRNA** always ends in '**G**', and the other in CCA, where **amino acids** are attached. There are **20** different types of **tRNAs** for the **20 amino acids**. The **anticodon** sequence of each **tRNA** is **related** to the **amino acid** it **carries**, and is **complementary** to the **codon** that specifies the **amino acid on mRNA**. The amino acids are carried to the **ribosome** by their respective tRNAs.

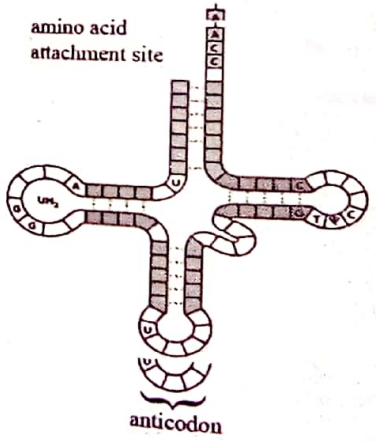

Figure 8.22. A proposed model for the structure of transfer RNA (RNA). The whole molecule is composed of about 80 nucleotides, but only 21 show complementary base pairing.

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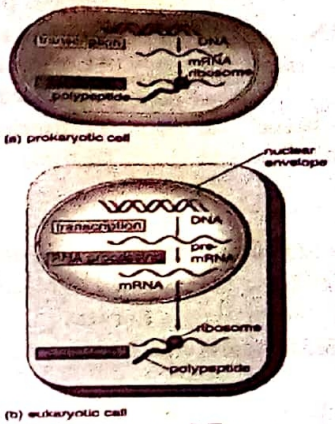
Ribosomes are composed of a **small** and **large sub-units**. They are made from **ribosomal RNA (rRNA)** associated with **protein molecules** in the cytoplasm. The **ribosome** holds in place the **mRNA**, **tRNA** and **enzymes** needed for **translations** until **peptide bond** forms between adjacent **amino acids**. There are **3 sites** in ribosome: the **'A' site** where the **amino acid enters**, the **'P' site** where pepetide bond forms between amino acids and **'E' site** where the freed **tRNA** and **polypeptide** are released.

Steps of translation

a) The **first 2 codons** on **mRNA** enter the **ribosome**.
b) **Two tRNAs**, having **complementary anticodons** to the **two codons**, bring their respective **amino acids** and **bind** to the **mRNA** at the **codons** being **read**.
c) A **peptide bond** forms between the two **amino acids**, and a dipeptide released.
d) The **ribosome moves forward** to the **third codon** on **mRNA**, and tRNA with **anticodon complementary** to the **codon** comes in position.
e) A **peptide bond** forms between the **second** and **third amino acids**.
f) The **ribosome** moves to the **4th codon** on **mRNA**, and the **whole process repeats itself** until a **'stop' codon** is encountered at which point **translation ceases**. As translation proceeds, a **polypeptide** and freed **tRNAs** are released from behind. The process of **translation** uses **energy** from the **breakdown** of **Guanosine triphosphate (GTP)**, instead of **ATP**.

$$GTP \longrightarrow GDP + p_i + \text{energy}$$

Figure 8.23 simplified summary diagram of the major structures and processes involved in protein synthesis in the cell



Introns and Exons

Immediately after transcription, **eukaryotic mRNA** has alternating regions called **exons** and **introns**. Introns are **non-coding** regions, and they are **cut out** and **removed** by a **protein complex** called **'spliceosome'** in a process called **splicing**. The coding **exon** parts are put together and **translated** into a **polypeptide** (protein). Such **post transcriptional processing** of **mRNA** is not needed in **prokaryotes** (bacteria) as their **mRNA does not** contain **introns**. So, bacterial **mRNAs** are **translated** directly without being **splicing**.

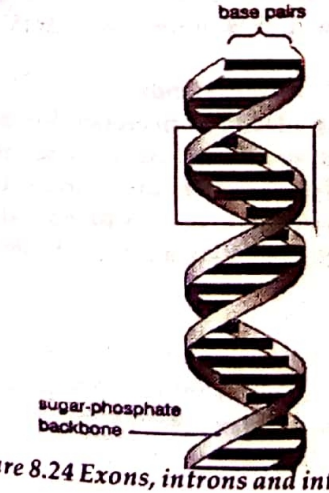


Figure 8.24 Exons, introns and intron splicing

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Transcription and translation compared in prokaryotes and eukaryotes

	Prokaryotes	Eukaryotes
1.	No nucleus	Nucleus present
2.	No splicing of mRNA	Splicing is required
3.	Transcription and translation are copied.	Translation & transcription are separated by time and space.
4.	Both processes occur in cytoplasm at the same time and place.	Transcription in nucleus; translation in cytoplasm.

Examples of proteins made in organism's body

a) **Structural proteins** - these are involved in **building body parts.**
Example: collagen (in **cartilage**), **keratin** (in **nails, feathers** and **hair**).

b) **Enzymes** - all **enzymes** are **proteins**, and they **catalyse** chemical **reactions** in **living things**.
Example: ATP synthase, DNA-polymerase, pepsin, etc.

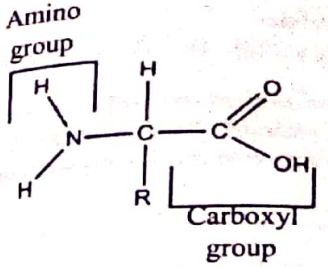
c) **Peptide hormones** - are **proteins** that act as **messengers** to keep internal body **conditions** of animals **constant**.

d) **Antigens** - act as cell **identify markers** (Example: determine blood groups), or allow **binding** of **hormones** or **foreign bodies** such as HIV.
Example: Antigens on **red blood cells, CD_4 protein** molecules on white **blood cells**, gp-120 on HIV, etc.

e) **Antibodies** - produced to **destroy foreign bodies** that may **hurt** the body.
Example: anti-A & anti-B antibodies which **agglutinate** (clot) **red blood cells** with **antigen A & B**, respectively; **HIV antibodies** (destroy some HIV antigens), etc.

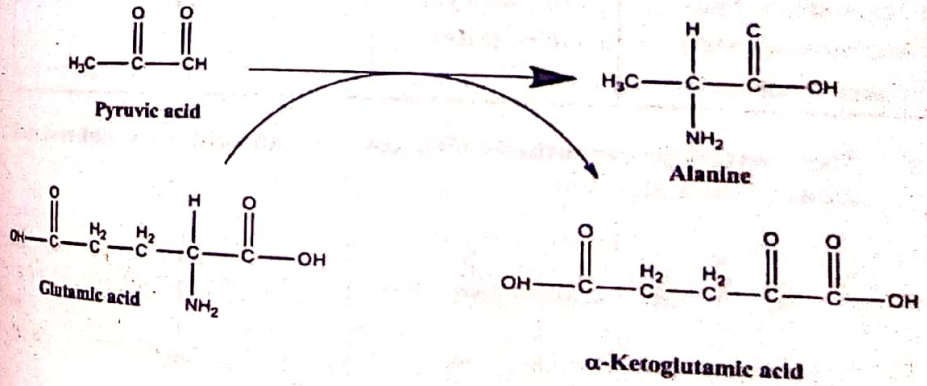
Amino acids

Amino acids are the **building blocks of proteins**. We obtain them from the **food** we consume. An average **adult** required about **50g protein per day**. The proteins are **digested (hydrolysed)** in to **amino acids** in the **gut** and absorbed by **active transport**. They are carried to **cells** by **blood** and put together into the **proteins** required by the body. The following is the **basic structure** of amino **acids**.



R = carbon chain and/or **rings** of **different forms**

There are **20** types of **amino acids** in proteins (table 8.3 & figure 8.25). They are **hydrophilic** or **hydrophobic**. They can also be **neutral, charged** or **polar**. Some of them are synthesized in the body by **transamination**. **Transamination** is the **transfer** of the **amino gruop** of **one amino acid** to a **ketoacid**. In the process, a **new** (a different) **amino acid** is formed from the **ketoacid**, and the original amino acid is converted into a **ketoacid**.
Example: Pyruvic acid (a ketoacid) is converted into **alanine** (an amino acid). In the process, **glutamic acid** (an **amino acid**) provides an **amino group** ($-NH_2$) and is converted into **ketoglutarate** (the new **ketoacid**).



The **amino acids** not **synthesized** by **transamination** must be **obtained** from food, and are called **essential amino acids**. They are 8 in **number**. These are **lysine, tryptophan, threonine, metheonine, phenylalanine, leucine,** isoleucine and **valine**.

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Sources of proteins

Animal sources: meat, fish, polutary, eggs, milk, etc. These are **balanced sources** of all **8 essential amino acids.**

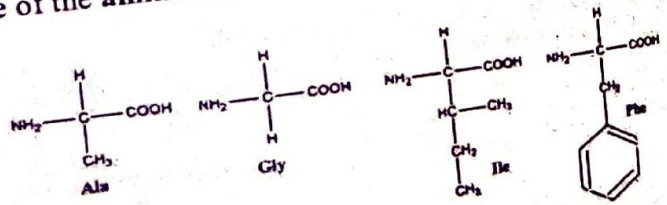
Plant sources: generally have **lower protein content** than **animal sources.**

- Quinoa, buck wheat, hempseed and amaranth (best in essential amino acids, but less overall protein content than some **cereals, nuts and pulses**).
- Cereals (wheat, rice, corn)
- Nuts, pulses (legumes)

Table 8.3 The **20 amino acids** found in **proteins** and their **properties**

Hydrophobic amino acids (neutral)	Hydrophilic Amino acids		
	Neutral & polar	Basic	Acidic
Alanine (Ala)*	Asparagine (Asn)	Arginine (Arg)	Aspartic acid (Asp)
Phenylalanine (Phe)	Serine (Ser)	Histidine (His)	Glutamic acid (Glu)
Isoleucine (Ile)	Cysteine (Cys)	Lysine (Lys)	
Proline (Pro)	Threonine (Thr)		
Leucine (Leu)	Glutamine (Gln)		
Tryptophan (Trp)	Tyrosine (Tyr)		
Metheonine (Met)	Glycine (Gly)		
Valine (Val)			

* The **3 letters** in **parenthesis** after each **amino acid** is the **abbreviated name** of the **amino acid.**



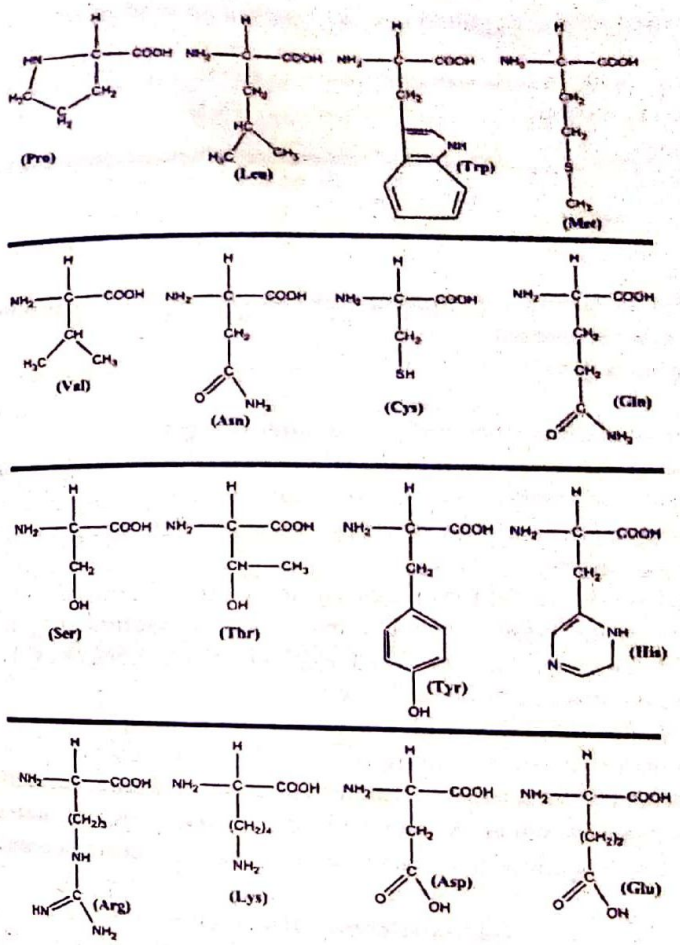


Figure 8.25. Structure of the 20 amino acids found in proteins.

Control of gene expression

Gene activity (expression) may depend on **age** and **sex** of the **organism** and/or the site of the **gene** in the **body**. Therefore, **not all genes** are **active always** and **anywhere**. **Some genes** are **expressed** at **childhood**, others at **adulthood**. **Example:** The **genes** that control **color** of our **iris** are found in **all cells**, but **expressed** in **iris cells only**. The **genes** controlling **milk production** in mammals

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are switched off during childhood and switched on at adulthood in females only.

Promoter - is **DNA base sequence** near a **gene** that **regulates** the **activity** of the **gene expression.**

Transcription factors - are **protein molecules** that bind to the **promoter** and **promote gene expression.**

Switching on a gene

Steps

a) **Transcription factors** bind to **promoter** of a **gene.**
b) **RNA - polymerase** binds to the **DNA-transcription factor complex.**
c) The **enzyme** is **activated** and moves **forward.**
d) The **enzyme transcribes antisense** strand of **DNA.** The **gene** is now said to be **expressed.**

A **fertilized egg** forms the **first transcription factors.** These pass to the newly formed **cells** of the **zygote.** Each cell makes its own **factors** using the factors passed to it **from** the first **fertilized egg cell.** Some **hormones** may act as **transcription factors**, and cause **cancer.**

Example: Oestrogen binds with a **receptor** in the **cytoplasm** and enters the **nucleus** of breast and **uterus lining** cells. This complex binds to **promoters** of **some genes** and activates them. The **activated genes** may cause **cell division** and lead to a type of **cancer.** A **drug** called **tamoxifen** binds with **oestrogen** and inhibits it. This complication occurs when oestrogen concentration increases in the body.

Gene switching off/silencing

There are **factors** that repress **gene action.** Some of them act **directly** on the **gene**, others act on **mRNA** and **silence** a gene and are called **post transcriptional silenceing factors.**

Example: short interfering RNA (siRNA). These are **unusually double stranded RNAs.** They are **very short (21-23** nucleotides long). Their mechanism of action is hypothesized as follows.

Steps

a) **Double stranded RNA (dsRNA)** is produced in the **nucleus** from different

b) This is **chopped** into **short fragments** by an **enzyme** called **'Dicer'**, and changed into **single stranded RNA.**
c) Then, the **antisense strand** of this **siRNA** binds with a **complex** of **molecules** called **RISC (RNA-induced silencing complex).**
d) The **siRNA** in the above complex binds with **mRNA** and allows RISC to **degrade**/cleave the **mRNA** into **small fragments.** As a result, **mRNA** is prevented from **translating** its **codons** into a **protein.** The **gene** is said to have been effectively **silenced** now.

Illustrative Examples

51. RNA differs from **DNA** because it is
(A) Linear (C) Single stranded
(B) Larger (D) All

Explanation:

- **RNA** differs from **DNA** because it is **smaller** and **single stranded.** It also contains **ribose sugar** and **uracil** in its **nucleotides** as opposed to **deoxyribose** and **thymine** found in **DNA.**

Answer: C

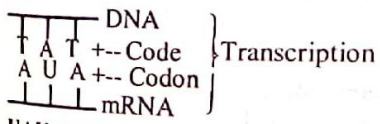
52. What are the **properties** of the **genetic code**? List them.

Explanation:

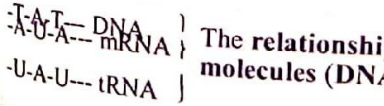
- The **genetic code** is **triplet code, non-overlapping, commaless, polar, degenerate** and **universal.**

53. The DNA triplet TAT would code for an amino acid carried by tRNA with the anticodon _____

Explanation:



- **UAU** would be the **anticodon** on **tRNA** that carries the **amino acid** coded for by **DNA** code **TAT.** This same code is copied in the form of **mRNA** as **AUA.**



The **relationship** between the **triplets** on the **molecules (DNA, mRNA** and **tRNA)**

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54. Which of the following is **removed** from **eukaryotic mRNA** during post transcriptional modification?
(A) Intron (C) Promoter
(B) Transcriptase (D) Exon

Explanation:
- **Introns** are **cut** out and **removed** from **eukaryotic mRNA** before it is translated.

Answer: A

55. Which of the following crops is considered to be the **best choice** for a human diet? **(EUEE, 2005 E.C)**
(A) Quinoa (C) Rice
(B) Maize (D) Wheat

Explanation:
- **Quinoa** is the best in its **essential amino acids**, though it is inferior to **cereals** in its **overall protein** content.

Answer: A

56. Gene silencing is the function of one of the following molecules. **(EUEE, 2005 E.C)**
(A) dsRNA (C) siRNA
(B) mRNA (D) tRNA

Solution:
- **Short infering RNA (siRNA)** is involved in **gene switching off (silencing)**.

Answer: C

57. Suppose the amino acid coding region in a **mRNA** is **1200** nucleotides long. How long is the **protein** interms of **amino acid number**? **(EUEE, 2005 E.C)**
(A) 1200 amino acids (C) 400 amino acids
(B) 600 amino acids (D) 300 amino acids

Explanation:
- Each **amino acid** is **represented** by a **triplet (3 bases)** in mRNA. So, if **1200** nucleotides are divided by **3**, the number of amino acids coded for by the **1200 bases** will be **400**.

Answer: C

58. Which group of **organisms** has a system of **protein synthesis** in which **transcription** and **translation** take place at separate times? **(UEE, 2004E.C)**
(A) In all eukaryotic organisms.
(B) In multicellular animals only.

(C) Only in prokaryotic organisms.
(D) In both prokaryotic and eukaryotic organisms.

Explanation:
- In **eukaryotes transcription** and **translation** are **separated** by both **time** and **space**. **Transcription** occurs in **nucleus**, and the **mRNA** so formed moves to **cytoplasm** where it is **translated** into a **polypeptide (protein)**.

Answer: A

59. How many different kinds of **amino acids** are there for **protein synthesis**? **(UEE, 2004 E.C)**
(A) 20 (B) 24 (C) 26 (D) 32

Explanation:
- There are **20 amino acids** for **protein synthesis**.

Answer: A

60. Which of the following is the **correct direction** of **transfer** of **genetic information** in most living things? **(UEE, 2003 E.C)**
(A) Protein --- DNA --- mRNA (C) Protein --- RNA --- DNA
(B) DNA --- Protein --- tRNA (D) DNA --- RNA --- Protein

Explanation:
- The information on **DNA** in nucleus is copied in the form of **RNA** and moves to **cytoplasm** where it is **translated** into a **protein**.

Answer: D

61. The building blocks of **DNA & RNA** are _____
(A) Monosaccharides (C) Amino acids
(B) Nucleotides (D) Glycerol & fatty acids

Explanation:
- **Nucleotides** are the **smallest structural units** of **nucleic acids (DNA & RNA)**.

Answer: B

62. What are the **differences** between the **nucleotides** of **DNA** and **RNA**?

Explanation:
- **DNA nucleotides** contain a **5-carbon sugar deoxyribose**, a **phosphate group** and a **base (adenine, guanine, cytosine** or **thymine)**. **RNA nucleotides** contain a **5-carbon sugar ribose**, a **phosphate group** and a base **(adenine, guanine, cytosine** or **uracil)**. So, **DNA** and **RNA** nucleotides differ from each other in their **sugar** content and **one base**.

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63. The process of **protein synthesis** is known as _____

Explanation:

- The process by which the **information** in **mRNA** is converted into **polypeptides** (proteins) is called **translation**.

64. The process of **RNA synthesis** is known as _____

Explanation:

- **Transcription** is the process by which **RNA** is **synthesized** to copy and **carry** the **information** on **DNA** in **nucleus** to the **cytoplasm**.

65. Which of the following is **true** about **DNA & RNA**?

(A) Both have pentose sugar
(B) Both are polymers of amino acids
(C) Both are single stranded
(D) Both are double stranded

Explanation:

- Both **RNA** and **DNA** contain a **5-carbon sugar (pentose)**. But, the **type** of **pentose** in each **differs**. **RNA** contains **Ribose**, and **DNA** **Deoxyribose**.

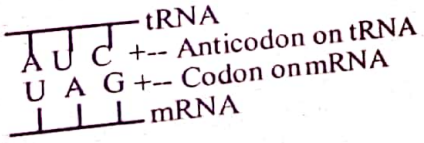
Answer: A

66. If the bases of an **anticodon** are **AUC**, what are the bases on the corresponding **codon triple**?

(A) CUA
(B) TAG
(C) UAG
(D) TUG

Explanation:

-



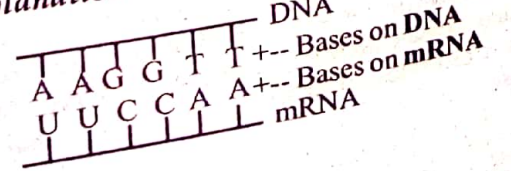
Answer: C

67. If the sequence of bases on DNA strand is **AA GG TT**, which of the following is the corresponding base sequence on the messenger RNA? **(CEE, 1995 E.C)**

(A) TT CC AA
(B) UU CC AA
(C) AA CC UU
(D) AA GG TT

Explanation:

-



Answer: B

68. **Transcription** is the **transfer** of the **genetic code** from a **DNA** molecule to a _____

(A) RNA molecule
(B) Ribosomal subunit
(C) Second DNA molecule
(D) Sequence of amino acids

Explanation:

- **Transcription** is the **transfer** of **genetic code** in **DNA** into RNA.

Answer: A

69. After transcription takes place, the DNA molecule:

(A) Disintegrates into its nucleotides
(B) Re-associates to form its original structure
(C) moves to ribosome
(D) Replicates itself.

Explanation:

- After **RNA** is **synthesized** on it (transcription), the **DNA** strands **rejoin** and **recoil** into their **original structure**.

Answer: B

70. After transcription takes place, the RNA molecule:

(A) Moves out into cytoplasm
(B) Attaches to the ribosome
(C) Replicates itself
(D) A & B are correct

Explanation:

- **RNA** molecule **moves** to the **cytoplasm** and **associates** with **ribosome** to be **translated** into a **polypeptide** (Protein).

Answer: D

71. The numebr of cytosine bases in a mRNA is equal to the number of _____

(A) Uracil
(B) Thymine
(C) Guanine
(D) Adenine

Explanation:

- **mRNA** is **complementary** to the sequence in the **DNA** where it was made. So, the number of **cytosine** in the **mRNA** is **equal** to that of **guanine**.

Answer: C

72. Transcription is most similar to:

(A) Chemiosmosis
(B) DNA replication
(C) Translation
(D) Facilitated diffusion

Explanation:

- **Transcription (RNA synthesis)** is most similar to **DNA replication**.

Answer: B

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73. Most genes of **eukaryotes** have both **introns** and **exons**. The _____ are **non-coding** sequences that are **not expressed** as **polypeptides**.

Explanation:

- **Introns** are the non-coding parts of the **DNA**. The **exons** are **translated** into **polypeptides** (Proteins).

74. How does **RNA polymerase** know when it has reached the end of a gene and that **transcription** should **stop**?

Explanation:

- A **stop signal** of **base triplet**, called the **terminator**, is built into the **DNA** molecule at the end of each gene.

75. What are the **3** major types of **RNA** transcribed from **DNA**?

- These are **messenger RNAs** (mRNAs) that **copy information** from **DNA** and take it to the **cytoplasm**, **ribosomal RNA** (**rRNA**) which is associated with **proteins** and form the **ribosome**, and **transfer RNA** (**tRNA**) that carries **amino acids** from **cytoplasm** to **ribosomes** during **translation**.

76. What are the **primary** and **functional mRNA**, and how do they differ?

Explanation:

- **Primary mRNA** is the **original mRNA transcript** synthesized on **DNA**. It has regions called **introns** (non-coding) and **exons** (coding). **Functional mRNA** is a **processed (spliced) mRNA** in which the **introns** are **removed**. It is the **functional mRNA** that is translated into a **polypeptide**.

77. Which of the 3 types of **RNAs** are **long lived**?

Explanation:

- **mRNAs** are produced **rapidly**, and **tRNAs** & rRNAs are produced **slowly**. Whatever proteins are produced, the functions of tRNA and **rRNA** are always the same. Therefore, they are **long-lived**. mRNA is **short-lived** because it is **immediately degraded** after translation as the cell needs **different proteins** at **different times**. At any given time, the total amount of **mRNA** is very **low** in the cell, but tRNA and **rRNA** are abundant.

78. Translation is the transfer of the genetic code from _____ to _____

(A) mRNA to a sequence of amino acids in a polypeptide.
(B) DNA to tRNA
(C) DNA to rRNA
(D) DNA to mRNA

Explanation:

- Translation is the conversion of the genetic code in **mRNA** to a sequence of amino acids in protein. **Answer: A**

79. A polypeptide (protein) is assembled on a _____

(A) DNA molecule
(B) Nuclear membrane
(C) Cell membrane
(D) Ribosome

Explanation:

- Translation occurs on the ribosome. **Answer: D**

80. Translation is composed of _____, _____ & _____ stages.

Explanation:

- Translation is composed of 3 stages namely **initiation**, **elongation** & **termination**.

8.4. Mutations

Mutation - is any **spontaneous change** in the **amount**, **arrangement** or **structure** of the **DNA** of an organism. They are **accidental mistakes** during **DNA replication**. They are **spontaneous** and **random** meaning that they can occur in **any gene** at **any time**. They are **rare** with **varying rates** between organisms. Their rate of occurrence is estimated to be about **one** in **fifty million** (**1 in 50×10^6**) base pairs. Each cell contains **six billion** (**6×10^9**) base pairs.

$$\Rightarrow \frac{6\times10^9 \text{ base pairs / cell}}{50\times10^6 \text{ base pairs/mutation}} = \frac{120 \text{ mutations}}{\text{cell}}$$

This implies that **120 mutations** per **newly emerging** cell can occur. Although these seems worrying, there are **2 major reasons** why they do **not affect genes** very **frequently**. These are:

a) Most of these **mistakes** (**mutations**) are **detected** and **repaired** by enzymes in the cell, and
b) Since **95%** of our DNA is **non-coding**, most mutations **do not fall** in the coding regions, and **do not** affect the **genes**.

But, mutation rates can be increased by **mutagenic factors** such as **chemicals** and **high energy radiations/particles**. **Chemicals** such as **mustard gas**, **colchicine**, **caffeine**, **formaldehyde**, chemicals in **tobacco**, some **drugs**, food **preservatives**, **pesticides**, etc. have been shown to be **mutagenic**. **Radiations/particles** such as **x-rays**, **y-rays**, **a & 1 particles**, **neutrons**, and **cosmic radiation** are also **mutagenic**. **Mutations** may involve **single base**, **parts** of

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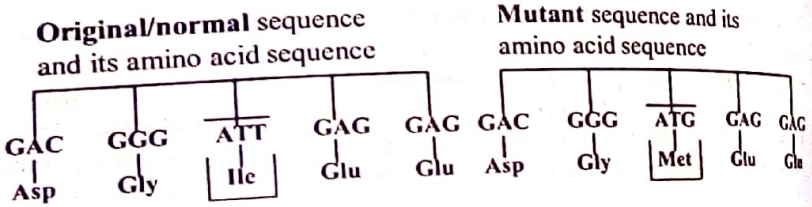
a **chromosome** or the **whole chromosome(s)**, and may be **serious**, or with varying levels of **negative** or **positive effects**. Two major **types** of **mutations** occur. These are **point** and **chromosome mutations**.

Point mutations - are **changes** that occur and affect a **gene**. It involves single **locus** (gene). It maybe due to **base substitution, addition** or **deletion**.

Chromosome mutations - are changes in the **number, arrangement** or **structure** (size) of **chromosomes (s)**. These have more **profound effect** on the organism than **point** (gene) **mutations**. **Example: Inversion, deletion, insertion, duplications, translocation, aneuploidy** and **euploidy**.

Point mutations

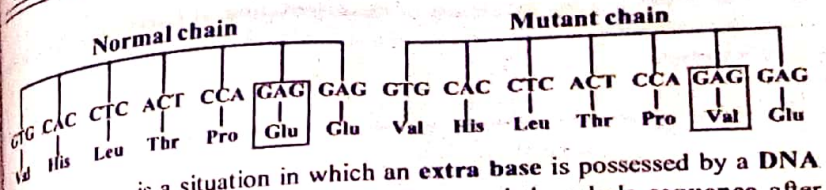
a) **Substitution** - is the replacement of a **base** with another **base**.
Example:



The replacement of 'T' by 'G' in the **mutant gene** has caused the production of a **different polypeptide** that has **methionine (Met)** instead of isoleucine (Ile) at the **third** position. This polypeptide **may** or **may not** be significantly **different** from the **original** one **functionally**. **Substitutions** may also result in a **termination** ('**stop**') triplet code that leads to the **ceasing** of **transcription** or production of a **shorter mRNA**. Such a **mRNA** may be **non-functional**, or may result in a **shorter polypeptide**. A **substitution** mutation that leads to a 'stop' code is called a **non-sense substitution**. Substitutions affect a single triplet only.

Sickle-cell anaemia

Haemoglobin molecules found in **red blood cells (RBCs)** are used to **transport oxygen** in our body. They are **globular proteins** made from 4 **polypeptide** chains folded on one another. When **valine** replaces **glutamate** in one of the polypeptide chains at the 6th position, the **haemoglobin** formed will be **abnormal**. **Enough oxygen** can **not** be **carried** by such a molecule. The **RBCs** in which such molecules are found will have the shape of a sickle } sickle cellS anaemia. The sickled cell often **fracture** and **stick together**, and **block capillaries**. Due to this, **enough oxygen fail** to be **transported** around the **body**, and the individual suffers from **anaemia**.



b) **Addition** - is a situation in which an **extra base** is possessed by a **DNA** sequence. There will be **one more** base, and the whole **sequence** after the added base will **change**. This leads to **frame shift** mutation. A totally different **mRNA**, or **no mRNA**, and a **non-functional**, or **no protein** is produced.

c) **Deletion** - is the **loss** of a **base** during **DNA replication**. Its outcome is **similar** to that of **base addition**. **Addition** and **deletion** are more **destructive** and **significant** than **substitutions**. There will be **one more** or **one less** base , and the **whole sequence** then after will be **altered**. Sometimes, a **triplet** is **added** (inserted) or **deleted** (missed). This leads to **one more** or **one less codon** in the **mRNA**. In turn, this leads to **one extra** or **one less amino acid** in the **polypeptide** (protein) chain so formed.

Mis-sense code - is a code that has **lost** its **sense** or **meaning** due to **deletion** or **insertion**.

Example: If a letter is **deleted** or **added** at some point in the following sentence, it will lose its **meaning**.

THE MAN WAS HOT AND RUN FOR HIS HAT. (Meaningful)

a) Delete **'A' from 'MAN'**, and **rearrange the letters in triplets**.
⇒ **THE MNW ASH OTA NDR UNF ORH ISH AT**
- The 1st and 3rd **triplet words** only have **English meaning** here, i.e. **'THE' & 'ASH'**

b) Add **'B' between 'W' & 'A' in 'WAS'**, and rearrange in similar way.
⇒ **THE MAN WBA SHO TAN DRU NFO RHI SHAT**
- Only the words **'THE'**, **'MAN'** and **'TAN'** seem to have meaning. Others lost their meaning due to the **frame shift** mutation.

Chromosome mutations

These refer to any **change** in the **number arrangement** or **structure** of the **chromosomes**. They most often occur during **meiosis** at **crossing-over** in **prophase-I**, and more **significant** than **gene (point)** mutations. They **affect** the organism if **essential** parts of **DNA** are involved, and usually result in **cell death**.

Example: Inversion, deletion, insertion, duplication, translocation, aneuploidy, polyploidy.

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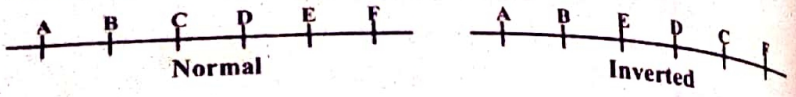
a) **Invertion** - is the **reversion** of the **orientation** of **parts** of DNA on the **chromosome.**

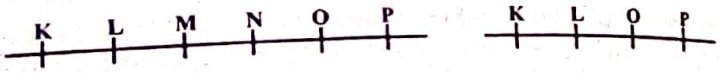
Example:



Genetic problems such as **leukaemia** (due to inversion on chromosome-16); **embryo miscarriage, growth failurity,** etc are caused by **inversion.**

b) **Deletion** - is the **loss of larger portion** of a **chromosome.** It causes decrease in **gene number.** Parts represented by 'M' and 'N' arelost below, for example,

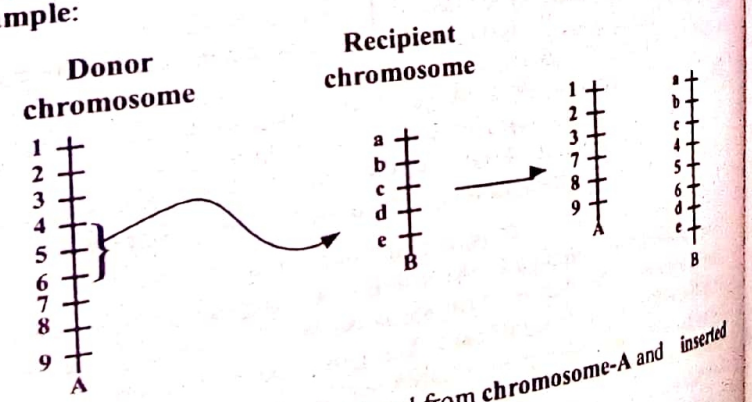
Example:



Chromosomal deletion may lead to disorders such as **Prader-Will;** syndrome-malfunctioning of **hypothalamus** (endocrine organ at the base of brain - controls **hunger, satiety, temperature, pain, fluid balance, emotions** and **fertility**).

c) **Insertion** - is the **addition** of a **piece** of **chromosome** from one chromosome into **another**. The **donor** chromosome **shortens**, and the **recipient** becomes **longer.**

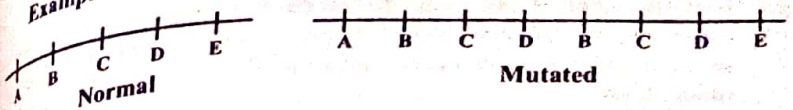
Example:



Portions **4, 5, & 6** have been deleted from **chromosome-A** and **inserted** between portions 'c' & 'd' on **chromosome-B.**

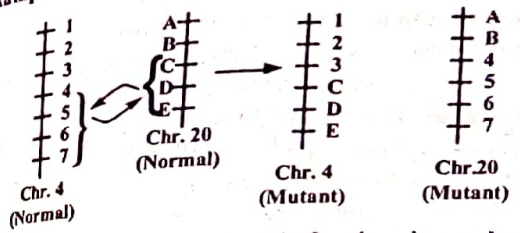
d) **Duplication** - is the occurrence of **multiple copies** of a **piece** of **chromosome.** These are **usually harmless,** but sometimes cause **abnormality.**

Example:



Portions **B, C, D** were **duplicated** in the **mutant chromosome.**

e) **Translocation** - refers to a condition in which two **non-homologous** chromosomes **exchange a piece** of chromosome **among themselves** (**Example**: between **chromosome-4** & **chromoseme-20**).



Translocation is often responsible for chronic **myelogenous leukaemia** (cancer of white blood cells).

f) Aneuploidy - is the **gain** or **loss** of **one** or a **few chromosomes** by a **cell.** If the cell is **diploid** (i.e 2n), an aneuploid organism will be characterized by **2n+1, 2n+2, 2n-1, 2n-2,** etc.

Example: **trisomy** of chromosome **16** & **21.**

It occurs as the result of chromosome **non-disjunction** which refers to the inability of one or **more homologous chromosomes** to **separate** at anaphase-I of **meiosis-I.** Both homologues move to the **same pole.** Later when they **separate** in **meiosis-II, some gametes** will be **devoid** of a copy of the **chromosome** in **focus,** while the **others** possess **an extra** of it. In **trisomy** of **chromosome-16** of **humans,** there are **3** copies of chromosom-16 instead of **2.** This causes **some disorders.**

Example: **poor foetal growth, muscular** and **skeletal anomality,** congenital **heart defects** and **underdeveloped lungs. Trisomy of chromosome-21 (Down's syndrome)** occurs as a result of the presence of 3 copies of **chromosome-21** instead of **2.** It causes **mental retardation,**

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heart defects and stunted growth. Trisomic individuals will have 47 chromosomes in every cell of their body instead of 46 (i.e. they are 2n+1 characteristically).

g) **Euploidy (polyploidy)** - is the increase in entire haploid sets of chromosmes in a cell leading to the possession of multiple haploid sets of chromosomes. Prefixes tri-, tetra-, penta-, etc are used to indicate the ploidy level.

Example: 3n is triploid, 4n is tetraploid, 5n pentaploid, and so on.

A triploid human cell, for example, will have 3n=23x3=69 chromosomes, Tetraploid will have 4n=23x4=92 chromosomes. Human haploid chromosome set is n=23; diploid set is 2n=46 chromosomes, and the later is the normal chromosomes number in our body.

Polyploidy is more common in plants than in animals, and is associated with advantageous features such as increases size, hardiness and disease resistance (this is called hybrid vigour). Most of our crops with large fruits, storage organs, flowers and leaves are polyploids. There are 2 forms of polyploidy: autopolyploidy and allopolyploidy.

Autopolyploidy arises naturally or artificially as a result of chromosome doubling with in the same species. Allopolyploidy arises from chromosome doubling of sterile hybrid, for example. The haploid chromosome sets, coming from each species giving rise to the hybrid, are doubled, and the hybrid will have 2 sets of chromosomes from each of its parents instead of the usual single set. It will then be a tetraploid, and this restores its fertility.

Consequences of mutations

Mutation consequences depend on two important factors: type of cell or gene in which they occur. There are two major types of cells in organisms: normal (somatic) body cells and gamete (sex) cells. Mutations that occur in somatic cells do not affect another individual, and will not pass to the next generation. Probably, these are more common, but go unnoticed. Sometimes, somatic mutations may increase cell division and growth. Such cells may give rise to a tumour (mass of cells produced due to uncontrolled cell division) which may be benign (does not affect other tissues) or malignant (parasitizes other cells) that becomes cancerous and distribute in the body. Somatic mutations have 4 possible outcomes. They may totally be harmless, damage the cell, kill it or make the cell cancerous. Cancer may kill the organism.

Mutations that occur in gamete cells or in pro-gamete cells (cells giving rise to gamete cells) can pass to the next generation and affect it. Similarly, mutations in different genes have different effects. There are 2 important genes that control and regulate cell division: proto-onco genes and tumour suppressor genes. Proto-onco genes stimulate cell division, and tumour suppressor gene prevents tumour formation that may lead to cancer.

Porto-onco genes $\xrightarrow{\text{Factors}}$ Onco genes $\longrightarrow$ Cell division

Normally, some factors activate the genes and cell division continues. Tumour suppressor gene suppresses uncontrolled cell division. But, if a mutation that activates proto-oncogenes and inactivates tumour suppressor genes occur, uncontrolled cell division and tumour formation will take place.

Benefits of mutation

Crossing-over, segregation and random assortment in meiosis, and random fusion of gametes during fertilization reshufle existing genetic material in the organism. They do not produce new genes. Mutations produce new genes. If a mutated allele (gene) is advantageous to the organism, its frequency increases in the successive generations. The number of organisms with the new gene will also increase as natural selection favours them. Therefore, mutations are the raw materials of evolution.

Example: Mutations in bacterial plasmids can provide antibiotic resistance to them. These mutant genes may pass from bacterium to bacterium and make all of them resistant to the antibiotic. Such mutations are advantageous to the bacteria in the presence of the antibiotics. Conjugation, transduction and transformation can help the passage of mutant genes between the bacteria.

Conjugation - plasmid transfer via "conjugation" tube.

Transduction - plasmid transfer from bacterium to bacterium using viruses as vectors.

Transformation -absorbing plasmids from dead bacteria, and being transformed by the genes on it.

Illustrative Examples

81. Which of the following is not the characteristics of a mutation?
(A) Randomness (C) Accidental mistakes (E) None
(B) Spontaneous (D) Rareness

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Explanation:

- All refer to the properties of mutations.

Answer: E

82. Although the **number of mutations** that occur in a cell is relatively high (120), why is it **not worrying** that much?

Explanation:

- **Most** of the **mutations fall** in the **non-coding** regions of DNA and have **no drastic effect** even if they are **not corrected**. **Secondly**, there are **inbuilt** mechanisms of **corrections** using **enzymes**.

83. What percent of our DNA is coding on average?

(A) 90% (B) 5% (C) 95% (D) 10%

Explanation:

- The **non-coding** part of **DNA** is about **95%**. The ramming **5%** is coding.

Answer: B

84. Which of the following is the cause of mutation?

(A) a-particle (C) Pesticide (E) Accidental mistakes
(B) X-ray (D) Some drugs (F) All

Explanation:

- **Accidental mistakes** during **DNA replication** are the basic causes of **mutations**. **Mutagens** such as **a-particle**, **X-rays**, **pesticides**, drugs and the like **increase** the **rate** of such **mistakes** and complicate the matter.

Answer: E

85. In some human liver cells there are **92** chromosomes per cell. What is the **ploidy level** of such cells? **(EUEE, 2005 E.C)**

(A) Haploid (C) Tetraploid
(B) Diploid (D) Hexaploid

Explanation:

- Since human haploid chromosome set contains **23** chromosomes, **23x4 = 92** is a **tetraploid** ploidy level.

Answer: C

86. What exactly happens during a **substitution** mutation? **(EUEE, 2005 E.C)**

(A) One base is replaced by a different base.
(B) An extra base is added during DNA replication.
(C) A base is missed out during DNA replication.
(D) Changes occur in the arrangement or structure of a chromosomes.

Explanation:

- **Substitution** mutation refers to the **replacement** of a **base** by **another.**

Answer: A

87. Which one of the following **is not** a mutation **(UEE, 2004 E.C)**

(A) DNA replication to form 2 daughter molecules.
(B) Gain of an extra chromosome by a cell.
(C) Deletion of a base pair from DNA.
(D) Loss of a chromosome by a cell.

Explanation:

- **DNA replication** is a **normal life process** that should take place when needed. It is **not mutation** (mistake).

Answer: A

88. Which one of the following refers to a **change** in a **gene**? **(UEE, 2003 E.C)**

(A) Mutation (C) Protein synthesis
(B) Replication (D) Transcription

Explanation:

- **Mutation** refers to **gene change**; **replication** refers to **DNA synthesis** on itself; **translation** is **protein synthesis** (conversion of the information residing in **DNA** into **protein**), and **transcription** is the **copying** of the **information** in **DNA** by synthesizing **RNA**.

Answer: A

89. Which of the following types of mutations is responsible for **sickle cell anaemia**? **(EUEE, 2005 E.C)**

(A) Addition of a base pair.
(B) Substitution of a base pair.
(C) Deletion of a base pair.
(D) A shift in the reading frame of the genetic code.

Explanation:

- **Sickle-cell anaemia** is caused by a **substitution** mutation that results in the **replacement** of **glutamate** by **valine** in one of the **4** polypeptide chains of **haemoglobin**. Consequently, **RBCs** with such a haemoglobin molecule will **attain sickle-shape**, and **fail** to carry **enough oxygen**.

Answer: B

90. Why is it that the typical diploid chromosome number if many organisms including man is an **even number**? **(UEE, 2003 E.C)**

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(A) It is only a coincidence.
(B) Chromosomes duplicate before cell division.
(C) Both parents contribute equal numbers of chromosomes.
(D) Meiosis reduces chromosome number.

Explanation:

- Whatever number the two **parents** contribute (**even** or **odd**), they contribute **equal number** of **chromosomes** that add together to give an **even diploid number (2n)** after **fertilization**.

Answer: C

91. Why is an individual with sickle cell anaemia suffer from oxygen shortage?

Explanation:

- His/her **sickle-shaped RBCs fructure, stick together** and block **capillaries**. The **haemoglobin** in them also, can **not** carry enough **oxygen**.

92. What is the type of mutation that arises from a change in the nucleotide sequence of the DNA at a particular locus of the chromosome?

(A) Gene mutation
(B) Numerical mutation
(C) Structural mutation
(D) Chromosomal mutation

Explanation:

- **Mutations** in a **locus** of a **chromosome** are called **point (gene)** mutations.

Answer: A

93. Which of the following types of **gene mutations** results in a replacement of just **one amino acid** by **another** in a **protein? (CEE, 1995 E.C)**

(A) Addition of a base pair
(B) Deletion of a base pair
(C) Substitution of a base pair
(D) Same sense mutation

Explanation:

- **Base-pair** substitution may cause the **replacement** of one amino acid by another. **Addition** and **deletion** of **base-pairs** results in profound change called **frame-shift**. **Same sense mutation** does not bring any change in amino acid type.

Answer: C

94. Which of the following types of chromosome mutations involves loss of a **chromosome** segment? **(CEE, 1995 E.C)**

(A) Aneuploidy
(B) Translocation
(C) Duplication
(D) Deletion

Explanation:

- **Loss of a chromosome segment is deletion**

Answer: D

95. Which of the following is **not** an example of **point mutation?**

(A) Substitution
(B) Deletion
(C) Addition
(D) Translocation

Explanation:

- Translocation is a chromosome mutation in which **2 non-homologous** chromosomes exchange a segment.

Answer: D

96. The more chromosomes an organism has, the more genetic variability it gets from_____.

Explanation:

- **Independent random assortment**, because the **number** of **possible arrangements** on meta-phase plate **increases**.

97. The longer the chromosomes of an organism, the more genetic variability it gets from_____.

Explanation:

- **Crossing-over**, because the **probability** of **recombination** of**linked genes** on **homologous chromosomes increases** if the chromosomes are **longer**.

98. When a **base pair** has been **inserted** into or **deleted** from the **genetic code**, the **triplet groupings** may be **altered**. Such a mutation is called a_____mutation.

Explanation:

- **Frame-shift**, because the triplets after the mutation will be rearranged again.

99. *Polyploidy* is most common in the_____kingdom.

(A) Bacterial
(B) Plant
(C) Fungal
(D) Animal

Explanation:

- **Euploidy (polyploidy)** is most **common** in the **plants**. Polyploidy may be induced in other organisms too, but usually they do not survive.

100. **Down's syndrome** is an example of _____

(A) Euploidy
(B) Polyploidy
(C) Aneuploidy
(D) A & B

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Conceptual Summary Example

1. List out the types of dominance with their explanations and examples

Explanation:

There are three major types of dominance. They are:

i. **Complete dominance** - is a condition where the effect of one of the allele of the gene masks the other.

Example: Allele for tallness was dominant over shortness in Mendels experiment.

ii. **Incomplete dominance** - is a condition where both alleles of a gene get expressed at the same time but not enough for dominance.

Example: four ockok flowers having Red flower crossed with White flower and produced Pink flower.

iii. **Codominance** - is a condition where both alleles remain dominant and express themselves.

Example: In humans when alleles A and allele B appear together both get expressed and forms AB blood type.

2. Clearly explain the distinguishing features between cross breeding and in breeding.

Explanation:

- **Cross breeding** - breeding between unrelated groups of organisms.
- **Inbreeding** - involves breeding animals or plants with their close relatives.

3. What is crossing over? How is it important in Meiosis?

Explanation:

- Crossing over is the process in which homologous chromatids exchange materials during meiosis I.
- It is important because it ensures greater variety of gametes.

4. Clearly explain the differences between sex linked and sex influenced traits. Use some examples for better illustration.

Explanation:

- **Sex - influenced traits** - are characteristics that are dominant in one sex and recessive in the other.

  Example: Baldness in Men

- **Sex - linked traits** - are those characteristics that are determined by genes on the X-chromosome.

  Example: Red-green color blindness.

- **Sex - limited traits** - are phenotypic characteristics that appear only in individuals of one sex.

  Example: face hair in male humans

5. Shortly discuss how the structure of DNA is constructed and its components.

Explanation:

DNA (Deoxy Ribo Nucleic Acid) is an amazing molecule in the science of biology. It is composed of two chain of nucleotide is composed of two chain of nucleotides that are joined by their bases. A nucleotide is composed of phosphate unit, deoxyribose and nitrogenous bases (adenine, thymine, cytosine and guanine).

6. What is DNA replication? Explain how it takes place.

Explanation:

DNA replication is the process by which the genetic material (DNA) of the cell reproduces itself prior to its distribution to the next generation.

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7. What is cloning? Explain the usefulness and drawbacks of cloning

Explanation:

- **Cloning** - is an asexual reproduction in which a single somatic cell develops into a complete, identical organism
- **Gene cloning** - is the process of making multiple copies of a gene.

8. Explain the relationship between DNA and chromosome.

Explanation:

- DNA is the back bone of chromosome. It constitutes about 35% of the chromosome part.
- Chromosome is thread like structure which is made up of complex material called Chromatin, which consists of fibers containg about 60% protein, 35% DNA and 5% RNA.

9. Clearly explain the relationship between genes and chromosomes? What are the functions of genes?

Explanation:

- Genes are segments of chromosome. A single chromosome contains many genes.
- Genes are units of heredity that codes for specific RNA molecule.

10. What are the roles of messenger RNA, transfer RNA and ribosomal RNA in heredity?

Explanation:

- **Messenger RNA (mRNA)** - carries messages directly from DNA in the nucleus to the cytoplasm during the making of protein.
- **Transfer RNA (tRNA)** - carries amino acids to the mRNA at the ribosome in order to form a polypeptide.
- **Ribosomal RNA (rRNA)** - helps to make up ribosomes along with proteins.

11. Explain the meaning of:

- Translation
- Transcription

Explanation:

i. **Transcription** - The process by which DNA makes RNA, It is facilitated by an enzyme RNA polymerase and the process occurs in the nucleus.

The triplet in DNA is transcribed into a codon sequence in mRNA. The base pairing rule is:

- Adenine pairs with Uracil (AU).
- Cytocine pairs with Guanine (CG).

For example:

- If the sequence in DNA triplet is

  AAA TAA CCG GAC

- The complementary sequence of codons in mRNA is

  AAA TAA CCG GAC
  ||| ||| ||| |||
  UUU AUU GGC CUG

ii. **Translation** - is the process by which the mRNA sequence is converted into an amino acid sequence. The process occurs in the cytoplasm.

12. What is mutation? Explain how it can be caused.

Explanation:

Mutation - is any spontaneous change in the amount, arrangement or structure of the DNA of an organism.

Some mutations can be caused by mutagenic factors such as chemical and high energy radiations.

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- Chemicals include colchine, caffiene, formal dehyde, etc
- The high energy radiations include x-rays, rays, y-rays, a-rays, B-particles, neutrons, etc.

13. Describe the types of mutation with their explanations and examples.

Explanation: There are two major types of mutations, they are:

i. **Point mutations** - is a mutation in which one base replaces the other in the DNA chains. It may involve base substitution, addition or deletion.

Example: Sickle-cell anemia

ii. **Chromosomal mutation** - a change in the gene arrangement in a cell as a result of breaks in the DNA molecule or structural change within chromosomes which involves inversion, deletion, insertion and duplication.

14. Define the term, phenotype, genotype, locus, allele, dominant allele, recessive allele, homologous and heterozygous.

Explanation:

- Genes are in chromosomes; the site a gene occupies in the chromosome **locus**. Different forms of a particular genes are alleles; they occupy corresponding loci on homologous chromosomes. Therefore, genes exist as pairs of alleles in diploid individuals.
- An individual that carries two identical alleles is said to be homologous for the locus. If the two alleles are different, that individual is said to be heterozygous for that locus.
- **Dominant allele** - is the allele that masks the effect of the other allele which is called recessive allele.
- **Homozygous** - is a gene in which the alleles are the same.
- **Heterozygous** - is a gene in which the alleles are separate.

15. Describe Mendel's principles of segregation and independent assortment.

Explanation:

- According to Mendel's principle of segregation, during meiosis the alleles for each locus separate or segregate, from each other. When haploid gametes are formed, each contains only one allele for each locus.
- According to Mendel's principle of independent assortment, alleles of different loci are distributed randomly into the gametes. This can result in recombination, the production of new gene combinations that were not present in the Parental (P) generation.

16. Solve genetics problems involving monohybrid, dihybrid and test crosses.

Explanation: Cross between homozygous parent (P generation) that differ from each other with respect to their alleles at one locus is called a **monohybrid cross**; if they differ at two loci, it is called a **dihybrid cross.**

POST TEST

1. A red rose bush cross-pollinated a white rose bush. The offspring have white flowers with red streaks. What types of inhertinace is this?

A. Co dominance
B. Crossing over
C. Multiple alleles
D. Incomplete dominance

2. Which describes the use of pieces of chromosomes to replace faulty or absent pieces that cause disease?

A. Karyotyping
B. Gene therapy
C. Non disjunction
D. DNA fingerprinting

3. Which base sequence **cannot** occur in RNA?

A. TGCTAT
B. GCUAGG
C. CCUAAC
D. AAGCCC

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4. A DNA strand with the sequence of bases shown below is transcribed to mRNA. ATT CGA GTT which is the **correct** base sequence in the mRNA?

A. ATT CGA GTT
C. TUU GCT GUU
B. TAA GCT CAA
D. UAA GCU CAA

5. Which term describes the failure of homologous chromosomes to segregate properly during meiosis?

A. Polyploidy
C. Non disjunction
B. Recombination
D. Independent assortment

6. What is the complementary DNA sequence for the following segment DNA? CTG GAA CTA

A. CTG GAA CTA
C. TAT GAA GGT
B. GAG GTT GAT
D. GAC CTT GAT

7. Which statement correctly describes the number of chromosomes in body cells and gametes?

A. Body cells and gametes are both diploid.
B. Body cells and gametes are both haploid.
C. Body cells are diploid and gametes are haploid.
D. Body cells are haploid and gametes are diploid.

8. Current reproductive techniques can be used to alter the genetic composition of a population. Some of these methods were also used in the nineteenth century by Gregor Mendel. How did Mendel use reproductive techniques in his experiments?

A. He artificially inseminated the pea plants to achieve wrinkled seeds.
B. He cloned the pea plants with round seeds to increase their food supply.
C. He creates a transgenic species by mixing tall pea plants ans short pea plants.
D. He artificially pollinated the pea plants to test for different genotypes in the offspring.

9. In humans, a smooth chin and a straight hairline are inherited as autosomal recessive traits. The alleles for each of the genes involved are:

Chin line - S: cleft chin hairline
s: smooth chin
W: widow's peak
w: straight hairline

A mother and son each have a smooth chin and a straight hairline. The father of the boy has a cleft chin and a widow's peak. The father's genotype must by

A. SS Ww
C. Ss Ww
B. SS WW
D. Ss ww

10. Pea plants were particularly well suited for use in Mendel's breeding experiments for all of the following reasons **except** that

A. Peas show easily observed variation in a number of characters, such as pea shape and flower color.
B. It is possible to completely control mating between different pea plants.
C. Pea have an unusually long generation time
D. Many of the observable characters that vary in pea plants are controlled by single genes.

11. A plant with purple flowers is allowed to self-pollinate. Generation after generation, it produce purple flowers. This is an example of

A. True breeding.
B. Incomplete dominance.
C. The law of segregation.
D. Poly genetics.

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12. Which of the following statements about Mendel's breeding experiment is correct?

A. None of the Parental (P) plants were true-breeding.

B. All of the F2 progeny showed a phenotype that was intermediated between the two Parental (P) phenotypes.

C. Half of the Fl progeny had the same phenotype as one of the Parental (P) plants, but only some of the F2 progeny did.

D. None of the above

13. What is the difference between a monohybrid cross and a dihybrid cross?

A. A monohybrid cross involves a single parent, whereas a dihybrid cross involves two parents.

B. A monohybrid cross produces a single progeny; whereas a dihybrid crosses produce two progeny.

C. A monohybrid cross involves organisms that are heterozygous for a single character, whereas a dihybrid cross involves organism that are heterozygous for two characters.

D. A monohybrid cross is performed only, whereas a dihybrid cross is performed twice.

E. A monohybrid crosses results in a 9:3:3:1 ratio whereas a dihybrid crosses gives a 3:1 ratio.

14. A cross between homozygous purple-flowered and homozygous white-flowered pea plants results in offspring with purple flowers. This demonstrates.

A. The blending model of genetics.

B. True-breeding.

C. Dominance.

D. A dihybrid cross.

E. The mistakes made by Mendel.

15. How many unique gametes could be produced through independent assortment by an individual with the genotype AaBbCCDdEE?

A. 4 C. 16 E. 64

B. 8 D. 32

16. Two plants are crossed, resulting in offspring with a 3:1 ratio for a particular trait. This suggests.

A. That the parents were true-breeding for contrasting triats.

B. Incomplete dominance.

C. That a blending of traits has occurred.

D. That the parents were both heterozygous.

E. That each offspring has the same alleles.

17. Two characters that appear in a 9:3:3:1 ratio in the F2 generation should have which of the following properties?

A. Each of the characters is controlled by a single gene.

B. The genes controlling the characters obey the law of independent assortment.

C. Each of the genes controlling the characters has two alleles.

D. Only A and C are correct.

E. A, B, and C are correct.

18. A 9:3:3:1 phenotypic ratio is characteristics of which of the following?

A. A monohybrid cross

B. A dihybrid cross

C. A trial hybrid crosses

D. Linked genes

E. Both A and D

19. A sexually reproducing animals has two unlinked genes, one for head shape (H), and one for tail length (T). its genotype is HhTt. Which of the following genotypes is possible in a gamete from this organism?

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A. HT
B. Hh
C. HhTt
D. T
E. Tt

20. It was important that Mendel examined not just the F1 generation in his breeding experiments, but the F2 generation as well, because

A. He obtained vary few F1 progeny, making statistical analysis difficult.
B. Parental traits that were not observed in the F1 reappeared in the F2, suggesting that the traits did not truly disappear in the F1.
C. Analysis of the F1 progeny would have allowed him to discover the law of segregation, but not the law of independent assortment.
D. The dominant phenotypes were visible in the F2 generation, but not in the F1.
E. All of the above

21. Mendel accounted for the observation that traits which had disappeared in the F1 generation reappeared in the F2 generation by proposing that

A. New mutations were frequently in the F2 progeny, "reinventing" traits that had been lost in the F1.
B. The mechanism controlling the appearance of traits was different between the F1 and the F2 plants.
C. Traits can be dominant or recessive, and the recessive traits were obscured by the dominant ones in the F1.
D. The traits were lost in the F1 due to blending of the parental traits.
E. Members of the F1 generation had only one allele for each character, but members of the F2 had two alleles for each character.

22. Which of the following is false, regarding the law of segregation?

A. It states that each of two alleles for a given trait segregate into different gametes.
B. It can be explained by the segregation of homologous chromosomes during meiosis.

C. It can account for the 3:1 ratio seen in the F2 generation of Mendel's crosses.
D. It can be used to predict the likelihood of transmission of certain genetic diseases within families.
E. It is a method that can be used to determine the number of chromosomes in a plant.

23. The fact that all seven of the pea traits studied by Mendel obeyed the principle of independent assortment means that

A. None of the traits obeyed the law of segregation.
B. The diploid number of chromosomes in the pea plants was 7.
C. All of the genes controlling the traits were located on the same chromosome.
D. All of the genes controlling the traits behaved as if they were on different chromosomes.
E. The formation of gametes in plants occurs by mitosis only.

24. Black fur in mice (B) is dominant to brown fur (b) short tails (T) are dominant to long tails (t). what fraction of the progeny of the cross BbTt BBtt will have black fur and long tails?

A. 1/16
B. 3/16
C. 3/8
D. 1/2
E. 9/16

25. In certain plants, tall is dominant to short. If a heterozygous plant is crossed with a homozygous tall plant, what is the probability that the offspring will be short?

A. 1/2
B. 1/4
C. 0
D. 1/6

26. A 1:2:1 phenotypic ratio in the F2 generation of a monohybrid cross is a sign of

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A. Complete dominance.
B. Multiple alleles.
C. Polygenic inheritance.
D. Pleitropy.

27. Refer to the result below to answer the following questions.

Atall plant is crossed with a short plant, and the progeny are all intermediate in size between the two parental plants.

A. Incomplete dominance.
B. Polygenic inheritance.
C. Complete dominance.

CRITICAL THINKING

1. Could a gene be involved in the growth of both stem cells and some kinds of cancer? Explain your answer
2. Why is an understanding of gene regulation in eukaryotes crucial to an understanding of developmental process?
3. A common belief about human genetics is that an individual gene alone determine his or her destiny. Explain why this misconcenption is.

ANSWER TO POST TEST

1. A
2.
3. A
4. D
5.
6. D
7. C
8. D
9. C
10. C
11. A
12. D
13. E
14. C
15.
16. A
17.
18. B
19. A
20. B
21. B
22.
23. D
24. D
25. C
26. A
27. C

Solution:

Down's syndrome results when human chromosome-21exists as 3 instead of 2 in the **diploid** individual. Such chromosome number **abnormality** in which **one** or **more extra** or **less** chromosomes exist in the cells of an organism, is called aneuploidy.

Answer: C

REVIEW QUESTIONS

1. **Fill in the second column of the following table with the appropriate amino acid or a 'stop' by referring to figure 8.20.**
(**Hint**: an amino acid may be coded for or represented by more than one triplet)

No.	Triplet codes on DNA	Amino acid /'stop'
1.	3'T', TT, TTC	
2.	TTA, TTG, CTC, CTT, CTA, CTG	
3.	TCT, TCC, TCA, TCG, AGT, AGC	
4.	TAT, TAC	
5.	TAA, TAG, TGA	
6.	TGT, TGC	
7.	TGG	
8.	CCT, CCC, CCA, CCG	
9.	CAT, CAC	
10.	CAA, CAG	
11.	CGT, CGC, CGA, CGG, AGA, AGG	
12.	ATT, ATC, ATA	
13.	ATG	
14.	ACT, ACC, ACA, ACG	
15.	AAT, AAC	
16.	AAA, AAG	
17.	GTT, GTC, GTA, GTG	
18.	GCT, GCC, GCA, GCG	

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19.	GAT, GAC	
20.	GAA, GAG	
21.	GGT, GGC, GGA, GGG	

II. Fill in the blank spaces with the appropriate word or phrase

22. The set of alleles of an organism is_____, where as its physical **appearance** is its_____.
23. _____occurs more frequently between genes far apart on a chromosome, and less frequently between genes closer together.
24. _____is the exchange of a piece between non-homologous chromosomes.
25. _____ machine is used in *in vitro* DNA cloning.
26. DNA fragments are separated according to their size by_____in genetic fingerprint preparation.
27. The process of protein synthesis is_____.
28. The process of mRNA synthesis is_____.
29. During DNA replication, the 2 strands unzip as_____bonds between the bases are broken.
30. After RNA processing, mRNA travels from the_to the_____, where it joins with a_____.
31. After a mRNA has been translated and leaves the ribosome, it is_____.

III. Match the words or phrase under column "B" with their explanations under column "A"

"A"	"B"
32. Template for RNA	A. Ribosome
33. Base triplet on DNA	B. Anticodon
34. Composed of RNA and protein and has 2 sub-units	C. Code
35. Base triplet on tRNA	D. Antisense strand
36. Base triplet on mRNA	E. Codon

IV. Match the enzymes under column "B" with their functions under column "A"

"A"	"B"
37. Synthesizes DNA from DNA-nucleotides	A. DNA helicase
38. Tie together broken DNA fragments	B. RNA polymerase
39. Break hydrogen bonds between bases and unzip DNA double helix.	C. DNA ligase
40. Convert HIV RNA into DNA	D. Restriction endonuclease
41. Synthesize RNA from RNA nucleotides	E. DNA polymerase
42. Cut DNA molecules at a specific sequence	F. Reverse transcriptase

V. Answer the following questions by saying true or false

43. In the improvement of cultivated crops and domestic animals, genetic engineering is faster than conventional selection breeding.
44. Proteins can not act as hormones in the body of organisms as they are macro-molecules.
45. If a mutation is beneficial for an organism, its frequency will increase in successive generations.
46. The genetic code is overlapping and degenerate.
47. In DNA double helix, purines always bind with pyrimidines, but not between themselves.
48. The two strands in the DNA double helix are anti-parallel to each other.
49. The law of independent assortment (Mendel's second law) holds true for linked genes.
50. In multiple allele inheritance, an individual will have more than 2 alleles at a locus.

VI. Answer the following questions by choosing the correct answer from the given alternatives.

51. Two parents of genotype RW were cross-bred. The alleles show codominance. What proportion of the progeny will look like their parents?
(A) 50% (C) 25%
(B) 75% (D) All will resemble their parents
52. An animal has a genotype GgRr. There is **no linkage** between the two genes. Which gametes can be produced by this animal?
(A) Gg, Rr, gR, gr (C) Gg, Gr, gR, gr
(B) GR, Gr, gR, gr (D) Gr, gg, rr, gr

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53. In multiple allele inheritance.
(A) A gene has more than two alleles.
(B) There are more than two alleles in the population.
(C) The alleles may be dominant, recessive or co-dominant.
(D) An individual will have only two alleles of the gene.
(E) All of the above are true.

54. Which of the following **is not** true?.
(A) Diploid individuals have two copies of a gene.
(B) True-breeders produce the same phenotypes generation after generation.
(C) Albinism is a recessive trait.
(D) Widow's peak is a recessive trait.

55. Assuming that E (gene for brown eye) is dominant over e(gene for blue eye) and that the father and the mother have both genes in their sex cells, what would be the proportion of homozygous children for eye color? **(CEE, 1995 E.C)**
(A) 100% (C) 75%
(B) 90% (D) 50%

56. In human ABO blood group system, genotype AA produces antigen-A, genotype BB produces antigen-B where as genotype AB produces both antigens. Which type of dominance is involved? **(CEE, 1995 E.C)**
(A) Complete dominance (C) Codominance
(B) Partial dominance (D) Over dominance

57. Color blindness is due to an X-linked recessive gene. Suppose a woman heterozygous for color blindness marries a normal visioned man. What proportion of their children will be color blind? **(CEE, 1995 E.C)**
(A) All the daughters (C) Half the daughters
(B) All the sons (D) Half the sons

58. Which bacterial species is used to transfer recombinant DNA to plants? **(CEE, 1995 E.C)**
(A) *Escherichia coli* (C) Nitrogen fixing bacteria
(B) *Agrobacterium tumefaciens* (D) Denitrifying bacteria

59. Regarding the ABO blood group inheritance, in which of the following matings a child of blood type 'O' is produced? **(CEE, 1995 E.C)**
(A) AAxOO (C) AOxAB
(B) BBxOO (D) AOxBO

60. What is the least number of alleles of a gene in a diploid organism?
(A) 1 (B) 2 (C) 3 (D) 4

61. The point at which 2 sister chromatids are attached together is
(A) Telomere (C) Centromere
(B) Histone (D) Cyctomere

62. All of the following are frame-shift mutations except_____.
(A) Deletion (C) Addition
(B) Substitution (D) Insertion

63. The process that produces new genetic material for evolution is_____.
(A) Segregation (C) Recombination
(B) Mutation (D) Crossing-over

64. In the life cycle of sexually reproducing organisms, which of the following processes restores the chromosome number from **haploid** to **diploid** state? **(UEE, 2003 E.C)**
(A) Fertilization (C) Chromosome replication
(B) Meiosis (D) Mitosis

65. In plant hybridization experiment, which one of the following is the best way to prevent self-pollination? **(UEE, 2003 E.C)**
(A) To remove the stamens (C) To cross pollinate flowers
(B) To cover flowers with bags. (D) TO make reciprocal crosses

66. What proportion of the number of chromosomes found in a normal cell of an angiosperm plant is expected in its endosperm tissue? **(UEE, 2003 E.C)**
(A) Half (C) Two times
(B) Same (D) Three times

67. If due to incomplete dominance, the F_1 plants from a cross of red flowered and white flowered parents are pink, which of the following ratios are expected in the F_2 generation? **(UEE, 2003 E.C)**
(A) 1 red : 2 pink : 1 white (C) 1 red : 1 pink : 1 white
(B) 2 red : 1 pink : 1 white (D) 3 red : 1 pink : 1 white

68. When two heterozygotes are crossed, the proportion of the progeny that exhibit the recessive trait is
(A) 25% (C) 75%
(B) 50% (D) 100%

69. Which of the following enzymes is used to put together DNA-nucleotides into DNA?
(A) DNA ligase (C) DNA gyrase
(B) DNA polymerase (D) DNA helicase

70. Which of the following bases is not found in DNA?
(A) Thymine (C) Guanine
(B) Uracil (D) Cytosine

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71. As was shown by Gregor Mendel in garden pea, what percentage of the F_1 generation of a monohybrid cross has the recessive phenotype?
(A) 75% (B) 50% (C) 25% (D) 12.5%

72. What do geneticists call the genotype in which the 2 alleles of a pair are identical? **(UEE, 2004 E.C)**
(A) Dominant (C) Homozygous
(B) Recessive (D) Heterozygous

73. The only source of new genetic material to act as a raw material for evolution is_____.
(A) mutation (C) Recombination
(B) Synapsis (D) Segregation

74. Which one of the following is refered to as the first law of Mendel? **(UEE, 2004 E.C)**
(A) The occurrence of alleles in pairs
(B) The dominance of one allele over the other
(C) The equal contribution of alleles by both parents.
(D) The separation of alleles during gamete formation.

75. In the genetic finger-print preparation, DNA fragments are made visible by
(A) Autoradiography (C) Electrophoresis
(B) Blotting (D) Fragmentation

76. Which of the following is the best way to check whether an individual having a dominant phenotype is homozygous or heterozygous for the trait? **(UEE, 2004 E.C)**
(A) To self the individual
(B) To cross it to a heterozygous individual.
(C) To cross it to homozygous recessive individual.
(D) To cross it to a homozygous dominant individual.

77. A genetic cross between two F_1 hybrid pea plants having yellow seed (dominant) will yield what percent green-seeded (recessive) plants in the F_2 generation **(UEE, 2004 E.C)**
(A) 0% (B) 25% (C) 50% (D) 75%

78. Which part of the angiosperm flower are both essential for the success of hybridization experiments? **(UEE, 2004 E.C)**
(A) Sepal and petal (C) Pollen and filament
(B) Stamen and petal (D) Gynoecium and androecium

79. Traits which are controlled by genes found on the sex chromoses are called
(A) Sex-influenced (C) Sex-limited
(B) Sex-linked (D) Sex-dimorphic

80. Which of the following bases is not found in RNA?
(A) Adenine (C) Thymine
(B) Cytosine (D) Uracil

81. Which of the following statements is **true**?
(A) DNA strands in the double helix are parallel.
(B) Bacterial DNA is associated with histones.
(C) Sense strand of DNA is transcribed.
(D) Antisense strand of DNA is transcribeed.

82. Color blindness is due to an x-linked recessive gene. Suppose a man with normal vision marries a color blind woman. What percent of their children will be color blind?
(A) 10% (B) 75% (C) 25% (D) 50%.

83. In the garden pea, the allele for round seed (R) is dominant to the allele for wrinkled seed (r), and the allele for yellow seed (Y) is dominant to that for green seed (y). An Rr Yy plant was self-pollinated and produced 1600 seeds. How many of these seeds are expected to be round-yellow?
(A) 900 (B) 600 (C) 300 (D) 200

84. A man with blood group O married a woman with blood group B. They produced a child with blood type 'O'. What is the genotype of the mother?
(A) BO (C) BB
(B) OO (D) AB

85. What is the term that describes all the alleles of all the genes in a population?
(A) Gene pool (C) Karyotype
(B) Genotype (D) Phenotype

86. In a certain plant genotype, RR produces red flowers, Rr pink flowers and rr white flowers. Which of the following ratios is expected in the offspring if a pink-flowered plant is crossed with a white-flowered plant?
(A) 1 pink : 1 white (C) 1 red : 1 pink
(B) 1 red : 1 white (D) 3 red : 1 white

87. Which of the following does not describe the genetic code?
(A) Overlapping (C) Degenerate
(B) Universal (D) Triplet

88. The chart that shows the hereditary history of humans over many generations is called.
(A) Punnett square (C) Pedgree
(B) Karyotype (D) Pyramid

89. What are the most likely causes of varatitions within species? **(UEE, 2004 E.C)**

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(A) Mitosis and asexual reproduction.
(B) Overpopulation and over production.
(C) Vegetative propagation and cloning
(D) Mutations and sexual reproduction.

90. One of the following is an important accelerator of gene mutation. **(EUEE, 2005 E.C)**
(A) Old age
(B) Alcoholic drinks
(C) Smoking
(D) Radiation

91. Genetic information in a DNA molecule is coded in the
(A) Base pairings
(B) Sequence of nucleotides
(C) Proportion of each base
(D) Turning pattern of the helix

92. The two strands of a double helix of DNA are held together by hydrogen bonds between
(A) Sugar and phosphate groups
(B) Sugars and nitrogenous bases
(C) Phosphate groups and nitrogenous bases
(D) Nitrogenous bases.

93. If the sequence of bases along one side of a DNA molecule is AAGCT, then the complementary sequence of bases on the other side of the DNA molecule is_____.
(A) TTCGA
(B) AAGCT
(C) UUCGA
(D) GGTAC

94. Which process is held responsible for chronic myelogenous leukaemia? **(EUEE, 2005 E.C)**
(A) Translocation
(B) Translation
(C) Transcription
(D) Duplication

95. In which of its contents RNA differs from DNA? **(EUEE, 2005 E.C)**
(A) Deoxyribose & guanine
(B) Ribose & uracil
(C) Ribose & thymine
(D) Phosphate & adenine

96. Among the following matings, where the ABO blood genotypes of the parents are shown, identify the mating in which all the children will have the same blood type. **(EUEE, 2005 E.C)**
(A) AOxBO
(B) AAxOO
(C) ABxBO
(D) BBxAO

97. During which stage of meiosis does crossing-over between homologous chromosomes take place?
(A) Telophase
(B) Anaphase
(C) Metaphase
(D) Prophase

98. Which of the following pairings between genetic disorders and mutation that causes them is **correct**?
(A) Sickle cell anaemia - Susbstitution
(B) Down's syndrome - Aneuploidy (trisomy of chromosome-21)
(C) Prader-willi syndrome - Deletion
(D) Myelogenous leukemia - translocation
(E) All are correct

99. Which of the following is chromosome mutation?
(A) Aneuploidy
(B) Euploidy (polyploidy)
(C) Translocation
(D) All

100. How many possible phenotypes are there for the ABO blood groups?
(A) 16 (B) 8 (C) 4 (D) 6

Answer for Review Questions

I. Filling table

1. Phenyl alanine	8. Proline	15. Asparagine
2. Leucine	9. Histidine	16. Lysine
3. Serine	10. Glutamine	17. Valine
4. Tyrosine	11. Arginine	18. Alanine
5. 'Stop'	12. Isoleucine	19. Aspartic acid
6. Cysteine	13. Methionine	20. Glutamic acid
7. Tryptophan	14. Threonine	21. Glycine

II. Fill in the blanks

22. Genotype, phenotype
23. Crossing-over
24. Translocation
25. PCR
26. Gel-electrophoresis
27. Translation
28. Transcription
29. Hydrogen
30. Nucleus, cytoplasm, ribosome
31. Degraded

III & Iv Matching

32. D 33. C 34. A 35. B 36. E 37. E
38. C 39. A 40. F 41. B 42. D

V. True/false

43. True 44. False 45. True 46. False
47. True 48. True 49. False 50. False

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VI. Multiple choice

51. A
52. B
53. E
54. D
55. D
56. C
57. D
58. B
59. D
60. B
61. C
62. B
63. B
64. A
65. A
66. D
67. A
68. A
69. B
70. B
71. C
72. C
73. A
74. D
75. A
76. C
77. B
78. D
79. B
80. C
81. D
82. D
83. A
84. A
85. A
86. A
87. A
88. C
89. D
90. D
91. B
92. D
93. A
94. A
95. B
96. B
97. D
98. E
99. D
100. C

UNIT NINE

EVOLUTION

In general, **evolution** means **change** over a period of time. It proposes that living things might have evolved from **relatively simple chemicals**, and explains what happens to organisms over time. Its present **working definition** is as follows:

> "**Evolution** is the change in **genetic composition** of a **population** over successive generations, which may be caused by **meiosis, hybridization, natural selection or mutation**. This leads to a sequence of events by which the population diverges from other populations of the same species, and may lead to the origin of a new species."

9.1 Theories of the origin of life

There are **5** main **theories** that try to explain the **origin of life**. These are:

a) Special creationism
b) Spontaneous generation
c) Eternity of life
d) Cosmozoan theory
e) Biochemical origin

a) **Special creationism** - attributes the creation of life to a **supernatural event** at a particular time in the past, and is **supported** by most **religions** and **civilizations** of the world. It is almost always linked to **religion** and **faith** as opposed to **evolution**, which is linked to **scientific thinking**. **Science** describes the world using **observations** and **empirical testings** that result in the development of **scientific theories**. Religion is less concerned with **empirical observable facts** and **testable hypothesis**. It mainly focuses on spiritual matters that can not be seen, **touched** or **measured**. According to this theory, some supreme being (**God**) created life on earth. But, there are different **versions** of **creationism**. These are: **Young earth creationism, old earth creationism, day-age and gap creationism, progressive creationism, theistic evolution/evolutionary creationism** and **intelligent design.**

Young earth creationism - suggests that the earth is only a **few thousand** years old. Earth was created in **six 24-hour days**. The earth is **round** and revolves around the **sun**. According to this theory, **geological events** are interpreted in light of **Noah's flood.**

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Old earth creationism - accepts that life was created by **God** as it was put in **Genesis** but explains that the **earth** is **very old**.

Day-age and gap creationism - considers that each of the '**six-days**' in **Genesis** represent a **billion** or so years of geologic time, making the 'days' just **symbolic**. It also discusses a **large gap** between the formation of **earth** and creation of **organisms**. The gap may be in **millions** or **billions** of **years**. Both accept that the **creation story** actually took **longer** than **6-earth days**. Though it **does not accept evolution theory**, it **accepts** the scientific evidence that the earth is several billion years old.

Progressive creationism - accepts **fossil record** of a series of organisms, but **denies genetic relationship** between **ancient** and **modern species**. It sees them as **unique creations**.

Theistic evolution/evolutionary creationism - proposes that God '**invented**' **evolution** and takes **active part** in its **process**. This theory is promoted by **catholic church pope** and main line **protestants**.

Intelligent design - latest **version** of **creationism** that claims that life **developed** due to a **combination** of **natural forces** and the **intervention** of **supernatural forces**. **Supporters** of this theory give **sophisticated arguments** based on **mathematics** and **cell biology**. Their idea is to show that the theory of **evolution** is **incorrect**.

b) **Spontaneous generation (abiogenesis)** - suggests that life can be obtained from **non-living objects** '**spontaneously**'. Few hundred years ago, people believed that **rotting meat** can give rise to **flies**, and **souring** wine, **bacteria**. The former was disproved by **Francisco Redi**, who placed a **piece** of **meat** in 2 jars. **Redi covered** one of the jars, and left open the other. Flies entered into the open jar and laid their **eggs**. The eggs hatched into **larvae (maggots)**, and then into **adult flies** through **pupae** within 3 **weeks**. **No flies** were seen emerging from the **closed** (covered) **jar**. The '**generation**' of **bacteria** from **broths** and **souring wine** was disproved by **Louis Pasteur**. He showed that such **mixtures** (liquids) go sour if and only if they are spoiled by **micro-organisms**. For this, he sterilized **broth** (made it free of microbes) by **boiling**. He used an **S-necked flask** to allow **air** in and out, but settle out **dust** and **bacteria** in the **bended part** of the flask's neck. When the sterile broth was moved to the **bended** part by **tilting**, it **interacted** with the **bacteria**, and went **sour** because bacteria grew and multiplied in it. The **two scientists proved** that both **micro-organisms** (bacteria) and **macro-organisms** (flies) arise only from pre-existing organisms. So, they disproved the theory of **spontaneous generation**. **Cells** come from **pre-existing cells** now a days! Though this is

the case, **scientists believe** that the **first cell** was formed as a result of **complex biochemical reactions** which occurred in the **primitive sea** and/ or **atmosphere**. This is a kind of **spontaneous generation**. According to **scientists** (evolutionists), once **life** was **formed** in that way, then after cells come from **pre-existing cells**.

c) **Eternity of life** - according to this theory, **life** is the **inherent property** of the **universe** and has **always existed** as the universe itself. It had **no beginning and end**. It is **eternal**.

d) **Cosmozoan theory** - claims that **life** on **earth** originally came from **elsewhere** in the **universe** (possibly from another **planet**). Life reached earth in the form of highly resistant **spores** (**Cosmozoa** = **germs** of the **cosmos**) carried by **cosmological structures** such as **meteorites** (fallen stone) and comets (star-like heavy bodies moving around the sun-sometimes fall on earth). The idea was **first proposed** by **Hermann Richter** in **1865** and **supported** by **Lord Kelvin** and **Herman Von Helmholtz**. Advocates of this theory accept that **life** is **eternal**. Later, in **1908** the **Swedish** physical chemist **Svante Arrhenius** supported **cosmozoan** theory by **forwarding** its **new version**. He proposed a different **mechanism** for **life** to **reach earth**. According to him, **bacterial spores** were carried between **planets** by **radiation pressure**. He named his theory **panspermia**. Most scientists do not accept both **Richter's** and **Arrheniu's** versions as the **high heat** of **meteorites** and **radiations** can **destroy** the **spores**.

Weak panspernia (or **pseudo-panspermia**) is the arrival of **organic compounds** from outer space. These were added to the **chemicals** on **earth** and gave rise to the **first life**, according to the **theory**. This theory is supported by two **evidences**: the landing in Australia of a **12% water meteorite** containing traces of **18 amino acids**, and the detection of complex **organic molecules** in **star-forming clouds**.

e) **Biochemical theory (abiogenesis)** - suggests that life on earth originated as a result of a number of **biochemical reactions** producing **molecules** which associated to form cells. This current idea is based majorly on the work of two scientists, **Alexander Oparin**, a **Russian** biologist who proposed the idea first in **1924**, and **John Haldane**, an **English** biologist who forwarded identical idea independently in **1929**. Both suggested that:

a) The **primitive atmosphere** had **no oxygen**.
b) There was **supply** of **energy**. Example: lightning, UV light, etc.
c) These forms of **energy** might have **driven reactions** that would synthesise **organic compounds** such as **amino acids**, **sugars** and **fatty acids**.

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According to **Oparin**, simple **organic compounds** were formed in the **atmosphere** (aqueous environment) and reacted to form more **complex molecules**. These formed **colloidal aggregates** (**'coacervates'**) which **absorbed** and **assimilated organic compounds** from their surrounding like a **matabolizing cell**. The **coacervates** were subject to **natural selection**, and eventually gave rise to the **first true cell**. **Haldane** proposed similar event in the **sea**. A **'hot dilute soup'** of **organic monomers** and **small polymers** were formed in **sea** according to him. He called this **'soup'**, **'prebiotic soup'**. So, **Oparin's 'coacervates'** and **Haldane's 'prebiotic soup'** gave us the **combined biochemical theory** of **origin** of **life (Oparin-Haldane view)**.

Evidence for biochemical origin of life

Stanley Miller conducted his famous experiment known as the **'spark discharge experiment'** in **1953**. He passed **electric sparks** repeatedly through a **mixture** of **gases** containing **methane (CH4)**, **ammonia (NH3)**, **water (H2O)** and **hydrogen (H2)**, thought to represent **primitive atmosphere**. He was able to produce **organic molecules** such as **amino acids**, **pentose** and **hexose sugars** and **hydrogen cyanide(HCN)**. **Amino acids** are building blocks of **proteins**, **pentose sugars** are used in **nucleotide formation**, **hexose sugars** are needed in **respiration** and **polysaccharides** (cellulose, starch) formation. It has been shown that **HCN** can act as a precursor of **nitrogenous bases** found in **nucleotides**.

Problems of Oparin-Haldane hypothesis

a) Both **'left-handed'** (**levoratatory**) and **'right-handed'** (**dextrorotatory**) **amino acids** can be produced, but only **left-handed amino acids** are found in **living things**.

b) **Purines** (adenine and guanine) could not be produced in the laboratory under the same conditions as **pyrimidines** (uracil, thymine and cytosine).

c) **Miller** could not produce **polymers** from his **monomers**. But, recently it has been shown that **polymerization** can occur under appropriate conditions.

Biologist **John Desmond Bernal** suggested **3 stages in evolution of life.**

Stage 1: the origin of biological **monomers**

Stage 2: the origin of biological **polymers**

Stage 3: the **evolution** from **molecules to cell**

The first **2 stages** were **demonstrated**, and **extensive research** is being done on the **3rd stage**.

Evolution of autotrophs

According to **geochemists**, **earth** was **molten rock** with violent **volcanic activities**. Soon a thin **crust** of **rock** (the shell on which we live) solidified over the **hot core**. Many **gases** released from the **molten core**. These formed a **cloud** around the **earth**; and became the earth's **atmosphere**.

Geochemists - scientists who study the **condition** of **primitive earth**.

By studying **gases** released from **volcanoes**, most **geochemists** agreed on the composition of **primitive atmosphere** to be consisting of **nitrogen** (N_2), **carbon dioxide** (CO_2), **water** (H_2O), **hydrogen sulphide** (H_2S), **ammonia** (NH_3), **methane** (CH_4) and **hydrogen** (H_2). Such an atmosphere, where hydrogen is available, is called **reducing**, as opposed to today's **oxidizing** atmosphere with 21% oxygen. In the primitive atmosphere gaseous oxygen was **unavailable**.

The **abiogenetic Oparin-Haldane** evolution of life was hypothesized to have been occurred in such an atmosphere. The **level** of **oxygen** has **increased** after the evolution of **photosynthetic organisms**. The **age** of **earth** is estimated to be about **4.5 billion years**. The **first organisms** (heterotrophs, prokaryotes) appeared **3.5–4 billion years ago**. They had **no nucleus**, and may be, possessed **RNA** instead of **DNA** as their **genetic material**. These first life forms might have given rise to **3 lines**.

a) **Archeabacteria** - prokaryotes that survive locally, or in unusual habitats.
   Example: Thermophilic bacteria, sulpho-bacteria, methano-bacteria, halophilic bacteria (salt loving), etc.

b) **Eubacteria** - the vast majority of prokaryotes today
   Example: Ordinary bacteria, cynobacteria (blue-green algae)

c) **Eukaryotes** -modern **unicellulars** or **multicellulars** that are almost all **aerobic**
   Example: Protoctistans, fungi, plants and animals

The **reducing atmosphere** was gradually changed to **oxidizing** one about **2.4 billion years ago** as the level of **oxygen** started to rise because of **photosynthesis**. **Cyanobacteria** started releasing it about **3.5 billion years ago** but its level stayed low as it was absorbed by the huge **iron** existing on earth by then. It **rusted** the iron. Cyanobacteria are one of the earliest **autotrophs** (photosynthetic), but used **phycocyanin** instead of **chlorophyll** to absorb **sun light**. The cyanobacteria are termed **photoautotrophs**. **Chemoautotrophs** are the other categories of **autotrophs**. They used **chemical reactions** as a source of energy instead of **light**. They are also called as **archeabacteria** and inhabit **hostile** environments such as deep sea vents, hot springs, salty waters, etc. Chemo-autotrophs are belived to be one of the **first organisms** to inhabit **earth**.

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Example: Sulphobacteria use hydrogen-suphide (H_2S), methano-bacteria use methane (CH_4).

- **3.5-2.1 billion years ago: bacteria** were the only **life forms.**
- **2.4-2.1 billions years ago: oxygen** concentration started to **rise.**
- **2.1-1.9 billion years ago: eukaryotes** became **numerous.**
- 0.9 billion years ago: fungi-like organisms appeared.
- 0.6-0.7 billion (700-600 million) years ago: animals evolved.

The rise of oxygen in the atmosphere due to photo autotrophs led to the evolution of **aerobic respiration** as energy-releasing pathway. This release more energy than anaerobic respiration, and gave way to the a=evolution of more active organisms such as animals.

Illustrative Examples

1. Which group of organisms is found at a lower level of evolution according to the modern principles of biology? **(UEE, 2003 E.C)**

(A) Protists
(B) Fungi
(C) Animals
(D) Plants.

Explanation:

- **Protists** are **unicellular organisms.** They are evolutionarily closer to the common ancestor than **fungi, animals** and **plants.**

Answer: A

2. In the process of evolution of life on earth, which of the following four processes evolved last? **(EUEE, 2005 E.C)**

(A) Photosynthesis
(B) Aerobic respiration
(C) Chemosynthesis
(D) Photo-autotrophism.

Explanation:

Chemosynthesis evolved first and gave rise to **life. Photosynthesis** (Photo-autotrophism) evolved later and released **oxygen,** which led to the evolution of **aerobic respiration.**

Answer: B

3. During the course of evolution, which of the following events came before all the others? **(EUEE, 2005 E.C)**

(A) Origin of dinosaurs and other reptiles
(B) Availability of free oxygen in the atmosphere
(C) Origin of the oldest eukaryotic organism
(D) Origin of the first multicellular animals and plants

Explanation:

- Oxygen availability started to rise **2.4-2.1 billion** years age (b.y.a); **eukaryotes** became available **2.1-1.9 billion** years ago (b.y.a); and animals evolved 0.7-0.6 billion years ago.

Answer: B

4. From the times when oxygen was first produced on the plant, it took approximately_____years for the level to begin to rise.

(A) 1 billion
(B) 0.6 billion
(C) 0.1 billion
(D) 2 billion

Explanation:

- **Oxygen** was being produced by **cyanobacteria** since **3.5 billion** years ago, but its concentration started to rise by **2.4 billion** years ago. So, the **gap** between the times **oxygen started** being **produced** and the times its **concentration** started to **rise** is about **1 billion years.**

Answer: A

5. Which of the following are the first autotrophic organisms in the evolution of living things?

(A) Cyanobacteria
(B) Blue-green algae
(C) Sulphur bacteria
(D) Green algae

Explanation:

- **Cyanobacteria** (blue-green algae) are the **first photo-autotrophs. Archaebacteria** such as **sulphur bacteria** are the **first autotrophs,** but **not photosynthetic,** rather **chemosynthetic.**

Answer: C

6. All forms of the **cosmozoan** (panspermia) theory suggest that:

(A) Life forms arrived on earth from other celestial bodies
(B) Life forms arrived on earth due to radiation pressure
(C) Organic molecules arrived on earth in meteorites
(D) Other celestial bodies have been important in the origin of life on earth.

Explanation:

- The **core** point of **cosmozoan theory** suggests that other **celestial bodies** have **contributed** to the **origin** of **life** on **earth** in one way or another. Choice "A" is **not correct** because there is a **version** of the **theory** that suggests the **arrival** of **organic molecules** on earth instead of **life forms.** Choice "B" is **incorrect** because the **other version** of the **theory** speculates the arrival of life forms in **meteorites.** Choice

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"C" is **incorrect** as the other forms of **panspermia** suggest the arrival of **life forms** instead of **organic molecules**.

Answer: D

7. A genetic change in population that passes on through successive generations is known as:
   (A) Mutation (C) Selection
   (B) Evolution (D) Adaptation

Explanation:
- Any **change** in a **gene** in an **individual** is called **mutation**, but the **change** in the **genetic composition** of a **population** over successive generations is **evolution**.

Answer: B

8. Which of the following scientists developed the biochemical (abiogenetic) theory of the origin of life?
   (A) Haldane & Bernal (C) Miller & Bernal
   (B) Oparin & Haldane (D) Miller & Oparin

Explanation:
- **Alexaner Oparin** (a Russian) & **John Haldane** (an English) independently suggested the theory of **abiogensis** at different times.

Answer: B

9. Which of the following statements is **incorrect**?
   (A) Only 'left-handed' amino acids are found in organisms.
   (B) All stages of the evolution of life were demonstrated in the laboratory as an evidence for abiogenesis.
   (C) Stanley Miller could not produce purines and pyrimidines under similar laboratory conditions.
   (D) Not all versions of creationism reject evolution.

Explanation:
- Not all stages of the biochemical theory of evolution of life were demonstrated so far. The first and the second (i.e. monomer & polymer formation) were demonstrated in the laboratory. But, the evolution from molecules to cells (i.e. the third stage) is not shown.

Answer: B

10. What are the **3 versions** of **cosmozoan theory**?

Explanation:
- The **first version** suggests that **life** in the **form** of **spores** arrived on earth in **meteorites**. The **second** proposes the involvement of **radiation pressure** in the mechanism of carrying the **spores** to earth.

The **third** is called **weak panspermia (pseudopanspermia)**, and argues that **organic molecules** instead of **life forms** (spores), have reached the earth in **meteorites**.

9.2 Theories of evolution

These try to explain a **mechanism** by which **species** gradually **change**. For many years, **Europeans** believed that **earth** and **organisms** had been **created** about **6000 years ago**. Since then, they stayed **unchanged**. This idea was challenged for the first time in mid **1700s** by **George Buffon**, who suggested that:
- The **earth** was **older** than **6000 years**.
- **Organisms changed** over time in response to **environmental pressure** and **random events**. However, the following are **major theories** of **evolution**. These are:

A) **Lamarckism** - ("the theory of transformation") - was proposed by a French biologist **Jean-Baptiste Lamarck** at the beginning of the **19th** century. It claimed that the traits acquired by organisms during their life-time can pass to subsequent **generations**. In **1809**, he published a paper entitled "**Philosophie zoologique**". In this paper he described a two-part **mechanism** by which **species** gradually **change** (transform) and pass the change to the coming **generation**. The **2 parts** of his **theory** are:
   i) **Use** and **disuse**, and
   ii) **Inheritance** of **acquired traits**

Use and disuse - according to **Lamarck**, continuous use of a **structure** or a **process** leads to the **enlargement** of the **structure** or to the **development** of the **process**, and **vice versa**. **Little use** or **disuse** of a **structure** or a **process** leads to **size reduction** or **less development**. He forwarded 3 examples to illustrate his argument.
- The elongated **neck** of **giraffe**
- The elongated, webbed **toes** of **water birds**, and
- The smaller and undeveloped **wings** of **penguins**.

According to him, **giraffe's** neck became **longer** due to the **strain** over its life time to reach high branches, and this is a "natural tendency towards perfection". The **toes** of **water birds** such as **ducks**, became **longer** and **webbed** again due to the **strain** imposed on them to **swim** through water. As an example for **disuse**, he used **penguin wings**. Penguin wings became **smaller** and **underdeveloped** because they were **not continuously used** for **flying**.

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Inheritance of acquired traits - Lamarck believed that traits **acquired/changed** during an individual's lifetime, could be passed down to its offspring. The body of an individual may be built by exercise as muscle size increases, but this can **never pass** to the **offspring**. **Somatic** (body) **changes** resulting in **phenotypic changes** do not affect the **genes** in the **gametes** which give rise to the **new generations**. Only **genetic changes** that occur in the **gamete cells** can pass to the **offspring**. So, Lamarck's inheritance of acquired traits was disproved by **genetics**. But, his belief that **species** start as **simple** and move towards **complexity** (perfection) is still acceptable, though explained through **genetic changes**.

B) **Darwinism** - was forwarded by **Charles Darwin** and **Alfred Russell Wallace** in **1858**. They jointly published a paper called "*origin of species by natural selection*". **Darwin** visited **5** of the **Galapagos Islands** found about **600** miles away from the coast of **Ecuador** in **South America**. He collected **specimen** of different organisms and made **drawings** of them. Particularly, he studied the **finches** (types of birds) found on the **Islands**, and observed **similarities** and **differences** between them. According to him, an **"ancestral finch"** colonized the Islands from South American mainland and evolved into different species of **finches** such as **seed eaters**, **insect eaters** and others with their own respective **beak shapes**. After **150** years, **geneticists** confirmed this and made **family tree** of the finches based on **DNA similarities**. Darwin proposed that those organisms that were **best adapted** to their environment would have an **advantage** and be able to **reproduce** in **greater numbers** than other types, and pass on their advantageous **adaptations** to their **offspring**. Darwin's theory of **natural selection** (adaptive radiation) has **4 core ideas**.

a) **Fecundity** - all species tend to produce **more offspring** than can possibly survive.

b) **Variation** - there are **differences** among the offspring.

c) **Struggle for survival** - as there is **limited resource**, there will be **struggle** among the offspring for existence.

d) **Survival of the fittest** - those members of a species that are better **adapted** to the environment will **survive**.

So, Darwin's theory can be restated as follows:
"Those **members** of a **species** which are **best adapted** to their environment will **survive** and **reproduce** in greater numbers than others **less adapted**".

Table 4.1 comparison of Lamarckism and Darwinism

Aspects of theory	Lamarckism	Darwinism
Variation	Environmental changes force organisms to change	Natural heritable variations occur in organism
Survival	New features develop for survival. Example: long neck	Environment selects organisms with adaptable features
Inheritance	Acquired traits pass to the offspring	Adaptable (advantageous) variations (traits) pass to the offspring
Evolution	New species form over time	New species form over time

C) **Neo - Darwinism** - is a **revised version** of **Darwin's theory** of **evolution** by means of **natural selection**. It combines **Darwin's original theory**, **genetic** theory and **ethology** (animal behaviour). **Alleles** of **genes** determine **features** of organisms. A **population** evolves into **new species** when its **gene pool** is changed. **Gene pool** refers to **all alleles** of **all** the **genes** found in the organism. If an **allele** is **advantageous** in an **organism** in a given **environment**:

- The **organism** will **survive** and **reproduce** more.
- The **advantageous allele** will **pass** to the **offspring** of the organism.
- **Frequency** of the **advantageous allele** (gene) will **increase** in the **gene pool** of the **population**.

Mutations introduce **new variations** into **populations**. Any **gene mutation** can confer one of the following **3 fates** to the organism. The **gene** can be:

a) **Neutral** - the **mutation** will have generally no **harm** or **benefit** to the organism. Depending on the environment, its frequency may **increase, decrease** or **stay constant**.

b) **Advantageous** - **frequency** of an **advantageous gene** (allele) **increases** in the **gene pool** of a population.

c) **Disadvantageous** - **frequency** of such a gene will **decrease**, and eventually **disappear** from the **gene pool**.

Behaviour patterns of organisms can also be **neutral, advantageous** or **disadvantageous**, and may be used as a **factor** by **natural selection**.

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Modern example of selection in action

Mutations may give **resistance** to an **antibiotic** in **bacteria**. Most bacteria are **killed** by **penicillin** the first time it is used against them. **Resistant strains** may exist naturally because of their **genetic make up**, or **mutation**, **which** may confer resistance after the **application** of the **antibiotic** (penicillin).

If **penicillin** application **continues**, frequency of the **mutant bacteria** will **increase** as the **non-resistant** ones are **eradicated**. But, if penicillin application **stops**, frequency of mutant bacteria stays very low.

Illustrative Examples

11. The idea that the earth and organism were created at some specific time and stayed unchanged was challenged for the first time by
(A) Charles Darwin (C) Jean-Baptiste Lamarck
(B) George Buffon (D) Gregor Mendel

Explanation:
- Although he **could not convince** people at the time because he could not forward evidence, **Buffon** was the **first** to suggest that the **earth** is **very old** and **organisms change** over time

Answer: B

12. According to the theory of evolution as proposed by Charles Darwin, which one of the following is the most important deriving force of evolution? **(UEE, 2003 E.C)**
(A) Mutation (C) Genetic recombination
(B) Over-reproduction (D) Natural selection

Explanation:
- Organisms naturally tend to **over-reproduce**. **Mutation** and genetic **recombination** increase **variation** among individuals of a species. There is **scare resource**. Therefore, not all individuals of a species can survive as there is over-reproduction. So, **natural selection** operates on the **individuals** favouring those that **best fit** the **environment**. The selection is in different forms. **Examples: food** or **space** shortage, **diseases**, **natural calamities**, etc.

Answer: D

13. From which of the following sources did Charles Darwin get the idea that selection can change living organisms? **(UEE, 2003 E.C)**
(A) From the work of plant and animal breeders.
(B) From his observation of the Galapagos birds.

(C) From the evolution theory of Lamark
(D) From the book by Malthus

Explanation:
- Darwin first convinced himself that selection can change organisms, by observing the work of plant and animal breeders because they could modify the traits of organisms.

Answer: A

14. Which of the following is consistent with the concept of the gene pool **(UEE, 2003 E.C)**
(A) Species with large gene pool have low potential of adaptation.
(B) Endangered species usually have small gene pool.
(C) large gene pool is a hindrance for livestock improvement.
(D) Species with large gene pool are prone to diseases.

Explanation:
- **Endangered species** have **small gene pool**, species with **large gene pool** have **high potential** of **adaptation**, are **less prone** to **diseases**, and can easily be **improved**.

Answer: B

15. Which of the following theories is more accepted by scientists?
(A) Darwinism (C) Neo-Darwinism
(B) Lamarkism (D) A & C

Explanation:
- **Neo-Darwinism** which incorporates **genetics** and **ethology** (animal behaviour) in to the first **Darwinian** theory of **natural selection** is more acceptable among the scientists.

Answer: C

16. The paper entitled "**Philosophie zoologique**" was written in 1809 by _____
(A) Darwin (C) Aristotle
(B) Miller (D) Lamarck

Explanation:
- "**Philosophie zoologique**" describes the theory of **use**, **disuse** and **inheritance** of **acquired characteristics** and was written by **Lamarck**.

Answer: D

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17. Which of the following contributes to the theory of natural selection?
(A) Fecundity and variation (C) Survival of the fittest
(B) Struggle for survival (D) All

Explanation:
- Since there is scarce resource, there is always **struggle for survival**. There is also natural tendency of organisms to **over-reproduce** (fecundity). Natural variation also exists among organisms. All these lead to natural selection which favours survival of the fittest.

Answer: D

18. What are the possible features of any mutation that occurs in a gene?

Explanation:
- There are **3 possibilities**: the gene change (mutation) may be **neutral** imposing no observable effect on the organism, it may also be **advantageous** or **disadvantageous**. Frequency of a neutral **mutation** may **increase, decrease** or **stay stable** depending on the **environment**. Frequency of **advantageous** mutations increases, while that of the **disadvantageous** ones decreases.

19. Which of the following ideas is not part of Darwin's theory of evolution? **(UEE, 2004 E.C)**
(A) Over reproduction
(B) Use and disuse of body parts
(C) Existence of heritable variation
(D) Competition for scarce resource

Explanation:
- **Use** and **disuse** of **body parts** is part of **Lamarck's** theory of **transformation**.

Answer: B

20. What is the most probable selection pressure responsible for the evolution of green skin color in frogs inhabiting tropical rain forest? **(UEE, 2004 E.C)**
(A) Climate (C) Infection by pathway
(B) Reproduction (D) Predation

Explanation:
- The **environment** in **tropical rain forest** is **evergreen**. If there are **frog predators** in such an environment, those **frog variates** with **greenish color** will be **camouflaged** and **escape predation**. The number of other types of frogs will **decline**. Greenish colored frogs will increase in number over long period of time.

Answer: D

21. Which of the following is **true** about the evolutionary origin of groups of organisms? **(UEE, 2004 E.C)**
(A) The dinosaurs appeared before the origin of land plants.
(B) The earliest *Homosapiens* appeared before flowering plants
(C) The first photosynthetic organisms appeared before the oldest eukaryotes.
(D) The first animals appeared before the formation of free oxygen in the atmosphere

Explanation:
- The **evolutionary clock** goes as follows: formation of **earth** — oldest fossils of **prokaryotes** --- the **first photosynthetic** organisms — evolution of **free oxygen** --- **oldest eukaryotic** fossils --- first **animals** --- oldest **multicellular fossils** --- land **plants** --- age of **dinosaurs** --- first appearance of *homosapiens*.

Answer: C

22. Which of the following theories explains evolutionary changes of living things in terms of changes in their allele frequencies? **(EUEE, 2005 E.C)**
(A) Darwin's natural selection. (C) Neo-Darwinism
(B) Inheritance of acquired characteristics (D) Abiogenesis

Explanation:
- The theory of **New-Darwinism** which includes **genetics** and **animal behaviour** explains **evolution** in terms of **allele frequency changes**.

Answer: C

23. If a new mutant allele arises in a certain population, which of the following factors determines if the allele is going to be adaptive or non-adaptive? **(EUEE, 2005 E.C)**
(A) The environment in which the population lives.
(B) The rate at which the gene mutates.
(C) The population in which the gene is found
(D) The use and disuse of the gene by the population

Explanation:
- The **environment** in which the population lives determines the **fate** of a **mutation**. **Resistance** against **penicillin**, for example, becomes **adaptive** (advantageous) when penicillin is constantly utilized. In a **penicillin-free** environment, that mutation is **non-adaptive**.

Answer: A

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24. Which one of the following factors is **not important** for evolutionary change of a population? **(EUEE, 2005 E.C)**
   (A) Over population
   (B) Survival of all that are born
   (C) Existence of heritable variation
   (D) Insufficiency of natural resources

Solution:

- **Survival** of all **offspring** that are born in a population is **not** important for evolutionary change because it means that **natural selection did not operate** on the population.

Answer: B

25. At which of the following aspects of evolutionary theories do Lamarckism and Darwinism agree?
   (A) Source of variation among organisms
   (B) Mechanism of survival
   (C) Types of inherited traits
   (D) Formation of new species over long time.

Explanation:

- ***Both Darwinism and Lamarckism agree on the issue that new species form from populations of existing species over long period of time.*** But, they **disagree** on **source** of **variation** among individuals of the same species, **mechanism** of **survival** and **inheritance** of **traits**. According to **Lamarckism**, environment forces organisms to change, **new features** develop for existence, and **acquired traits** (somatic traits included) are inherited. On the other hand, **Darwinism** suggests that **inbuilt heritable genetic traits** are selected by the environment for the organisms existence. Such genetic traits pass to the next generation.

Answer: D

9.3 The evidence for evolution

Palaeontology, comparative anatomy, comparative embryology, comparative biochemistry, and plant and animal breeding experiments are used as evidence for evolution.

Palaeontology

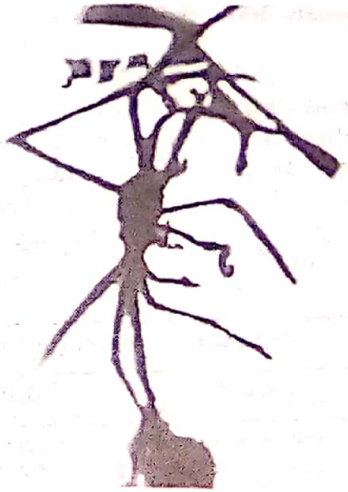
Palaeontology (Greek words: **Palaios** = ancient; **logos** = study) - is the study of **ancient life**, and is mainly based on **fossils** (the main direct evidence of past life).

Fossils (Latin words: **Fossus** = having been **dug up**) are **remains** or **traces** of organisms from the **remote past**. They can be categorized in to **two**. These are:

a) **Catagory-1 - Remains** of the **dead organism** or the **imprint** left from the remains.
   Example: Bones, teeth, hair, shells, skin impressions, whole organism impressions, etc.

b) **Catagory-2** - something made by the organism while it was **living** but **hardened** into **stone** since the death of the organism.
   Example: Foot prints, burrows, coprolite (animal faeces), etc.

The first type of fossils can be **actual organism** or its **parts** preserved in one way or another. A **spider** for example, can be trapped in **tree resin** and protected from the effect of **air** and **microbes**. In such cases, the organism may **not decompose** for **millions** of **years**. It is **fossilized** into **amber** and **preserved**. Sometimes, the organisms soft parts may decompose away and only the **exoskeleton** may remain.



But, most of the time fossils are **imprints** of whole organisms or their parts after death. **Death** is almost always associated with **decomposition**. But sometimes, dead bodies escape from this.

Stages of fossil formation

a) **Death without decomposition** - the organism dies in/close to **water** and covered with it. Microbes **decompose** its **soft parts** and leave hard body parts unchanged.

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b) **Sedimentation** - the hard parts are **buried** by **tiny particles** that settle out of the **water** over long period of time. This **sedimentation** further **insulates** the remain from complete decay. **Mudslides** and **landslides** into the area help the process. **Nature** of the **sediment** determines **nature and quality** of **fossil**. In this case, **particle size** and **chemical composition** play role. A **more detailed fossil** is formed by fine grained sediments. **Reddish** or **darker** fossils are created by **iron-rich** sediments or **phosphate-rich** ones, respectively.

c) **Permineralization** - the **sediments** are changed into rock because of pressure. If water seeps (percolates) through the **sediments** into the **hard part** remains, minerals in the water can dissolve and replace **molecules** of the **remains** of the organism. This is then compressed into **rock** which is somehow different from the surrounding sedimentary rock. It takes the **shape** of the **organism** or the **remain**. **Calcite** (calcium carbonate) is one of the **minerals** that can play important role in the fossilization process.

d) **Uplift** - earth movements such as **folding** and **earth quake** expose rocks that may contain **fossils**. **Rain** and **wind** can now **erode** the rock and expose the **fossils**.

Fossil dating

There are **2** ways of **fossil dating** (i.e. determining the age of the fossil): **stratigraphy** (relative dating) and **absolute** (actual) dating.

1) **Stratigraphy (relative dating)** - in **sedimentary** rock **layers** (strata), the **oldest** rocks (fossils) are found at the **bottom**, and **younger** (recent) ones are at the **top**. Geological time periods are divided into **eras** and **periods**. **Periods** are subdivisions of **eras**.

Table 9.2 An example of geological time table

Eras	Periods	Millions of years ago
Cenozoic era	♦ Quaternary ♦ Tertiary	65 million years ago up to now
Mesozoic era	♦ Cretaceous ♦ Jurassic ♦ Triassic	250-65 million years ago
Palaeozoic era	♦ Permian ♦ Carboniferous ♦ Devonian ♦ Silurian ♦ Ordovician ♦ Cambrian	560-250 million years ago.

So, **paleozoic era fossils** or **rocks** are **older** than **mesozoic** or **cenozoic** fossils /rocks and they are located at the **bottom most** in the **strata** of sedimentary rocks. On the other hand, **tertiary fossils** are older than **quaternary fossils** within the **cenozoic era**. The **most recent fossils** and **rocks** are found **nearest** to the **surface**.

2) **Absolute (actual) dating** - uses **radioactive elements** to determine the **actual age** of **rocks** or **fossils**. Basically, the **amount** of a **radioactive** element in a **fossil** is **measured** and **compared** to the estimated amount in the organism **before death**. The **difference** is **translated** into **age** of the fossil. There are **two** techniques. These are:
(i) Radiocarbon dating, and (ii) Potassium-argon dating

Radiocarbon dating - radioactive carbon (**C-14**) decays into **non-radioactive nitrogen** (**N**). **Fifty percent** of the existing **C-14** will decay after each successive **half-life**.

Half-life - is the **time needed** for **half** of the **atoms** of a **radioactive element** to **decay**. After one half-life, **50%** of the original element will have **decayed**, and **50%** remains. After 2 half-lives, **75%** will have been **decayed**, and **25%** remains, and so on.
In living things, the ratio of **C-14** to **C-12** (ordinary carbon) is **1 to 1 trillion atoms**, and this is maintained as the **loss** and **gain** of both carbon types is **balanced** through **respiration** and **photosynthesis**, or **feeding**. After death, **C-14** is lost through **decaying** without replacement. The half-life of **C-14** is **5730** years. After 5730 years, the amount of **C-14** that remains in the fossil will be **50%**, and the ratio of **C-12** to **C-14** will be **2:1**. After **11,460** years (second half-life), **25%** of the original **C-14** will remain, **C-12** to **C-14** ratio will be **4:1**. In that way, the amount of **C-14** **decreases** and **decreases** and finally **eliminated**.

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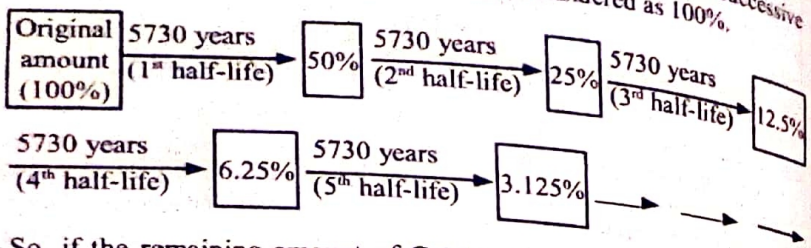
The following shows the amount of C-14 that remains after each successive half-lives starting from the original amount considered as 100%.



So, if the remaining amount of **C-14** is given, the **approximate age** of the **fossil** can be **calculated** by **adding together** the **half-lives** that have **passed**, or by **multiplying** the **half-life** by the **number** of **half-lives** that have **passed**.

Example: If it was found that a fossil contains only **3.125%** of the original amount of **C-14**, what will be the age of the fossil?

- The above illustration shows that **3.125%** is reached at after 5 half-lives. So, **5x5730 years = 28,650 years** will be the approximate age of the **fossil**.

Radiocarbon dating is **accurate** with **fossils** that are up to only 60,000 years old. For older fossils and rocks, a **radioactive** element with a longer **half-life** should be used. **Example: Potassium-40**.

Potassium - Argon dating - radioactive potassium (**K-40**) decays into **argon**. Its **half-life** is **1.3 million years**. This **dating** is suitable for fossils older than **60,000 years** though it can also be used for younger ones. It works in similar way as C-14.

Comparative anatomy

There are **two types** of **structures** of organisms: **analogous** and **homologous** structures. **Analogous structures** have **similar functions**, but different **morphology** and **anatomy**.

Example: Bird's wing and insect wing, mammalian legs and insect legs, etc (used for **flying** and **walking**, respectively).

They have **different evolutionary origin**, and hence **do not show common ancestor**. But, **homologous** structures have the **same basic anatomy** and a **common evolutionary origin**. They may have the **same** or **different** functions.

Comparative anatomy assumes that organisms with **homologous structures** are evolutionarily related, and probably share a **common ancestor**.

Example: Pentadactylity (having 5-digits) on fore limbs of mammals.

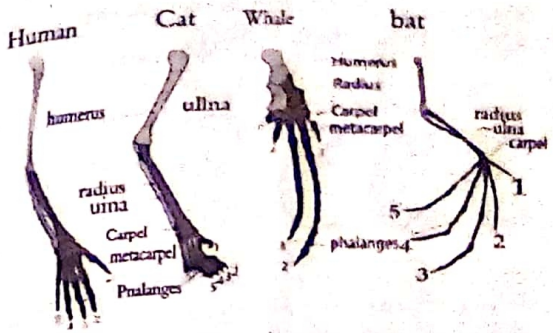


Fig 9.2. Forelimbs of some vertebrates

Table 9.3. The name & function of the forelimbs in some of the mammals

Mammals	Name of the forelimbs	Functions
Humans	Arm (hand)	- Manipulation (grasping, operation, etc)
Cats	Leg	- Walking, running
Whales	Flipper	- Swimming
Bats	Wing	- Flying

Comparative embryology

Comparative **embryology** studies **embryo development** in **vertebrates** before they **hatch** or are **born**. Vertebrate embryos **resemble** at early stages, and this shows a **common ancestor** according to the theory of **evolution**. They all have **gill slits** and **tails** at early stages, for example: The **gill slits** connect the throat to the external environment, and they are not gills. In **fish** and **tadpoles** (amphibian larvae), they contribute to **gill development**. In others, they close later in development of the embryo. The **embryonic tail** develops into **coccyx** (tail bone) in **humans**, and disappears in most species.

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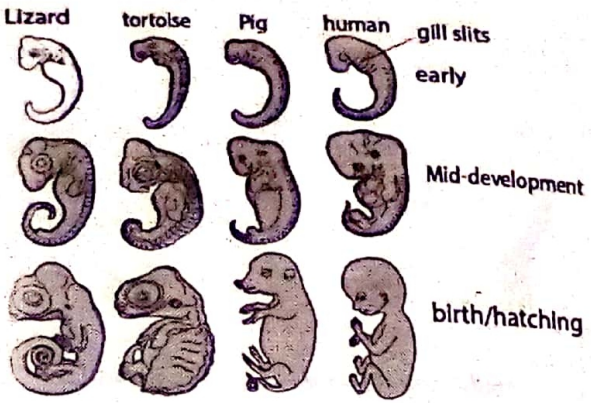

Fig 9.3. Similarities between embryos of vertebrates at early stages

Comparative biochemistry

Organisms that share **similar molecules** and **chemical pathways**, are closely related evolutionarily. **Comparative biochemistry** studies this. **Two** types of **molecules** have been used so far. These are:

a) **DNA** - base sequence of different organisms is compared.

b) **Proteins** - amino acid sequences in different individuals are compared. Example: **Cytochrome C** found in electron transport chain of respiration, and **haemoglobin** found in red blood cells to transport oxygen, are protein molecules that can be sequenced.

More closely **related species** have **most similar DNA** and **proteins**, and vice versa.

Example: We humans share 98% of our DNA with **Chimpanzees**, but only **50%** with **bananas**.

To **measure** the **similarity** between **two species, DNA** is **extracted** from both species and **heated** to change it into **single stranded** one (**ssDNA**). Then, they are **mixed** and **cooled**. The **DNAs** of the **2 species bind (hybridize)** at areas where their **base-pairs** are **complementary**. There will be **mismatch** in the areas where their bases are not complementary. The percentage of **binding** (hybridization) is measured, and used to determine the **extent** (degree) of **similarity** between the two species.

Haemoglobin molecule consists of **4 polypeptide chains** except in **lamprey** (primitive fish-like animal) where it consists of **single chain**. The 4 chains have **(show)** amino acid sequence **differences** in **different species**.

Example: Haemoglobin **amino acid sequence** of **humans** is different from the following **animals** by the **given number**. (aa, = amino acids).

- Macaque --- 8 aa,
- Bird --- 45 aa,
- Lamprey --- 125 aa,
- Dog --- 32 aa,
- Frog --- 64 aa,

Based on this information, **macaque** is **evolutionarily** more **related** to **humans** than the others. **Lamprey** is more **distantly related** to **humans**. The information also gives us some **insight** about **when** the given animals **diverged** from the **evolutionary line** that led to **humans**. Lamprey diverged very **long ago**, but **macaque** diverged very **recently**, relatively because the difference is not that much exaggerated.

Cytochrome amino acid sequence is used in similar way. **Not all molecules** can be used for all organisms in **comparative biochemistry**.

Example: Haemoglobin **can not** be used for **plants** and **algae** because they do not possess it; **molecules** which are **common** to **most organisms** are usually used.

Plant & animal breeding

Selective breeding - is the technique of choosing organisms with **desired traits** (**Example:** high milk yield, many seeds, etc) and **allowing** them to **reproduce** (breed).

Humans improved the yields of their stock **animals** and **crop plants** by **selective breeding**, and were able to produce **new varieties**.

Example: Different **varieties** of **domestic pigs** and **cattle** were the outcome of **selection** pressure by **humans** over many generations.

According to the theory of **evolution**, if **humans** can produce **new varieties** by **selective breeding, natural selection** (i.e. environmental pressure) can do the same and produce **new varieties**, and eventually **new species** over long period of time.

Illustrative Examples

26. What is the age of a fossil containing 25% of the original carbon-14 atoms?

Explanation:

- **Twenty five** percent of the **original carbon-14** atoms remain after the **second half-life** has passed. The half-life of **C-14** is **5730** years. So, **2x5730** years = **11,460** years. The fossils will be about **11,460** years old.

27. What is the age of a fossil containing **3.125%** of the original **carbon-14** atoms?

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Explanation:

- The above investigated amount of carbon-14 remains in the fossil after 5 successive **half-lives** have passed. So, **5x5730 = 28,650** years. The fossil will be **28,650** years old.

28. Why do you think that radiocarbon dating becomes less accurate with older fossils?

Explanation:

- The **half-life** of **C-14** is relatively **shorter (5730 years)**. With fossils that are **millions** of **years old**, its amount becomes **immeasurable** as it decays rapidly, relatively. **Radiocarbon dating** will be accurate with fossils that are up to **60,000** years old only.

29. Which of the following statements is **incorrect** about selective breeding?

(A) It shows that natural selection can produce variation.
(B) It can not provide evidence for evolution.
(C) It can produce new varites of crops and stock animals
(D) Organisms with a better yield can be created by it.

Explanation:

- All the others are correct except "B". **Selective breeding** which is used by **humans** in **plant** and **animal breeding**, is one of the **evidences** for **evolution** because, if humans can **modify organisms**, nature can do the same by environmental pressure.

Answer: B

30. We use protein biochemistry to classify organisms because

(A) Similar organisms will have similar proteins because they have similar DNA.
(B) The more similar the proteins, the more closely related the organisms.
(C) The extent of the similarity of plants measures how long ago species diverged.
(D) All are correct

Explanation:

- All **these theories** lie behind our utilization of **protein biochemistry** in classification.

Answer: D

31. What percentage of the amount of **carbon-14** originally present in a fossil would be left after its second half-life? **(UEE, 2003 E.C)**

(A) 75% (B) 50% (C) 25% (D) 12.5%

Explanation:

- After the **first half-life, 50%** will be left. After the **second half-life**, **half** of this will **decay**, and **25%** of the **original** will be left.

Answer: C

Choose the breeding method used by humans to produce new and improved varieties of plants and animals? **(UEE, 2003 E.C)**

(A) Crossing-over (C) Artificial selection
(B) Natural selection (D) Independent assortment

Explanation:

- Humans improve varieties of organisms by **selective breeding** **(artificial selection)**.

Answer: C

Why are fossils of soft-bodied organisms usually relatively rare in the environment? **(UEE, 2004 E.C)**

(A) They are generally small in size
(B) Their bodies decompose rapidly
(C) They all lived in environments where sedimentation did not occur.
(D) They were never common in environments in which they lived.

Explanation:

- **Soft-bodied** organisms **decompose rapidly**, and there is **less chance** of **fossilization**.

Answer: B

34. Carbon-14 has a half-life 5730 years. Suppose a fossil contains only 12% of the amount of carbon-14 normally present in living organism, how old is the fossil? **(UEE, 2004 E.C)**

(A) 5730 years (C) 17,190 years
(B) 11,460 years (D) 22,920 years

Explanation:

- After 3 half-lives, **12.5%** of the original **C-14** remains. So, **3x5730 = 17,190** years. What remained in the fossil is 12%. The fossil is, therefore, about **17,190** years old.

Answer: C

35. Which of the following pairs of molecules can give information about how much two species are evolutionarily related to one another? **(UEE, 2004 E.C)**

(A) DNA & proteins (C) Lipids & carbohydrates
(B) Starch & cellulose (D) Carbohydrates & proteins

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Explanation:

- **Molecules** found in **all**, or at-lest in **most organisms** are used in **comparative biochemistry**. In this regard, **DNA** and **proteins** are the best molecules to be used.

36. When two species are compared, which of the following sources of evidence is **least** informative about the degree of relationships between the species? **(EUEE, 2005 E.C)** Answer: A

(A) Nucleotide sequence of DNAs
(B) Amino acid sequence of proteins
(C) Glucose sequence of polysaccharides
(D) DAN-DNA hybridization.

Explanation:

- The repeating units of **polysaccharides** is mostly **glucose** alone. Therefore, it **does not** show **relatedness** or **unrelatedness**. Repeating units in **DNA** are **4 types**, those in proteins are **20 types**. So, they can bear some information about the **relationship** between species.

Answer: C

37. Which of the following characteristics can show the evolutionary relationships among organisms? **(EUEE, 2005 E.C)**

(A) Structures having similar functions
(B) Structures having same size
(C) Structures having common origin.
(D) Structures having different origins.

Explanation:

- **Structures** having **common origin** (i.e. **homologous structures**) show **common ancestory**.

Answer: C

38. Which of the following is the best source of fossils? **(CEE General science, 1995 E.C)**

(A) Volcanic rocks
(B) Sedimentary rocks
(C) Sand dunes
(D) Forest floors

Explanation:

- **Sedimentary rocks** are the **best sources of fossils** since fossilization process goes hand-in-hand with **sedimentation**.

Answer: B

39. What are the 5 evidence of evolution?

Explanation:

- **Palaeontology** (the study of **ancient life** using **fossils**).
- **Comparative anatomy** (the study of the relationship between organisms using homologous structures).
- **Comparative embryology** (the study of the **similarity** between embryos of **vertebrates**).
- **Comparative biochemistry** (**similarity** between **molecules** such as **DNA** and **proteins**).
- **Plant and animal breeding experiments**.

40. Explain the difference between **catagory-1** and **catagory-2** fossils, and show the outline of the stages of fossil formation.

Explanation:

- **Catagory-1** fossils are **remains/impressions** of the **dead organism**. **Example**: teeth, shell, feather, hair, bone, etc. **Catagory-2** fossils refer to something that was **made by** the **organism** when it was **living**.
  Example: burrows, coprolite, foot prints, etc.
  The **outline** of **fossilization** of any organism is as follows.
  Death without decomposition **Sedimentation** **Permineralization** **Uplift**.

9.4 The process of evolution

Natural selection is the **deriving force** behind **evolution**. It gradually brings about **change** in **populations** of the **same species** that may eventually develop into **different species**. The **modern concept** of **natural** selection is as follows.

"Those **members** of a **species** which are **best adapted** to their **environment** will survive and **reproduce** in **greater numbers** than others **less** well adapted. They will pass on their **advantageous alleles** to their **offspring**, and consequently the **frequency** of these **alleles** will **increase** in the **gene** pool. The **frequency** of the **individuals** with the **advantageous alleles** will also increase in the **population** in **successive generations**."

- Species - is a **group** of **related organisms** with **similar biochemistry**, physiology and **evolutionary history** that can **interbreed** and **produce** fertile **offspring**.
  Example: Humans, horses, donkeys, etc. are different species.

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Mule is a **hybrid** produced between **horses** and **donkeys**. It **can not** be considered as a **species** because it is **sterile** (non-fertile) meaning, it **can not produce** its own **offspring**. And, this also shows us that **horse** and **donkey** are different species since they **could not** produce a **fertile offspring** between them.

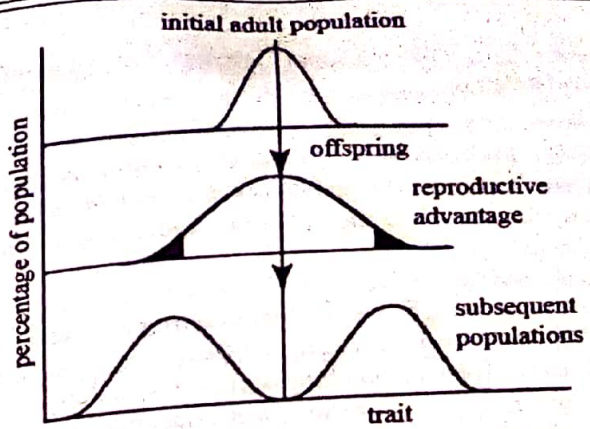
Types of natural selection

All types of **natural selection** work in **similar way**, but based on their effect on a **population**, there are **3 types** of **natural selection**. These are **directional**, **stabilizing** and **disruptive** selection.

a) **Directional selection** - is a type of selection that **works in favour** of one **extreme feature** of a population. It **works against** the **other features**.
Example: If a **fox population** has to establish in a very cold environment, those individuals with **longer** and **thicker fur** will be **favoured**. The **mean** and **range** of **frequencies** of **alleles** (features) will **shift** towards the **favoured** feature (i.e. longer and thicker fur). The foxes with the favoured feature must be the result of **mutations** or **new combinations** of alleles.

b) **Stabilizing selection** - works in **favour** of the **mean feature**, and **works against** the **extreme features**. It **operates** in a **stable environment**.
Example: Birth mass in humans: **very heavy** or **very light** babies usually **die** at or **after birth**.
Medium masses are **favoured**, and the **number** of **heavy** and **light** babies born is **decreasing**.

c) **Disruptive selection** - is a type of selection that **works in favour** of the **extreme features**, doing away with the **mean feature**.
Example: A disease wiping out **heterozygous** organisms but leaving the **recessive** and **dominant homozygotes** because they were **resistant**. Through time, **homozygotes** will **increase** in **number** in the area. The **number** of **heterozygotes decreases**, and may eventually become **extinct** if the selection pressure (the disease) **persists**.



The three types of selection: different selection (A), stabilizing selection (B), disruptive selection (C)

Mechanism of speciation

Speciation - is the **process** by which **new species** are formed from **populations** of a **species**. If **populations** of a **species** are reproductively **isolated** from each other by some mechanisms, they start to **mutate independently**. Genetic differences increase. Mutations that occur in one population do not pass to the other. They eventually **fail** to **interbreed**, and evolve into **distinct species** over very long time. There are **2 mechanisms** of **speciation**: **allopatric** and sympatric speciation.

Allopatric speciation - is a process in which different **populations** of a **species** develop into **distinct species** due to **physical/geographical isolation** that leads to reproductive isolation.
Example: - A river changing its direction of flow
- Creation of new land mass within a water body separating it into two (**Example:** the **Isthmus** of **Panama** between South and North America).

Isthmus - narrow strip of land connecting two larger land masses.
Sympatric speciation - is a process of **formation** of **new species** as a result of **reproductive isolation** caused by **seasonal**, **temporal** and **behavioural** isolation mechanisms. It does not involve geographical barrier. **Seasonal** isolation operates when the different **populations** of a species start **reproducing** at different **seasons** of the **year**. **Temporal isolation** refers to **mating** and

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reproduction during **different times** of the **day**. **Courtship differences** lead to **behavioural isolation**.

Polyploidy and its importance

Haploid cell - has **single set** of **homologous chromosomes**, while **haploid organism** has **single set** of **chromosomes** in all its **body cells**.
Diploid cell - has **2 sets**, and **diploid organism** has **2 sets** of **chromosomes** in all its body cells.
Polyploid cell - has **more than 2 sets** of **chromosomes**, and **polyploid organism** possesses **more than 2 sets** of **homologous chromosomes** in **all** its **body cells**.
Example: Triploid, tetraploid, pentaploid, hexaploid, heptaploid, etc contain 3, 4, 5, 6, & 7 **sets** of homologous chromosomes, respectively.
Hybridization and **polyploidization** played important role in **plant evolution**. Usually, **hybrid organisms** are **sterile**. When their **chromosome number** is **doubled**, they become **fertile** because **bivalents** can now easily form in **meiosis**. **Gametes** can form and **reproduction** continues.
Example: Modern **wheat** is produced by **hybridization** and **polyploidization**. In addition to **fertility restoration**, **polyploidization** also results in bigger **plants** with **bigger seeds** and **higher yield** (more seeds).

Divergent and convergent evolution

Divergence or **convergence** refers to the **lines** along which **speciation** proceeds in a situation.
Divergent evolution - is a process by which **populations** of a species evolve into **different species** due to **differing selection pressures** on each of them in their respective environments. It is **adaptive radiation**.
Example: Finches on Galapagos Islands, pentadactyl limbs in vertebrates, etc.

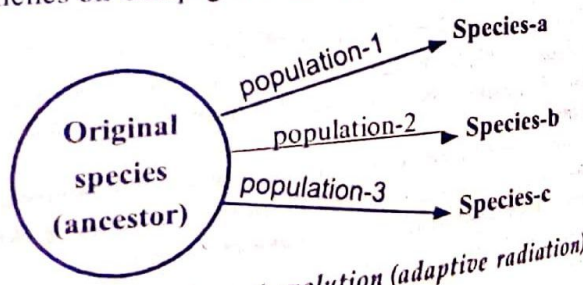

Figure 9.5. Divergent evolution (adaptive radiation)

Convergent evolution - is a process in which **unrelated species evolve similar structures** adapted for the **same function**. It occurs when **similar selection pressure** is imposed on **different species** occupying **similar niches**.

Niche - role of an organism in the **habitat**. The organisms , in this case, are evolutionarily **unrelated**, but they develop **similar adaptation**.
Example: (a) Wings of a **bird** and **bat** show **convergence**.
(b) The **giant armadillo** (North America), **giant pangolin** (Africa), spiny anteater (Australia/Oceania) and **giant** anteaters (South America) are evolutionarily **different**, but dig narrow cracks in the ground and eat **ants**.

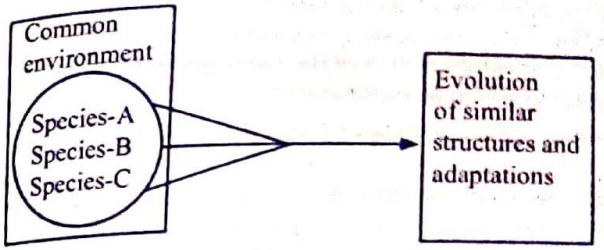

Figure 9.6. Convergent evolution

Illustrative Examples

41. The wings of a bird and a bat are the result of
(A) Divergent evolution
(B) Convergent evolution
(C) Stabilizing selection
(D) Disruptive selection

Explanation:
- **Birds** and **bats** are **unrelated**, they are **different species**, but they have **wings** to **fly**. So, they have evolved similar structures forced by similar selection pressure in the environment.

Answer: B

42. How does the concept of descent with modification explain both the unity and diversity of life?

Explanation:
- **Organisms** share **characteristics** (unity) because they share common ancestor. The great **diversity of life** occurs because new species repeatedly formed when descendants gradually adapted to different environments forming different species.

43. Explain how the following statement is **inaccurate**: "Anti-HIV drugs have created drug resistance in the virus."

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Explanation:

- An **environmental factor** such as **drug** does not create new traits such as drug **resistance**, but rather **selects** for **traits** among those that are already present in the population.

44. The forelimbs of humans, bats and whales are **homologous structures** with **similar** basic **anatomy**, but **whales** forelimb is very different from that of bats and humans. Explain why?

Explanation:

- **Natural selection** in an **aquatic environment** resulted in significant changes to **whale forelimb** anatomy.

45. Which of the following pairs of structures is **least** likely to represent homology?

(A) The wings of a bat and the arms of a human
(B) The haemoglobin of a baboon and that of a gorilla.
(C) The mitochondria of a plant and those of an animal
(D) The wings of a bird and those of an insect
(E) The brain of a cat and that of a dog

Explanation:

- Bird & insect wings are analogous, not homologous.

Answer: D

46. Sparrows with average sized wings survive severe storms better than those with longer or shorter wings. This illustrates:

(A) Disruptive selection
(B) Directional selection
(C) Stabilizing selection
(D) A & B are correct

Explanation:

- If the **2 extremes** (very long and very short) wings are selected **against**, this would be **stabilizing selection.**

Answer: C

47. What is the **difference** between **allopatric** and **sympatric** speciation? Which is more common? Why?

Explanation:

- In **allopatric** speciation, a **new species** forms while in **geographic isolation** from its parent **species**. In **sympatric** speciation, a **new species** forms in the absence of geographic isolation. Geographic isolation greatly reduces **gene flow** between populations, whereas ongoing **gene flow** is **more likely** in **sympatry**. As a result, **sympatric speciation** is much **less common** than **allopatric speciation**.

48. Allopatric speciation is more likely to occur to occur on an Island closer to the mainland than on an Island that is **more isolated**, but of the same size.

Explanation:

- **False, allopatry** is more **likely** to occur on the **more isolated** (more remote) Island because there will be **less gene flow** from the **mainland.**

49. Which of the following processes of evolution best explains the phenomenon known as **industrial melanism? (CEE General science, 1995 E.C)**

(A) Directional selection
(B) Disruptive selection
(C) Adaptive selection
(D) Stabilizing selection

Explanation:

- **Industrial melanism** is the phenomenon that occurred in **Europe** during the **industrial revolution** when surfaces and the environment were covered by **soot**. **Dark colored** moths were **favoured** by the environment. **Pale colored** ones were **selected against** because they could easily be **picked** by **predatory birds**. So, the **allele** for **dark color** has **increased** in the **population**. Individuals with the **dark color** also increased.

Answer: A

50. In spite of their ability to form **hybrids** when mated, which one of the following factors is more important to keep the **horse** and the **donkey** as separate species? **(UEE, 2003 E.C)**

(A) Their hybrid is not viable
(B) Their hybrid is sterile
(C) They can not form hybrid zygote
(D) They are ecologically isolated

Explanation:

- Even though they can form a **hybrid** (i.e. the **mule**), mule is **sterile**; it **can not produce** its own **offspring**. Therefore, **donkey** and **horses** stay separate species.

Answer: B

9.5 The evolution of humans

According to the theory of **evolution**, there has been a '**line of evolution**' for millions of years. This '**line**' gave rise to **old** and **new** world **monkeys**, the **great apes** and the different **species** of **humans**.

Homo sapiens - is the **modern** and **latest human species**. We are different from **primates** (great apes) in 2 major ways: by having **large brain** and by being **bipedal** (truly walking on 2 legs). **Chimpanzee** and **bonobo** (great apes)

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diverged from the evolutionary line that gave rise to humans about **6 million** years ago; **gorilla** about **8 million** years ago, and **orangutan** some **13 million** years ago. So, **chimpanzee** and **bonobo** are more related to **humans** than gorilla and **orangutan**. **Man** did not evolve from the **great apes**, but he had **common ancestor** with them. **Humans** and **chimpanzees** had **common ancestor** that lived about **6 million** years ago. There are **2 groups** of humans including pre-**humans**. These are **hominids** and **hominins**.

Hominids - includes **all species of humans** in the genera *Homo*, *Australopithecus* and *Ardipithecus*.

Hominin - encompasses all human species in the genus **Homo**.

Example: *Homo sapiens*, *Homo erectus*, *Homo habilis*, etc.

Lucy and Ardi

Both are important in the explanation of modern **man** and **chimpanzees**.

Lucy (*Australopithecus afarensis*) - was discovered by **Donald Johanson** and **Tom Gray** in **1974** at **Hadar** in **Afar** (Ethiopia). the fossil was about **3.2 million** years old. She was an **adult female** of about **25 years** of age. Her skeleton was about **40%** complete, and she was **bipedal** and partly **arboreal** (tree-dwelling). She was about **107**cm tall and about **28kg** in mass. **Lucy's brain** was about the **same size** as that of a **chimpanzee**. **Chimpanzee** is **not a bipedal**, but Lucy is **bipedal**. So, **bipedalism** evolved before **large brain** in the evolution of **humans**.

Ardi (*Ardipithecus ramidus*) - was first discovered in the **Afar desert** in **1992**. But, analysis was finalized in **2009**, and research paper published. **Ardi** was also a **female** fossil that is about **4.4 million** years old. She was also **bipedal**, but **closer** to the **common ancestor** of **humans** and **chimpanzees**.

Human evolution and brain size change

As **humans** evolved from **pre-humans** to **modern man**, **brain size** was **increasing** generally, and as compared to their **body size**. Species of *Australopithecus*, for example, had **brain sizes** ranging from **0.7% to 1.0%** of their body masses, while that of **modern man** ranges from **1.8% - 2.3%** of their body masses. Human (*H. sapiens*) brain uses **25%** of the resting energy **requirement**, while **brain** of **great apes** uses only **8%** of the **resting energy requirement**. This is another indication that **human brain** is **larger** and more **busy**. In humans, **large brain**, **longer legs**, **nimble fingers** and **straight spine** allowed:

- **Faster running** in a more **upright** posture
- **Planning in advance** to avoid attack.

- **Development** and **utilization** of **tools** and **weapons**.

Are humans still evolving?

According to fossil evidences, *Homo sapiens* appeared in Africa (near Ethiopia), and **migrated** to other parts of the world about **100,000** years ago. Different **selection pressures** (environments) forced the different **human populations** to evolve along **different lines**. In **cold regions**, for example, humans developed:

- **Shorter**, **squatter** body shape to **reduce surface area to volume** ratio and **minimize heat loss**.
- **Increased** layer of **adipose tissue** and **hairiness** to **reduce heat-loss**.

Based on recent **genetic analysis**, humans are loosely classified into **3 races** which have their own **sub-divisions**. These are:

a) **Africans (Negroid)** - Africans and Melanesian of South pacific.
b) **Eurasians (Caucasoid)** - has 3 divisions
   i) **Nordic** - people from **Scandinavian** and **Baltic** countries, **Germany**, **France** and **British**.
   ii) **Mediterranean** - Southern France, Spain, Italy, Wales, Egypt, Jews, Arabs, Afghanistan, Pakistan, and India.
   iii) **Alpine** - countries from Mediterranean to Asia.
c) **East Asian (Mongoloid)** - most numerous of all, and has **3** sub-divisions.
   i) **Eastern Siberians**, **Eskimos**, and the **Northern American Indians**
   ii) **Japanese**, **Koreans** and **Chinese**
   iii) **Indonesians** and **Malays**.

- Central African **pigmies**, the **Bushmen** and the **Australoids** were not included in the above classification.

Thousands of years ago, humans **might have** started to **evolve** into **different** species because **physical** and **genetic** differences were **emerging** between the races. But, their **large brain** enabled them to **design** and **make** all **things** needed (Example: buildings, clothes, tools, drugs, etc) and modify their environment, instead of evolving to adapt to it. They were also able to develop different ways of **transportation** (**Example**: air, road, railway and water) which allowed the **intermingling** of people from **different races**. As a result, they interbred and **minimized genetic differences**.

Humans could still evolve, but the **technological development** that we are exercising has **halted** it. We, more or less, modify our environment in the way that suits us. Other organisms can not do this. They have to **evolve** and develop **adaptations** to their changing environment.

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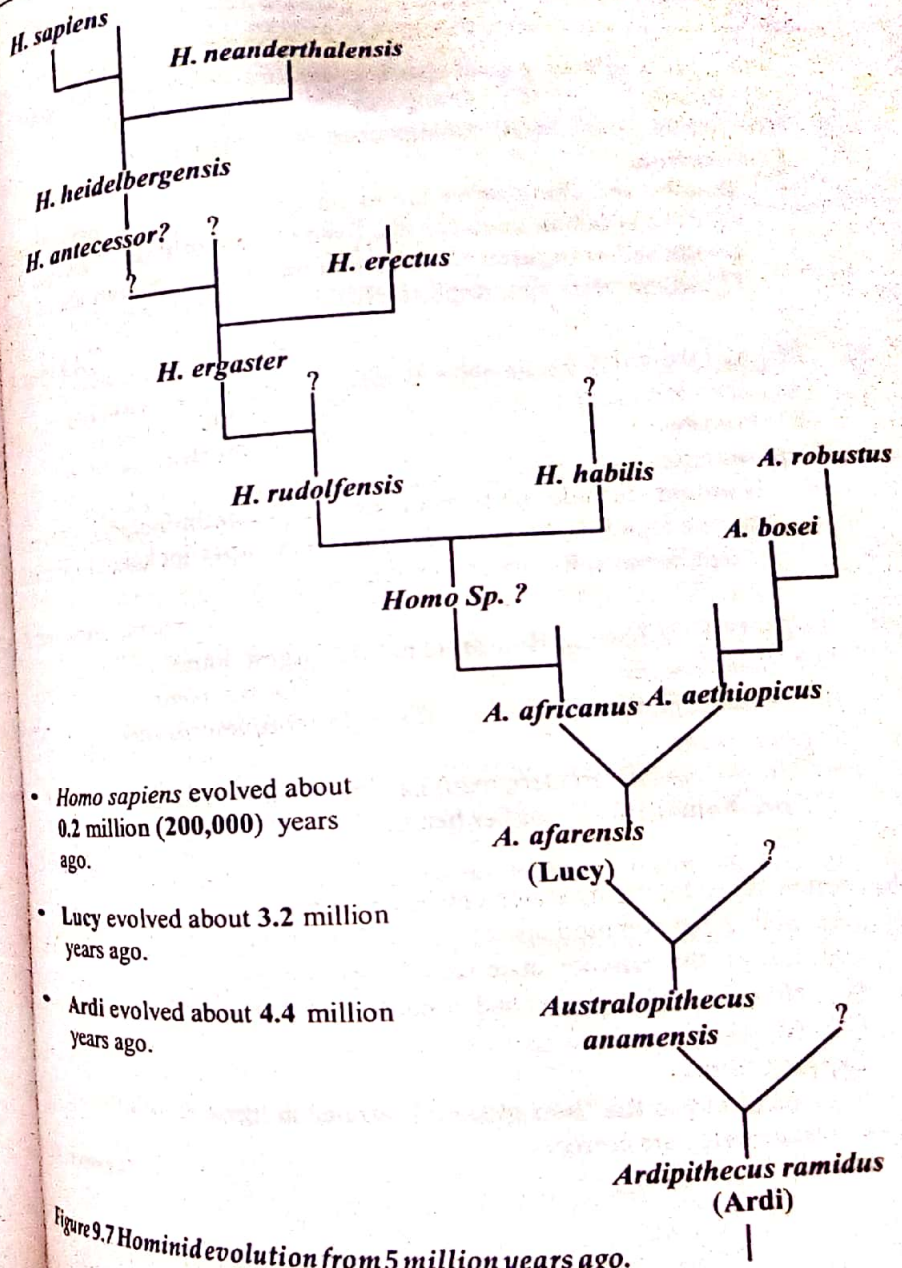
Hominid evolution

The **"best guess"** of *Hominid* evolution from **5 million** years ago shows that it started with **Ardi** (*Ardipithecus ramidus*). This species gave way to the genus *Australopithecus* consisting of **6** species (*A. anamensis, A. afarensis*/**Lucy**, *A. africanus, A. aethiopicus, A. bosei* and *A. robustus*). Then, species of genus Homo have desceneded from *A. africanucs* (direct descendent of Lucy). The *Homo species* include *H. rudolfensis, H. habilis, H. ergaster, H. erectus*, H. antecessor, *H. heidelbergensis, H. neanderthalensis* and *H. sapiens*. All the species in the 3 genera except *Homo sapiens* have been **extincted**, and are known by their **fossil records**. (Figure 9.7)



- *Homo sapiens* evolved about 0.2 million (**200,000**) years ago.
- Lucy evolved about **3.2** million years ago.
- Ardi evolved about **4.4** million years ago.

Figure 9.7 Hominid evolution from 5 million years ago.

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Illustrative Examples

51. Which of the following great apes is evolutionarily closer to humans?
(A) Bonobo (C) Orangutan (E) A & D
(B) Gorilla (D) Chimpanzee (F) B & C

Explanation:
- **Bonobo** and **chimpanzee** are evolutionarily closer to **man** because they had **common ancestor** that lived about **6 million** years ago. But, **gorilla** and **orangutan** has **common ancestor** with **man** about 8 and **13 million** years ago, respectively.

Answer: E

52. Which of the following includes all the others?
(A) Great apes (C) Hominis
(B) Primates (D) Hominids

Explanation:
- **Hominids** includes all **human species**; **Hominin** includes all human species found in the genus **Homo**. **Great apes** include all primates except humans. **Primates** includes all.

Answer: B

53. Which of the following **Hominids** has the largest brain?
(A) *Homo erectus* (C) *Homo sapiens*
(B) *Australopithecus* (D) *Ardipithecus ramidus*

Explanation:
- *Homo sapiens* (modern man) has the **largest brain**. Others are **pre-humans** with **smaller brain**.

Answer: C

54. Which of the following statements are **correct**?
(A) Ardi is the common ancestor of all Hominids.
(B) Lucy is the common ancestor of all Hominins
(C) *Homo neandertalensis* had common ancestor with *H. sapiens*
(D) All are correct

Explanation:
- According to the "**best guess**" illustrated in figure 9.7, all the statements are correct.

Answer: D

55. What are the 3 major human races?

Explanation:
- **Africans** (Negroids)
- **Eurasians** (Caucasoids), and
- **East Asians** (Mongoloids)

56. Which of the 3 major races of humans has the largest proportion of the world's population?

Explanation:
- **Eastern Asians** (Mongoloids) which includes the **Koreans, Japanese** and **Chinese**, and is most populus.

57. What is special about **Australoids**, the **Bushmen** and the **Central African pigmies**?

Explanation:
- These are groups of **human races** that were **not included** in the **classification** of human **races** into **Negroids, Caucasoids** and **Mongoloids**.

58. From among the following alternatives, select and indicate your own species. **(CEE- General science, 1995 E.C)**
(A) *Homo habilis* (C) *Homo sapiens*
(B) *Australopithecus afarensis* (D) *Homo erectus*

Explanation:
- Although all are **hominids**, *H. sapiens* is our **modern species.**

Answer: C

59. To which genus of human-like organisms does Lucy belong? **(UEE, 2004 E.C)**
(A) The genus ***Homo*** (C) The genus ***Sahelanthropus***
(B) The genus ***Ardipithecus*** (D) The genus ***Australopithecus***

Explanation:
- **Lucy's** scientific name is ***Australopithecus afarensis.***

Answer: D

60. Which of the fossils is the nearest to the **common ancestor** of the **hominids** and the **apes**? **(UEE, 2004 E.C)**
(A) *Homo habilis* (C) *Ardipithecus ramidus*
(B) *Homo erectus* (D) *Australopithecus afarensis*

Explanation:
- Ardi is closer to the **common ancestor**. Others are relatively recent.

Answer: C

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1. List out and explain the theories of origin of life.

Explanation:

The origin of life is explained based on different theories. They are:

ii. **Theory of special creation** - claims that different forms of life on the earth are created by a supreme being.

iii. **Theory of spontaneous generation** - claims that certain non-living thing can be directly transformed into living things.

iv. **Theory of eternity of life** - claims that the universe has always existed and that there has always been life in the universe.

v. **Cosmozean theory** - claims that life on earth originally came from elsewhere in the universe.

vi. **Theory of biochemical origin** - states that life on earth is originated as a result of a number of biochemical reactions organic molecules which are associated to form cells.

2. What is the essence of Oparin's theory of biochemical evolution?

Explanation:

Alexander Oparin a Russian Biologist worked intensively on a biochemical theory of evolution explained how life is originated on the earth. Accordingly he suggested that; simple organic compounds could have undergone a series of reactions leading to more and more complex molecules.

3. Discuss the historical development of the theory of evolution.

Explanation:

The science of evolution came up to this days through the contribution of various evolutionists. Among these the major ones are:

i. **Jean Baptiste Lamarks** - proposed that organisms undergo change over time as a result of some natural phenomenon rather than devine intervention. He thought organisms were endowed with a vital force drove them to change toward greater complexity over time.

ii. **Charles Darwin** - called the "Father of evolution" stated his observations while voyaging on the H.M.S Beagle, specially his observations on Galapagos Islands in South America. After his observation he wrote a book entitled 'the origin of species. Accordingly, he stated some major points as sources of evolution. They are:

- Survival of the fittest
- Fecundity
- Natural selection
- Adaptation

4. Summarize the evidence of evolution obtained from the fossil record.

Explanation:

Fossils are direct evidences of evolution that come from remains of ancient organism. Layers of sedimentary rock normally occur in their sequence of deposition, with the more recent layers on the top of the older, earlier ones **Index fossils** characterize a specific layer over large geographical area.

Radio isotopes present in a rock provide a way to accurately measure th rocks age.

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5. Summarize the evidence for evolution derived from comparative anatomy.

Explanation:

- **Homologous feature** - body parts of different species of organisms having structural similarities, but adapted to perform specific functions.

  For example: Birds wing and human hand.

- **Homoplastic features** - body parts that have similar functions in quite different, distantly related organism. They demonstrate convergent evolution, in which organisms with separate ancestries adapt in similar ways to comparable environmental demands.

- **Vestigial structures** - are non-functional or degenerate remains of structures that were present and functional in ancestral organisms.

6. Describe skeletal and skull differences between apes and hominids.

Explanation:

- Unlike ape skeletons, hominid skeletons have adaptations that reflect the ability to stand ereck and walk on two feet.
- The human brain is larger and has a foramen magnum at the base of skull.

7. Distinguish among the following members of genus Homo:

- *Homo habilis*
- *Home erectus*
- *Homo neandertals*
- *Homo sapiens*

Explanation:

- **Homo habilis** - the earliest known hominid with some of the human features. Has slightly larger brain. They have fashioned crude tools from stone.

- **Homo erectus** - had larger brain than Homo habilis.
  - made more sophisticated tools.
  - may have worn cloths, built fires and lived in caves or shelters.
- **Homo neandertals** - they had short, study builds, receding chins and fore heads, larger front teeth.
- **Homo sapiens** - modern human
  - The only species of genus homo remaining on the earth.

POST TEST

1. How is information from the field of embryology used as evidence for evolution?

   A. It shows that organisms from the same species have similar DNA.
   B. It shows that organisms from different species have similar DNA.
   C. It shows that organisms from different species have similar development patterns.
   D. It shows that organisms from the same species have similar developmental patterns.

2. Which idea about evolution was held by Darwin but **not** by Lamarck?

   A. Species are fixed and do not change.
   B. Species are moving toward perfection.
   C. Organisms best suited to their environment survive and reproduce.
   D. The use or disuse of an organ determines its appearance in a species.

3. According to natural selection, which of the following results would be considered a beneficial mutation to an organism?

   A. Resistance to a virus.
   B. Formation of cancer cells.

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C. Inability to undergo sexual reproduction.

D. Stopping the production of an essential protein.

4. Which is a conclusion drawn from the observed DNA similarities found between apes and chimpanzees?

A. Apes and chimpanzees are in the same genus.

B. Apes and chimpanzees are in the same species.

C. Apes and chimpanzees are products of artificial selection.

D. Apes and chimpanzees are thought to have evolved from a common ancestor.

5. What must be true of theories before they are accepted by the scientific community?

A. Practical applications must be suggested.

B. Similar explanations must be in existence.

C. They must be supported by empirical data.

D. Experiments must be conducted to refute the idea.

6. What assumption is made during the relative dating of fossils?

A. Remains of extinct organisms can be compared to organisms living today.

B. By comparing rock layers, a pattern is seen of organisms becoming simpler over time.

C. Upper rock layers contain younger remains and older remains are found in deeper layers.

D. The time it takes for known radioactive materials to break down is used to age the remains.

7. The effectiveness of a new insecticide was tested on a large population of mosquitoes over a number of breeding cycles. At first the population of mosquitoes was reduced dramatically by the use of the insecticide. After a

number of breeding cycles the population then began to increase until the insecticide to have little effect.

How would the Darwin/Wallace theory of evolution by natural selection explain these observations?

A. Some of the original populations were isolated from the insecticide as a control group.

B. Some of the original population had already reproduced before the insecticide was used.

C. Some of the original population were resistant to the insecticide and passed this on their offspring.

D. Some of the original population adapted to the insecticide and survived to produce offspring.

8. Biologists have suggested for a long time that reptiles evolved from fish-like ancestors. Recently, a 375 million year-old fossil fish (Tiktaalik roseae) was found in Canada. The fossil had fins, scales and a lower jaw like those of a fish but had a crocodile-like skull, a mobile neck relative to fish and crocodiles, this rare Tiktaalik fossil represents an example of:

A. A transitional form.

B. An index fossil.

C. Convergent evolution.

D. Co evolution.

9. According to modern evolutionary theory, changes to genetic material in mammals can only be inherited by the next generation if

A. They occur in somatic tissue.

B. They occur in reproductive tissue.

C. They make the species fitter.

D. They make the individuals fitter.

10. Comparison of sequences in mitochondrial DNAis often used to establish the degree of relatedness between organisms and thus to suggest

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evolutionary relationships, particularly in complex, higher-level organisms. Mitochondrial DNA is used because it

A. Is only inherited in males.
B. Has more genes than nuclear DNA.
C. Mutates more slowly than nuclear DNA.
D. Contains different nitrogen bases from those found in nuclear DNA.

11. Traces of starch from wild barley (Hordem) and wheat (Triticum) species were found on a 23,000-year-oldseed-grinding stone used by ancient humans in Israel. The method which would have been used to date these ancient starch grains is

A. Radioactive carbon - 14 dating.
B. Radioactive uranium - 235 dating.
C. Relative dating by comparing rock strata.
D. The use of a molecular clock.

12. Complex social activities require articulate speech. Evidence found with early Homo fossils suggests that this genus was the first to use articulate speech. This suggestion would be best supported by evidence of

A. Burial ceremonies.
B. Use of stone tools.
C. Living in groups.
D. Organized hunting of prey.

13. The use of pesticides on crops has been a common farming practice for decades. What has been the greatest effect of natural selection through the use of pesticides on certain insect populations?

A. Natural selection has been altered because the insects and their predators are killed.
B. The rate of selection is increased because the pesticides do not kill the insect that are naturally resistant to it.
C. The rate of selection has decreased because the pesticides kill only young insects.

D. The pesticides have altered natural selection by causing the insect DNA to spontaneously mutate.

14. The Giant Anteater of South America, the Giant Armadillo of North America, the Giant Pangolin of Africa, and the Spiny Anteater of Oceania share many characteristics. They all eat ants, have long narrow snouts, few teeth, and large salivary glands, yet they do not share a common ancestor. This is an example of which of the following?

A. Genetic drift
B. Founder effect
C. Adaptive radiation
D. Convergent evolution

15. Which of the following is a true statement about evolution?

A. Individuals evolve more slowly than populations.
B. Individuals evolve; populations do not.
C. Individuals evolve by changing the gene pool.
D. Populations evolve; individuals do not.

16. All of the following statements about evolution are true **Except**

A. Evolution occurs more rapidly at some times than others.
B. Evolution is directly by changes in the environment.
C. Evolution is not always from the simple to complex but may act in such a way that complex forms give rise to the complex but may act in such a way that complex forms give rise to simpler ones.
D. Single organisms, as well as populations, do commonly evolve.
E. Evolution is not always slow, gradual process

17. According to the Hardy Weinberg theory, which of the following represents homozygous dominant individual?

A. P B. Q C. Q2 D. 2pq

18. All of the following are part of the Hardy-Weinberg theorem, which describes a stable, non evolving population **EXCEPT**

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A. Natural selection exists.
B. Mating must be random.
C. No mutations can occur in the population.
D. The population must be isolated from others.
E. The population must be large.

19. Which of the following is a transition fossil?

A. Reptiles
B. Amphibians
C. Archaeopteryx
D. Fish
E. Apes

20. All of the following are part Darwin's theory of natural selection EXCEPT

A. Population tend to overpopulate
B. In any population there is unequal ability among individuals to survive and reproduce.
C. Only the best fit individuals survive and get to pass on their genes.
D. Only overpopulation leads to struggle for existence.
E. Organisms change in response to a need in the environment.

21. All of the following are true of homologous structures except:

A. They demonstrate common ancestory
B. An example is the wing of bat and persons arm.
C. An example is the wing of bat and the wing of fly.
D. An example is the whale's lateral fin and person's of fly.

Choose from the list of terms below.

A. Charles Darwin
B. Lamarck
C. Stephen J.Gould and Niles Eldridge
D. Oparin and Haldane
E. Hardy-Weinberg

22. Inheritance of acquired traits
23. Use and disuse
24. Theory of punctuated equilibrium
25. Populations tend to over populate
26. Hypothesized that under conditions that existed in early Earth, organic molecule could form.
27. Survival of the fittest.

28. Which of the following is a true statement about evolution?

A. Individuals evolve more slowly than populations.
B. Individuals evolve; populations do not.
C. Individuals evolve by changing the gene pool.
D. Populations evolve; individuals do not.

29. There is evidence that Homo sapiens and Homo neaderthalensis coexisted in Europe more than 30,000 years ago. Both of these species left signs of cultural evolution from this period. An example of evidence which would show that cultural evolution was occurring in these groups at this time is

A. Drawings and carving on rocks.
B. Animal remains close to a Homo skeleton.
C. Male and female skeletons in the same area.
D. Home sapiens and Home neanderthalensis skeletons in the same area.

30. Which process produces the most variation within a species?

A. Asexual reproduction
B. Sexual reproduction
C. Mitosis
D. Cloning

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31. What is the best explanation for the continual changes in the classification system of organisms?

A. All organisms struggle for existence become extinct.

B. All organisms compete to be at the top of the food chain.

C. Technological advances have allowed scientists to better compare organisms.

D. More species have been discovered, but scientists have not analyzed all the data.

CRITICAL THINKING

1. Charles Darwin once said, "It is not the strongest of the species that survive, nor the most intelligent, but the one most responsive to change" Explain what he meant

2. Write short paragraphs explaining each of the following statements

A. Natural selection chooses from among the individuals in a population those most suited to current environmental condition. It does not guarantee survival under future condition.

B. Individuals do not evolve, but population do

C. Evolution is said to have occurred with in a population when measurable genetic changes are detected.

3. Arguably, modern Homo sapiens have taken the manipulation and control of the environment to its highest level in history. Does this mean that our species will no longer physically evolve by the mechanism of natural selection? Justify your answer.

4. If you were evaluating whether other early humans exterminated the Neandertals. What kind of archeological evidence might you look for?

REVIEW QUESTIONS

I. **Choose the correct answer**

1. What is the best term used to refer to all of the genes present in a population? **(CEE General Science, 1995 E.C)**
(A) Biochemistry (C) Genome
(B) Gene pool (D) DNA

2. Which of the following modes of natural selection was responsible for the fast increase in the number of the black form of moths in Europe during Industrial Revolution? **(UEE, 2003 E.C)**
(A) Stabilizing selection (C) Normalizing selection
(B) Disruptive selection (D) Directional selection

3. Which one of the following is most important for speciation to occur? **(UEE, 2003 E.C)**
(A) Hybridization among populations
(B) Isolation of the gene pool
(C) Migration between populations
(D) Free flow of genes between populations

4. Animals without close evolutionary relations are sometimes seen to have similar structures adapted for the same function. Which evolutionary principle is illustrated by this observation? **(UEE, 2003 E.C)**
(A) Convergent evolution (C) Sympatric evolution
(B) Divergent evolution (D) Allopatric evolution

5. Which of the following descriptions is true about the action of selective pressures in populations? Selective pressure that acts:
(A) Around the mean value is stabilizing
(B) Around the mean value is disruptive
(C) At both ends of the distribution is directional
(D) At one ends of the distribution is stabilizing

6. What does the structural similarity between the flippers of whales and arms of humans show? **(UEE, 2004 E.C)**
(A) Whales evolved from the human species
(B) Whales are older than human species
(C) The human species began life in the ocean
(D) Whales and humans had a common ancestor

7. Whale fins and bat wings are anatomically similar. Which of the following does this suggest about the animals?

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(A) Whales and bats move in the same way
(B) Whales and bats have common ancestory
(C) Whales and bats have existed for the same amount of time
(D) Whales and bats were once adapted to the same environment

8. A population is separated into two groups by a geographic barrier, Overtime, enough differences develop between the two groups so that they can not interbreed when reunited. Which of the following terms best describes the process that has occurred?
(A) Extinction
(B) Hybridization
(C) Immigration
(D) Speciation

9. Which of these conditions should completely prevent the occurrence of natural selection in a population over time?
(A) All variation between individuals is due only to environmental factors.
(B) The environment is changing at a relatively slow rate.
(C) The population size is large
(D) The population lives in a habitat where there are no competing species present.

10. Heterozygote advantage should be most closely linked to which of the following?
(A) Sexual selection
(B) Stabilizing selection
(C) Diploidy
(D) Directional selection

11. Dog breeders maintain the purity of their dogs by keeping them apart when they are fertile. This kind of isolation is most similar to which of the following reproductive isolating mechanism?
(A) Hybrid sterility
(B) Hybrid breakdown
(C) Habitat isolation
(D) Gametic isolation

12. If two species of frogs mate and their offspring fail to develop and hatch, the mechanism keeping them separate is:
(A) Hybrid sterility
(B) Hybrid inviability
(C) Hybrid infertility
(D) Hybrid fertility

13. The formation of new plant species by hybridization and chromosome doubling of the hybrid is an example of:
(A) Sympatric speciation
(B) Polyploidization
(C) Allopatric speciation
(D) Habital selection

14. Plant species A & B have 12 & 14 diploid chromosome numbers. A new plant species C was formed by hybridization between the two. The diploid chromosome number of C would be
(A) 12 (B) 14 (C) 26 (D) 13

15. If chromosome doubling occurs in the above hybrid, what would be the polyploid chromosome number in the resulting polyploid plant?
(A) 26 (B) 52 (C) 28 (D) 24

16. Approximately how far back in time does the fossil **Ardi** (*Ardipithecus ramidus*) record extend?
(A) 3.2 million years
(B) 5.0 million years
(C) 4.4 million years
(D) 3.5 million years

17. Approximately how old is the fossil **Lucy** (*Australopithecus afarensis*)?
(A) 3.5 million years
(B) 3.2 million years
(C) 6.0 million years
(D) 4.5 million years

18. If scientists search for life on other planets, they are likely to focus on the presence of molecules containing
(A) Carbon
(B) Iron
(C) Potassium
(D) Sodium

19. In earliest fossils found in the geologic record are
(A) Fungi
(B) Seed plants
(C) Bacteria
(D) Protozoa

20. Individuals within a species tend to be generally different. The primary mechanism generating this individual variability is:
(A) Asexual reproduction (C) Meiosis (E) Poliploidy
(B) Sexual reproduction (D) Mitosis (F) B & C

21. In which hominid species did scientists find the smallest brain size (cranial capacity)? **(EUEE, 2005 E.C)**
(A) *Homo sapiens*
(B) *Homo habilis*
(C) *Homo erectus*
(D) *Homo neanderthalensis*

22. What is the reproductive isolating mechanism called if two species of frogs do not interbreed because they can not understand the mating calls of one another? **(EUEE, 2005 E.C)**
(A) Seasonal isolation
(B) Behavioural isolation
(C) Temporal isolation
(D) Isolation by distance

23. Among the following, which one is the best criterion to show that two populations belong to the same species? **(EUEE, 2005 E.C)**

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(A) Morphological similarity (C) Production of fertile offspring
(B) Physiological similarity (D) Inhabiting the same habitat

24. What kind of natural selection is at work in a situation where individuals at both extreme ends of the range get better survival advantages than those around the middle? **(EUEE, 2005 E.C)**
(A) Stabilizing selection (C) Disruptive selection
(B) Directional selection (D) Selection that removes the extremes

25. The hominid fossil known as **Lucy** belongs to the genus:
(A) Homo (C) Australopithecus
(B) Ramapithecus (D) Ardipithecus

26. Which of the following statements describes **Charles Darwin's** theory of evolution
(A) Acquired characteristics are inherited
(B) Evolution occurs by the use-&-disuse of body parts
(C) Natural selection leads to evolution and speciation
(D) Living things do not change through time.

27. Which of the following alternatives shows the gases that were believed to be present in the ancient atmosphere before the origin of life on earth?
(A) Ammonia, oxygen, methane (C) Methane, oxygen, hydrogen
(B) Hydrogen, ammonia, methane (D) Oxygen, hydrogen, ammonia

28. The half-life of a certain isotope is about 5000 years. What percent of the original amount of the isotope is expected to be present in a fossil that is 20,000 years old?
(A) 12.5% (B) 6.25% (C) 50% (D) 25%

29. The two scientists who developed the biochemical theory of the origin of life were:
(A) Ritcher & Arrhenus (C) Redi & Pasteur
(B) Haldane & Oparin (D) Aristotle & VanHelmont

30. In a certain plant population, nature favours the shortest and tallest plants while it eliminates plants with average height. What type of selection is operating?
(A) Stabilizing selection (C) Disruptive selection
(B) Normalizing selection (D) Directional selection

31. Life is said to begin on earth and transformed from simple forms to complex ones in millions of years. Which branch of biology is concerned with the study of this process?

(A) Microbiology (C) Paleobiology
(B) Astrobiology (D) Entomology

32. The process by which unrelated organisms living on different parts of the world evolve similar structures adapted for the same function is:
(A) Parallel evolution (C) Divergent evolution
(B) Adaptive radiation (D) Convergent evolution

33. Those animals that have similar basic structural pattern called the pentadactyl limb are assumed to have evolved form a common ancestor. Such structures are called:
(A) Analogous structures (C) Homologous structures
(B) Vestigial structures (D) Adaptive structures

34. Which of the following hominids is closest to the modern humans?
(A) *Homo habilis* (C) *Homo neandertalensis*
(B) *Homo erectus* (D) *Homo rudolfensis*

35. Which of the following animals are evolutionarily closest to *Homo sapeins*?
(A) Gorillas (C) Orangutans
(B) Chimpanzees (D) Gibbons

II. Give short answers to the following

36. The first hominids appeared on earth about 5 million years ago. They belonged to the genus_____.

37. What are the 4 factors that cause change in the genetic composition of a population, and lead to evolution?

38. What are the 5 theories about the origin of life?

39. What do you understand about the habitat or the nutrition of the following bacteria? Explain briefly
(A) Thermophilic bacteria
(B) Halophilic bacteria
(C) Sulphobacteria
(D) Methanobacteria

40. A mutation that occurs in a gene of an organism may be_____, _____ or_____to the organism.

41. What can you say about the evolution of humans presently?

42. Neo-Darwinism combines_____,_____&_____.

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Answer To Review Questions of Unit Nine

I. Multiple choice

1.	B	8. D	15. A	22. B	
2.	D	9. A	16. C	23. C	29. B
3.	B	10. B	17. B	24. C	30. C
4.	A	11. C	18. A	25. C	31. C
5.	B	12. B	19. C	26. C	32. D
6.	D	13. B	20. F	27. B	33. C
7.	B	14. D	21. B	28. B	34. C
					35. B

II. Short answer

36. Ardipithecus
37. Meiosis, hybridization, mutation and natural selection.
38. Special creationism, spontaneous generation, Eternity of life, Cosmozoan theory, and Biochemical origin of life.
39. (A) **Thermophilic bacteria** are **heat-loving** bacterial that live usually in **hot springs** where the **temperature** is very **high**.
    (B) **Halophilic bacteria** are **salt-loving** bacteria that reside in water **bodies** that are highly **salty**.
    (C) **Sulphobacteria** are **chemoautrophs** that use **hydrogen sulphide** (H_2S) as a source of energy.
    (D) **Methanobacteria** use **methane** (CH_4) as a source of energy in their nutrition.
40. Neutral, advantageous or disadvantageous.
41. We humans have the potential to evolve, but because of our large brain, our ability to influence our environment and intermingle easily, helped us to stay as a species rather than evolving into different species.
42. Original Darwinism, the knowledge of genetics and ethology (animal behaviour).

UNIT TEN

BEHAVIOUR

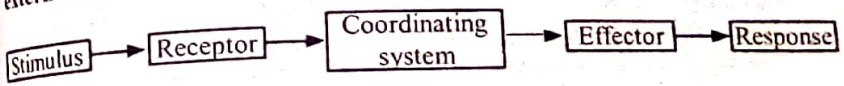
10.1 Introduction

Behaviour may be defined **traditionally** as:

- Observable **response** a person makes to any **situation**.
- **Manner** of acting or **conducting**.
- The way a person behaves towards others.
- **Actions** or **reactions** of an **organism** in **response** to **internal** or **external stimuli**
- **Response** or **reactions** or **movements** made by an organism in any situation.

Biological definition

Behaviour is the **coordinated response** of an **organism** to **internal** and **external stimuli**. It generally consists of the following **components**:



Stimulus - internal/external environmental change that may initiate coordinated response of an organism.
Example: Steeping on a sharp object/hot object, feeling sick, bad smell, encountering a dangerous situation, smell of delicious food, etc.
Receptor - is a **group** of **cells** or a **single cell** that receives a stimulus and processes it, and passes it to a central coordinating system.
Example: Taste buds in the tongue, pain/pressure receptors in the skin, rods and cones in eye's retina, cochlea in ear, etc.
Coordinating system - is an organized system that links **receptors** to **effectors** and coordinates **communication** between the two.
Example: Hormonal systems and/or nervous systems.
Effector - is any part of the organism that **produces** a **response** (reaction) to the stimulus.
Example: Muscle, glands, etc.

Plant responses

There are 3 major types of **plant responses**. These are **photoperiodism**, tropisms and **nastic movements**.

a) **Photoperiodism** - is generally the **developmental response** of plants to **photoperiod** (day-length). Particularly, it is the **flowering response** of

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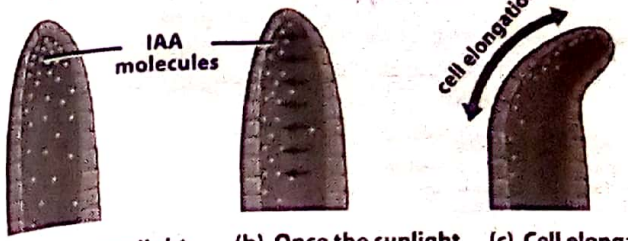
different plants to verifying **day-lengths**. Except in the **tropics**, **significant variations** occur in the **day-length** as one **moves** away from the **equator**. This affects the development (i.e. **vegetativeness**, **flowering** and **fruiting**) of plants. Based on this there are 3 types of **plants**: Short-day plants, long-day plants & day-neutral plants.

Short-day plants - bloom (**flower**) when the days are **shorter**. They stay in their **vegetative state** when the days are **longer**.
Example: Sweet Potato, Poinsettia, Strawberry, Chrysanthemum, Aster, Maryland mammoth (mutant tobacco), Saliva, Cannabis, Cotton, Rice, etc.

Long-day plants - are those plants that **flower** and **fruit** when the days are **longer**. Seasons of **long-days** in the **northern hemisphere** are seasons of **short-days** in the **southern hemisphere**. Under **very short-days**, long-day plants stay in their **vegetative stage**.
Example: Lettuce, oat, clover, ryegrass, potato, beet, raddish, spinach, etc.

Day-neutral plants - are indifferent of the day-lengths. They simply flower when they are ready, either during long or short photoperiods.
Example: Tomato, cucumber, roses, corn, strawberry, etc.

b) **Tropisms** - are **growth responses towards** or **away** from a **stimulus**. It mainly affects the **growing parts** of the **plant**.
Positive tropism - is a growth response **towards** a given **stimulus**, while **negative tropism** is growth **away** from a **stimulus**.
Phototropism - is plant **growth response** to a **unidirectional light**. **Shoots** are **positively phototropic** bending **towards** light. **Roots** are **negatively phototropic**.
Plant growth regulators (**hormones**) called **auxins** coordinate **phototropism** and other **tropisms**. They are produced in **shoot-tips** and migrate to other parts. Affected by ***a unidirectional light, they migrate to the darker side and stimulate cell division and enlargement on that side.*** The illuminated side grows less than the darker side. Consequently, the shoot bends to the light.



(a) When sunlight is overhead, the IAA molecules produced by the apical meristem are distributed evenly in the shoot.

(b) Once the sunlight shines on the shoot at an angle, the IAA molecules move to the far side and induce the elongation of cells on that side.

(c) Cell elongation results in the bending of the shoot toward the light.

Figure 10.1. Phototropism in the shoot: the increased concentration of auxin on the right side of the shoot enhanced growth on that side. The illuminated (left) side is devoid of auxin and grew slowly as compared to the right side, and hence the tip bends to the left (to the unidirectional light).

Gravitropism/geotropism - is a plant **growth response** to **gravity** which pulls down everything to the center of the earth. The roots are **positively geotropic** as they grow downwards. Shoots grow upwards against the direction of force of gravity, and hence are **negatively geotropic**. If a seedling is placed **horizontally** and **rotated**, both the **shoot** and **root** grow **horizontally**. But, if rotation is **slowed** down or **stopped**, the **shoot** grows **upwards**, and the **root** grows **downwards**. Gravity pulls down **auxin** in both the **shoot** and **root** causing **higher concentration** on the **lower side**. **Higher auxin** concentration stimulates growth in **shoots**, while it generally **inhabits growth** in the **roots**. As a result, the shoot bends **upwards**, and the root bends **downwards**.

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GEOTROPISM

("Geo" – Earth)

Geotropism is the growth of a plant in response to gravity.

Positive Geotropism

It is the growth of a plant towards the center of the earth-down with gravity.
Ex. roots growing down

Negative Geotropism

It is the growth of a plant away from the center of the earth-opposite from the pull of gravity
Ex. stems grow up

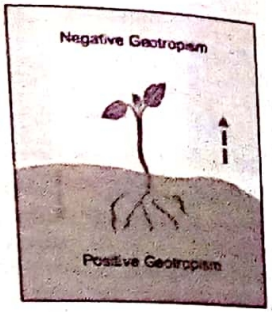

Figure 10.2. Effect of gravity on a seedling. The root and shoot upper sides are devoid of auxin and relatively grow rapidly in the root as the lower side is inhibited. In the shoot, higher auxin concentration stimulates growth in the lower side, and the upper side grows slowly. So, root bends down, but shoot bends up.

Hydrotropism - is growth response to **moisture (water)**. **Roots** are **positively hydrotropic**, and shoots are **negatively hydrotropic**.

Thigmotropism - is plant growth response to **touch** (contact).

Example: Tendrils touch solid object and wind themselves around it.

Chemotropism & aerotropism - are plant growth responses to **chemicals** and **gases**, respectively. If the **chemicals** or **gases** are **harmful**, the plant parts may tend to **grow away**, and if **useful**, tend to **grow towards** them.

c) **Nastic movements** - are **plant responses** to **uniformly distributed non-directional stimuli** such as **light intensity**, **temperature**, or **mechanical shock**. The plants **change** their **orientation** under such stimuli. These are not confined to growing regions only, they can occur in matured parts. They are mostly due to **changes** in **turgor pressure** and more rapid than tropisms, but biologically less important.

Example: Closing and **opening** movements of **flowers** and **leaves** in some plants during the **night** and **day** times, respectively, simple touch also stimulates closure of leaflets/flowers of some plants.

Response in simple animals
Taxis and kinesis

Taxis (Pl. taxes) - is **movement** of small animals **towards** (positive taxis), or **away from** (negative taxis) a stimulus.

Example: Euglena swimming towards light for **photosynthesis** (positive phototaxis).

Kinesis (Pl. kineses) - is **change** in **rate** of **movement** due to **change** in **stimulus intensity**. It does not involve change in direction.

Example: Increased rate of movement of woodlice in bright light to increase their probability of being in darker area (more humid).

Woodlice are **crustaceans** that live under logs, stones, bark and amongst leaves where it is **darker** and **humid** to minimize water loss as they have no waxy covering. Because of **smaller size**, they have **large surface area to volume ratio** and tend to **lose water quickly**.

Importance of the study of behaviour

Ethology - is the **study** of **animal behaviour** including **humans**.

Ethologist - a **biologist** who studies **animal behaviour**.

Behaviour should be studied as a branch of scientific knowledge, but also contributes to:

- The study of **neuroscience**
- The **human society**
- The **management** of **environment** and **resources**
- The **animal welfare**
- The **science education**

Impact on neuroscience

Neuroscience - the study of **brain** and **nervous system**

Neuroethology - the study of the **relationship** between **behaviour** and the **neural pathways**.

Neural pathways - sequence of **neurons** (nerve cells) involved in bringing about a specific behaviour.

The study of animal behaviour may help in the determination of:

- The **neural pathways**
- The **effect** of **behaviour** and **social organization** on **physiology** and **cellular processes**. **Example**: Ovulation and miscarriage may depend on social environment.
- The **relationship** between the quality of **social environment** and **immunity** (may be directly related).

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Impact on human society

The **interaction** between **behaviour** and **environment**, or between **behaviour** and **genetics** may be a cause to some society problems. So, **animal behaviours** may be **studied** and **interpreted** to understand human problems. It may help in the:

- **Assessment** of the **effect** of **cooperation** and **reconciliation**.
- **Understanding** of **theories** of **child development** and **attachment** formation.
- **Research** on **endogenous rhythms** (**Example**: circadian rhythms) to solve, for example, the problems associated with coping with **jet-lag** or **shift-working**.

Impact on environment and resources

Animal **behaviour changes** can give **clues** of **environmental damage** and help us to take quick action and save the environment. Knowing the **migratory patterns**, **home range size**, **interaction** with others, **foraging demands** and **reproductive behaviour** of an endangered organism helps in their **conservation** and **sustainable utilization**.

Example: Migratory pattern study helped **salmon** industry in **USA**, honey bee behaviour study revealed mechanism of **pollination** and helped **plant breeding**.

Impact on animal welfare

Farm animals, research animals, companion animals, and endangered species all need **care** and **well-being** in the **laboratory** and in their **natural environments**. To develop **reasonable** and **effective** standards for the **care** and well-being of these animals, **ethological information** is important.

Impact on science education

Sometimes, there is **lack of interest** in **science** on the part of students. Behavioural biology (**animal behaviour** and **behavioural ecology**) often attracts students. This may raise student interests to science education and can lead to wider scientific studies.

Review Examples

1. Which of the following statements is **true**?
   (A) Ethology is a valid study in its own right
   (B) Findings from the study of animal behaviour can help to predict human behaviour patterns.

   (C) Understanding animal foraging behaviour can help in conserving endangered species.
   (D) All are correct

Explanation:
- All the statements are true because they all talk about the importance of the study of **ethology**.

Answer: D

2. A response to a stimulus that involves movement towards or away from the stimulus is called
   (A) Kinesis (C) Taxis
   (B) Periodism (D) Tropism

Explanation:
- **Taxis** is a **directional response** to a stimulus in **small animals**.

Answer: C

3. Give some examples of behaviour.

Explanation:
- Hand withdrawal from sharp/hot object, mating, feeding, courtship, communication, nest building, and the like are examples of behaviour.

4. The larvae of most insects move away from light. What kind of behaviour is this?
   (A) Imprinting (C) Kinesis
   (B) Taxis (D) Tropism

Explanation:
- This is **negative phototaxis** because the larvae **move away** from the **light** stimulus.

Answer: B

5. Which of the following disciplines is studied in ethology? **(UEE, 2003 E.C)**
   (A) Heredity (C) Reproduction
   (B) Physiology (D) Behaviour

Explanation:
- **Ethology** is the study of **behaviour**.

Answer: D

6. The unicellular organism, euglena, swims using its flagellum towards areas of increased light intensity. What is this kind of response to stimulus called? **(UEE, 2003 E.C)**
   (A) Negative kinesis (C) Negative phototaxis
   (B) Positive kinesis (D) Positive phototaxis

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Explanation:

- Movement towards **light** in euglena is **positive phototaxis.**

Answer: D

7. What is the plant growth substance that causes them to bend towards light? **(UEE, 2003 E.C)**
   (A) Starch
   (B) Photosynthesis
   (C) Auxin
   (D) Chlorophyll

Explanation:

- **Auxin** is one of the **plant growth hormones** responsible for **tropisms** in plants

Answer: C

8. Woodlice increase their movement in different directions in response to increased light intensity. This is an example of which of the following? **(UEE, 2003 E.C)**
   (A) Positive taxis
   (B) Negative taxis
   (C) Kinesis
   (D) Phototropism

Explanation:

- Response in small animals such as **woodlice**, in which they increase their movement in all directions is **Kinesis.**

Answer: C

9. What are the three major types of responses in plants?

Explanation:

- **Tropisms, nastic movements** and **photoperiodism.**

10. What is the correct order of the components of behaviour

Explanation:

- Stimulus → Receptor → Coordinating system → Effector → Response

10.1 Innate behaviour

There are 2 types of animal behaviour: **Innate & learned**. Innate behaviour is potentially represent at **birth** or **hatching**. There is no gene that directly codes for innate behaviour, but some genes control development of neural pathways and mechanisms which cause the behaviour.

10.2 Types of innate behaviour

There are 3 major types of innate behaviour. These are:

a) **Reflex actions** - Are the **simplest forms** of innate behaviours, and are **single action performance** to **specific stimulus**. Nearly always, they are **protective.**

Example: hand withdrawal form hot/sharp object, blinking of the eye, sneezing, knee/elbow jerk, etc.

b) **Orientational behaviour** - is the **movement of simple animals away** from **unfavourable** conditions. It also includes the **movement towards** and remaining in favourable ones.
Example: Kinessi, taxis

c) **Instinctive behaviour** - is the **most complex innate behaviour.** It involves a **fixed action pattern** (FAP) for each **key stimulus.** This means that it is carried out to **completion** once started.
Example: Nest building, imprinting, web weaving, etc.

Reflex actions

These are of **two** types: **somatic** and **autonomic** reflexes.

i) **Somatic reflexes** - involve **special senses** (eyes, ears, pressure detectors, etc) and produce mostly **fast** and **protective** responses by **muscle.**

ii) **Autonomic reflexes** - involve **internal organs** and **sensors** in them to produce response.
Example: Reflexes controlling heart rate and breathing rate.

Structure of the nervous system (NS)

To understand how **reflexes work**, general knowledge of the structure of the **nervous system** is important. **Physically**, there are **2 types** of the system.

a) **Central Nervous System (CNS)** - consists of **brain** and the **spinal cord.** These coordinate the NS.

b) **Peripheral Nervous System (PNS)** - consists of **cranial** and **spinal** nerves which branch from **brain** and **spinal cord,** respectively.

Again, **functionally**, there are **two types** of **nervous system**: **Somatic** and automatic NS.

a) **Somatic Nervous System (SNS)** - integrates information from **sense organs** and produces **responses** in **skeletal muscles.**

b) **Autonomic Nervous System (ANS)** - integrates information from **receptors** found in **internal organs**, and **produces responses** in the **same** or **other organs** or **glands.** It is further subdivided into **3** types (divisions).

i) **Sensory division** - transmits **sensory impulses** to the **central nervous system** (CNS).

ii) **Sympathetic division** - transmits **impulses** from the **central nervous system** to **organs.** It generally, prepares the body for **"fight or flight"** in which the organism encounters a **dangerous** situation, and must **confront** or **run away** to save its life. Sympathetic autonomic nervous system may facilitate this by **increasing cardiac output** and

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pulmonary ventilation, for example. This will make the organism **more energetic** and prepares it for the "**fight or flight**" action.

iii) **Parasympathetic division** - acts **antagonistically** to sympathetic division. It prepares the body for "**rest and repair**" by **decreasing cardiac output** and **pulmonary ventilation**, for example.

Biological clocks

Biological clocks are **internal regulatory mechanisms** that **control** different **cyclical responses**.

Organisms show **genetically programmed yearly, monthly, daily** and other **cyclical changes**. The clocks evolved independently in different organisms and are examples of **convergent evolution**.

Circadian rhythms (**Latin**: 'circa' = about; 'dies' = a day) are daily cycles that persist with a period of **24 hours**, and can **synchronize** to a **24-hour** cues such as **light** and **darkness** (**entrainment**).

Example: Sleeping at night and becoming active during the day, or vice versa.

Circannual rhythms - are **yearly** cycles that synchronize to **seasonal variations** such as **summer** and **winter** with **longer** and **shorter** days, respectively.

Example: Reproductive cycles in many organisms.

In mammals and other animals, the biological clock is located in the **hypothalamus**, and is called **Suprachiasmatic Nucleus** (**SCN**). During night, SCN sends impulse to **pineal gland** which secretes **melatonin**, and the person feels sleepfulness. So, **sleep-wake cycle** is controlled by melatonin whose secretion is induced by SCN. Changes in light-dark ratio can also affect reproductive behaviour on a **yearly** basis because this affects melatonin concentration in the body. Consequently, some animals are **long-day** (**summer**) breeders, while others are **short-day** (**winter**) breeders.

Circannual (yearly) **rhythms** (cycles) can also be shown by many animals in such behaviours as **migration, hibernation, coat growth** and **camouflage coloring**.

Instinctive behaviour

These are **pre-programmed** behaviour patterns characterized by **complex sequence of actions**, as opposed to **simple reflexes** which are characterized by **single actions**.

Example: Web spinning by spiders-a perfect web is spinned at the outset.

- Characteristics
  - **Common** to all members of a species
  - **Perfect** at the outset
  - Triggered by a **key stimulus**

  - The **stimulus** and the **response** are linked by the **nervous** or **hormonal** system.
  - Show fixed action pattern (**FAP**)
  - **Adaptive-retained** by **natural selection** as it has **survival advantage**. **Examples**:
    - Feeding behaviour of young herring gall: pecks at the orange
    - spot of beak of adult gall and receives food.
      - **Key stimulus**: orange spot on the beak
      - **FAP**: pecking
    - Aggression in stickleback fishes: male **sticklebacks** are very territorial, and attack any male that invades.
      - **Key stimulus**: red belly of entering male
      - **FAP**: defensive attack
  - Some **FAPs** may be modified by **experience** (learning).

Imprinting

Imprinting is a type of **instinctive** (inborn) behaviour in **newly born** (hatched) organisms. It is the **process** by which animals acquire their first forms of behaviour, particularly their **attachment** to their **mother**.

FAP: To **imprint** (become attached to) the first thing that generally resembles their adult.

Key **stimulus**: sight of an **object** that resembles their adult, or the **adult itself** (Example: mother, father, etc).

There is a "**time window**" for **imprinting** to **develop** (occur). That may be the first **few days after hatching** in some **birds** such as geese. According to many **evolutionary psychologists**, similar behaviour occurs in **human infants**. Strong **emotional bond** (attachment) forms between the **infant** and its **primary** care giver (often the **mother**). This occurs in **3 stages**.

1) **Pre-attachment** (**0-2** months): infant prefers **people** to **objects** without discrimination.
2) **Indiscriminate attachment** (**2-7** months): infant starts to prefer **familiar people**.
3) **True emotional attachment** (from the 7th month onwards): initially a person is preferred, but soon, multiple attachments form.

Attachment formation in humans, like in other animals, has **survival value**, and **natural selection** makes it **wide spread** in the **species**.

- **Imprinting** is sometimes considered as **learned behaviour** as it requires **repeated exposure** to the **stimulus** to produce the **FAP** response. But, the development of **neural pathways** to facilitate the response is **innate**.

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Review Examples

Classify the following examples of innate behaviour as somatic reflex, autonomic reflex, orientational or instinctive behaviour.

11. Increasing breathing rate during exercise.
    - **Autonomic reflex** because it is controlled by the **autonomous nervous system.**
12. Maggots moving into a more humid environment.
    - Orientational behaviour
13. Geese moving eggs that have rolled out of the nest back to the nest.
    - Instinctive because it involves a FAP.
14. Producing tears when dust gets in to your eye.
    - **Somatic reflex** since it is controlled by the **somatic nervous system.**
15. Scratching behaviour in dogs.
    - **Instinctive behaviour** as it is more **complex** than other types of **innate behaviours.**
16. Small aquatic crustaceans accumulating at the surface of the water.
    - Orientational innate behaviour
17. A new born human grasping a rope strongly enough so that he can be lifted by it.
    - This is **imprinting**, which is a type of **instinctive behaviour.**
18. A bare-footed man stepping on a sharp object, and withdrawing his foot immediately.
    - Somatic reflex.
19. A wasp making its nest on a tree.
    - Instinctive behaviour.
20. Blood pressure rising in a pregnant woman.
    - Autonomic reflex.

10.3 Learned behaviour

Learning - is **strengthening** of **existing responses** or **formation** of new **responses** to the existing **stimuli** as a result of **practice/repeatation.**
Learned behaviour - is acquired through **experience** (Example: trial and error) or by **insight**. Usually, it is imperfect at the **outset.**

Table 10.1. Comparison of innate and learned behaviours

Innate behaviour	Learned behaviour
• Genetically determined and common to all members of a species	- Changed by, or develops through experience, may vary between individuals
• Fully functional at the outset	- Develops thorough experience (trial & error) or by insight
• Generally, no modification	- May be modified by new experience
• Adaptive, retained as a result of natural selection	- Learned anew by each member of the species, and may or may not be adaptive

Types of learned behaviour

The following **4** major types of **learned behaviour** are **known** so far.

a) Habituation
b) Sensitization
c) Insight learning
d) Associative learning - has the following subdivisions:
   - Classical conditioning
   - Latent learning
   - Operant conditioning

Habituation

Habituation - is a process that results in a **reduced response** to a **stimulus** after **repeated exposure** to it over a longer period of time. It can occur at the **sensory** or **brain level** in the **nervous system. Sensory habituation** occurs when **sensory system** stops sending impulses to the **brain.** In **brain habituation,** signals still reach the brain, but it stops giving attention to them.

Example: + People residing near to an **airport**, stop giving response to the **sound pollution** coming from an aeroplane **landing** or **taking off.**

- **Animals** in national **parks** stop running away from **humans** when they realize that humans are **harmless** to them.

Habituation depends on the **duration** of **training.** Habituations resulting from **single training session** lasts for **shorter period** of time, while those resulting

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from **repeated training sessions** last for **longer period**. This is due to the presence of **short** and **long-term memory**. Habituation may also result in the **change** of **nerve net-works** bringing about the response.

Sensitization

Sensitization is an increase in **response** to a **harmless stimulus** when it occurs after a **harmful** stimulus.

Example: + **Amoeba** becoming more reactive to a **touch** following an **electric shock**

♦ A **person** becoming more **sensitive to snake sight** during her/his **adulthood** preceded by a **snake bite** during **childhood**.

Sensitization is of **two types** in higher animals.

- **Peripheral sensitization** - results from **changes** in the **peripheral nervous system**.
- **Central sensitization** - results from **changes** in the **central nervous system**.

Associative learning

- Classical conditioning
- Operant conditioning
- Latent learning

Conditioning - is letting an animal to learn a particular behaviour by associating the behaviour with:

- An **event/stimulus** ⊄ **classical conditioning**, or
- A **consequence** (reward/punishment) ⊄ **operant conditioning**.

Classical conditioning (CC)

- A **natural stimulus** is associated to **another stimulus**.
- The **new stimulus** starts producing **similar response** as the **natural stimulus**.
- CC is a theory discovered by a **Russian physiologist Ivan Pavlov** who worked on dogs.
- The different **stimuli** and **responses** in CC are:

a) **The Unconditioned Stimulus (US)** - is the **natural stimulus** that automatically triggers a response.
Example: Smell of a favourite (delicious) food.

b) **The Unconditioned Response (UR)** - the original (natural) response to unconditioned stimulus.
Example: Secretion of saliva as the smell of delicious food is detected.

c) **The Conditioned Stimulus (CS)** - is a **neutral stimulus** that does not produce **unconditioned response** initially, but triggers it after association with the **unconditioned stimulus**.

d) **The Conditioned Response (CR)** - response to the **conditioned stimulus** in the **absence** of the **unconditioned stimulus** (CR is actually the same to UR).

Pavlov's experiment on dogs

He fed dogs at regular intervals (**routine established**). They salivated due to the **smell** of **food**, and obtained it immediately at regular intervals.
In this case, the **stimulus** and **response** are as follows:

- The food & its smell: US
- Salivation: UR

Then, as he fed them, he rang a bell a bell regularly for a longer period of time. The dogs salivates as food was provided with bell ringing. Later on, he rang a bell without food provision. They continued salivating. In this case, stimuli & response are as follows:

- Bell ringing: CS
- Salivation: CR
- **Humans** can also be **classically conditioned**. **Example**: "Little Albert's" study. "Little Albert" was an **11-month** old infant. He was afraid of **loud noise** but friendly with **white rat**. Psychologists (**Watson** and **Rayner**) showed him **white rat**, and **banged** a **steel pipe** with a **hammer**. After ?-pairings, Albert started to **cry** and **crawl away** from the **rat**, without the noise. The relevant **stimuli** and **responses**, here, are as follows:
  - Loud noise: US
  - Crying and crawling away from the noise: **UR**
  - White rat: CS
  - Crying and crawling away from rat alone: **CR**

  After not pairing the noise with rat repetitively for some time, "Little Albert" lost its fear of the rat. This is **extinction**.

Extinction - the eventual **loss** of the **conditioned response** when the **CS** is **not again associated** with the US any longer.

Operant conditioning

This is **training** an **animal** a **behaviour** by associating it with a **consequence** (reward/punishment). It is also learning thorough **trial** and **error**. It was first used by **B.F. Skinner**, a behaviour **psychologist**. He identified 3 responses called **operants** that can follow a behaviour.

a) **Neutral operants** - these **neither increase, nor decrease** the **probability** of a **behaviour** being repeated.
b) **Reinforcers** - increase the probability of a behaviour being repeated.
c) **Punishers** - decrease the likelihood of a behaviour being repeated.

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Skinner researched on **rats** and other animals using his box called **Skinner box** that has different signal stimuli including:

- Different **colored lights** and a **lever** to be pressed.
- A **speaker** to deliver **sound** stimulus.
- An **electric grid** in the floor to deliver a mild **electric shock.**

The animal can be conditioned in several ways:

Example: + **Pressing** the **lever** resulting in **food delivery** (here food delivery is **reinforcer**), and the act of pressing continues.
- Pressing the lever only when one of the lights is lit to get food (reinforcer). When the other lights are lit, and if the animal presses the lever, it gets nothing, neither it may be harmed. This situation will **neither increase, nor decrease** the act of the rat. So, the response that follows this pressing behaviour is **neutral.**
- **Pressing** the **lever** and being **attacked** by an electric shock. Here, as the reaction following the pressing act is a **punisher,** the rat reduces its act of pressing the lever.

- Many of human behaviours are affected by **reinforcers** or **punishers.**

Example: + Starting smoking to be accepted by peer group.
- **Smoking** is a **behaviour**
- Acceptance by peer group is **reinforcer.**

- Animal training (**shaping**) is based on **operant conditioning.**

Example: + Training guide dogs for the blind.
- Training horses, zoo animals, etc for different purposes.
- Training dolphins and killer whales at marine parks.

Latent learning

Literally, this refers to **hidden learning** that is not apparent as it occurs. Brain acquires knowledge that is used when needed, **without reinforcement,** and stores it.

Example: + A **person** passing by an institution many times on his way to work. Later, when the institution advertises a vacancy, the person will not face difficulty to relocate the site of the institution as he was building a "**cognitive map**" of the area (i.e. he was storing information about the area without reinforcement).
- A **student** studying very hard but not sure of his knowledge until he/she **answers** exam questions correctly.

Insight learning

This is **finding solutions** to **problems** without experience ("trial and error"). It involves mental "trial & error". Often, the solution is learned suddenly.

⊄ The "**eureka**" **moment.**

Example: Solving challenging calculations; solving challenging **social, economic** and **political** problems.

It is **confined** to animals with more **advanced brain, Example**: Primates. **Wolfgang Koholer** worked (experimented) with **chimpanzees** to demonstrate insight knowledge. A Chimpanzee was placed in an enclosed room where a **bunch** of **bananas** was hanging form the ceiling out of reach of the chimp. **Empty boxes** were also placed in the room. The chimpanzee **piled** up the **boxes,** and **climbed** up to reach the **fruits**. Another chimp ('Sultan') drew thinner hollow bamboo stick into a thicker one to reach the fruits.

⊄ The "**EUREKA!**" Insight leaning is based on **thought** and **reasoning.**

Review Examples

21. Which of the following behaviours is acquired through experience?
(A) Kinesis
(B) Sneezing
(C) Habituation
(D) Imprinting

Explanation:
- **Habituation** is learned behaviour that is obtained by **repeated exposure** (experience) to a **stimulus.** Others are **innate** behaviours.

Answer: C

22. Which of the following types of learned behaviour requires "mental trial & error"?
(A) Insight
(B) Operent conditioning
(C) Classical conditioning
(D) Sensitization

Explanation:
- **Insight learning** involves "**mental trial and error**" to solve problems without experience.

Answer: A

23. When a chimpanzee solves the problem of reaching a bunch of bananas using piled boxes, it is an example of insight because:
(A) It involved hours of trial & error
(B) It solved the problem mentally
(C) It was the result of conditioned reflex
(D) It demonstrated species specific behaviour

Explanation:
- The chimp reached the banana by piling boxes on top of another and that is the result of "mental trial & error".

Answer: B

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24. Which type of learning behaviour is promoted by the achievement of correct answers to items in this examination? **(CEE General Science, 1995 E.C)**
(A) Conditioning
(B) Insight learning
(C) Perception
(D) Trial & error learning

Explanation:
- **Insight** helps a **student** to solve **calculations** and answer other exam questions.

Answer: B

25. Which of the following is **true** about an innate behaviour? **(UEE, 2003 E.C)**
(A) It is non-adaptive behaviour
(B) Members of a species behave similarly
(C) It is learned behaviour
(D) It is environmentally determined

Explanation:
- **Individuals** of the same **species** perform an **innate** behaviour in **similar way**. Though **not determined** by **environment**, innate behaviour is **natural**, **inborn**, not learned, and it is **adaptive** to the organism (i.e it has **survival advantage** to them).

Answer: B

26. It is said that a goat mother accepts and nurses as her own kid any young that she smells at a certain critical period, and rejects all others. This is an example of: **(UEE, 2003 E.C)**
(A) Habituation
(B) Imprinting
(C) Insight learning
(D) Sensitization

Explanation:
- **Imprinting** requires a **critical time** after which its occurrence (development) disappears.

Answer: B

27. In Pavlov's classical conditioning experiments on dogs, which one of the following is the **unconditioned response? (UEE, 2003 E.C)**
(A) The sound of the bell before presenting food
(B) The sight of the food from far away.
(C) The salivation of the dog at the sight of food.
(D) The salivation of the dog at the sound of the bell

Explanation:
- **Unconditioned response** (UR) is a **reaction** to the natural (original) **stimulus**.

Answer: C

28. With which of the following is hibernation most associated in animals? **(UEE, 2003 E.C)**
(A) Migration
(B) Daily biological cycle
(C) Annual biological cycle
(D) Availability of food in the store

Explanation:
- **Hibernation** is **circannual cycle** (rhythm) that reoccurs during dry/ hot or very cold seasons.

Answer: C

29. What is the "eureka" moment?
(A) It is the critical time when imprinting occurs
(B) It is the time when an organism suddenly solves a problem without experience.
(C) It is the moment when an animal is conditioned
(D) It is the rime when the conditioned and unconditioned stimuli are associated.

Explanation:
- The **"eureka"** moment is the moment when an organism solves a problem by **"mental trial and error"**.

Answer: B

30. Which of the following is a reinforcer for a dog being trained in crime investigation?
(A) Hitting the dog after every action it performs
(B) Shouting at the dog every time it performs an action
(C) Doing nothing after the dog performs an action
(D) Giving the dog a piece of meat after every performance

Explanation:
- Giving **food** to the dog is a **reinforcer** as this **increases** the **probability** of the **performance** being repeated.

Answer: D

10.4. Examples of behaviour patterns

Three behaviours are treated here. These are **courtship, social** behaviour and **territorialism.**

Courtship behaviour

This is an activity that preceds and **results** in **mating** and **reproduction**. It allows members of a **species** to **recognize** each other, and **prevents** or **reduces** attempts of **interbreeding** between **different species.**

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Courtship behaviour may involve:

- Simple **chemical**, **visual** or **auditory** stimuli
- Or, **complex series** of **acts** by **two** or **more individuals** using several methods of **communication**.

Elaborate **courtship rituals** can also help to **strengthen** an already established **pair bonds**. The bonds may last either through **time** of **young rearing** or longer.

Methods of communication

- Use of **Pheromones** by some **female** insects.
- Use of **touch** by some animals. **Example**: painted turtle using courtship songs. **Example**: Songs of frogs, songs of whale, bird songs, etc.

Pheromones ("Hormone bearers") - are **chemicals**, usually secreted by **females**, to **trigger behavioural response** in the other **members** of the **species**.

Types of pheromones

There are some **4 types** of **pheromones**. These are:

a) **Alarm pheromones** - are secreted during an **attack**. They produce **flight** or **aggression** responses in members of a group of a species.

b) **Releaser pheromones** - are highly **volatile** chemicals used to attract **male** from a distance of two miles or more.

c) **Territorial pheromones** - are chemicals used to **mark boundaries** of an owner's territory.
Example: Dogs **demarcate** their territory by **urine** which contains powerful pheromone.

d) **Sex pheromones** - are **chemicals** that signal the **availability (readiness)** of a **female** for **mating**.

In most animals, **courtship** behaviour is **innate**. It consists of a preprogrammed fixed action patterns (**FAPs**) in response to a **key stimulus**. A fixed action pattern in one animal (**Example**: the female) serves as a stimulus for another fixed action pattern in the other animal. **Example**: the male. **Courtship ritual** (a set procedure of acts making up a kind of **ceremony** or **routine**) continues until it becomes **successful**, or until one of the pair **tires**.

Territorial behaviour

This behaviour refers to the act of **defending a territory**. In connection to territoriality, **two terms** are important: **home range** and **territory**.

Home range - is a **larger area** over which the animals **roam** during their daily activities. It is **common** to all **members** of the **species**, and is **not defended**.

Territory - is any smaller space/area that an animal or a group of animals **defend** against **intruders** of the **same species**. A **territory**:

- Provides areas to **forage** for **food**.
- Increases chances of **attracting** a **mate**.
- Decreases **vulnerability** to **predators**.
- Shown by almost **all animal** species, including humans.
- A **territory** contains a **nest**, a **den** (mating site) and/or sufficient **food**.

Conspecific conflicts - are contests between individuals of the same species for a territory. Usually, **males** are **territorial**, but sometimes, **females** can also defend a territory.

- **Conspecific conflicts** usually involve **ritualistic displays** (fighting), and rarely involve **actual fighting**.
- **Territory demarcation** and **defence** may involve **visual**, **auditory** or **olfactory** displays
- **Territories** are **defended** by expending **longer time** and **energy**.
- **Example**: Sunbirds use up to **13,000** kilo joules/hour in defending its territory. This is more than the daily recommended energy intake of average adult man.

The **Ethiopian wolf** (***Canis simensis***) is a social animal that hunts in group in parks. It maintains group territory by marking with **urine** which contains **pheromones**. All **adult wolves** and some **subadult** (younger) **males** contribute to the territory marking. But, **subadult females** never mark.

Defending a territory

- **Marking** to "**warn off**" intruders.
- **Threats** (**vocal**, **smell** or **visual** displays).
  Example: Bird songs, loud calls of monkeys, etc.
- If an intruder **ignores warnings** and **approaches** the **territory**, it is **threatened** by intensive **visual displays**. There are **2 ways** of visual displays in this case.

a) **Exaggerating** an animal's **size** by the **fluffing up** of **fur** or **feathers**, and
b) **Showing off** the animal's **weapons**. **Example**: Porcupine raises its spines before it attacks the enemy.

- Ritual fighting, not actual fighting, can also be displayed.

Ritual fighting - is showing (displaying) **acts** of **fighting** without any physical contact.

- **Intensity** of ritual fighting **varies** with the **place** of the **intruder**.
- **Less intense**, if the **intruder** is **nearer** to the **border**.
- **More intense**, if the **intruder** is **further in** the **territory**.

The **variation** in the **intensity** of **ritual fighting** helps **researchers** to define **territorial boundaries**.

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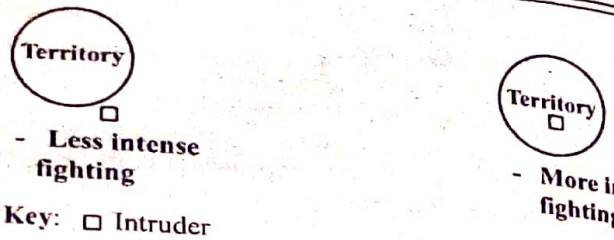

- **Less intense fighting**
- **More intense fighting**

Key: □ Intruder

Figure 10.3. Positions of an intruder and intensity of ritual fighting in animals

- Territorial **boundaries** may be found where the **fighting displays** of neighbours are **about equal** in **intensity**, or
- Where the **tendency** to **attack** and the **tendency** to **retreat** are about **equally** balanced.
- **Actual fighting** usually occurs in **over crowded** conditions where **resources** are **scarce**.
- **Injuries** and **death** may result during **actual fighting** (the **old** and sick animals are **affected most**).
- Usually, **territoriality** is an **effective** ways of **maintaining** a **healthy population** which is **biologically fit** and more **balanced**.

Social behaviour

Sociality is **interaction** between **two** or **more individuals** of a species. It **modifies** the individual's **behaviour**. It is **beneficial** to the **individual** and the **group** as a whole.

- It **serves** many **purposes**.
- Found in different animals.

  Example: Fish, birds, humans, mammals, etc.

Benefits of Social behaviour

- **Reduces** intra-specific **aggression** as **hierarchies** are formed leading to **stable groups**.
- **Improves effectiveness** of **reproduction** and/or parenting through courtship behaviour.
- Improves **efficiency** of **foraging**

  Example: Lions and wolves hunt in groups, dolphins surround fishes and trap turn by turn.
- Gives **protection** against attack
- Increases **survival chance** during migration flight.

Example: gees fly in a "V" shape which reduces **wind resistance** (the **lead position**, receiving the most resistance, is **rotated** among members).

- Social behaviour also allows animals to **communicate** across **long distances**.
- It also increases the **chance** of **surviving extreme conditions**.
- **Example**: Some birds, such as penguins, huddle together in a very cold weather ∝ **surface-area-to-volume ratio** is **reduced**, and this **minimizes heat loss** by up to **50%**. Here, position is constantly changed.

Social behaviour in bees

- **Honeybees** and **bumblebees** show **eusociality** (true social behaviour).
- **Eusociality** has 3 main features:

a) Cooperation in **caring** for **offspring**.
b) Presence of **several generations** in a colony. This ensures **sustainability** and parent assistance.
c) **Division of labour**. Not all members of social group is reproductively active.

Example: In honeybees:
- Queen - reproductively active
- Drone } - represent the 1st & 2nd castes
- Workers - females that are more or less sterilie (represent the 3rd caset)

Honey bees:
- Build their **nest** in **hollowed-out trees**, **beehives** or other **cavities**.
- Make **vertical sheets** of **hexagonal honeycomb** from **wax** secreted from their **abdominal glands**.
- Store **honey** and **pollen**, or **rear larvae** in the individual hexagonal cells.

Caste - a group within a social structure.

Table 10.2. Summary of the roles of different castes of bee colony

Caste	Role	Number in honeybee colony	Number in bumblebee colony
Queen (female)	Lay eggs	1	1
Workers (female)	Take care of larvae, build & clean nest & collect food	10,000 - 50,000	<50 to > 400 depending on species

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Drone (male)	Leave nest for mating & dies	100 - 500	0 - 50 depending on species & season

- **Queen** is larger in both **bumblebees** and **honeybees**.
- **Drones**: medium sized
- **Workers**: smaller
- **Honeybees** are **darker**, and **bumblebees** are **paler** in color.
- **Queen** produces **pheromones** within the nest
- **Pheromone** controls the **behaviour** of **worker bees** that are found at different stages of development.
- The **pheromones maintain** and **sustain social structure**.
- If the **queen** fails to produce enough **pheromone** and **eggs**, worker bees start feeding larvae **special food** so that they mature into a **new queen**.
- This **new queen** replaces the **older one**.
- Honey bee **colony** may last for **several years**.
- **Drones** are driven out of the nest during **winter** to conserve resource.
- **More drones** emerge the following **spring**
- At the **end** of **colony** cycle, **queen**, **drones** and most **worker** bees **die**.
- Few **large workers**, remain and assume the **status** of **queens**.
- These **fly away** to establish new **colonies** in the **coming spring**.

Roles of worker bees at different developmental stages

1. **First week after emergence**
   - **Clean** out **dirty** honeycomb
   - Feed larvae
2. **Second week after emergence**
   - Build honeycomb
   - Receive food from incoming forager
3. **Third week after emergence**
   - Guard the hive
   - Remove corpses
4. **Fourth week & afterwards**
   - Forage for nectar, pollen and water
   - Dies at 5-7 weeks.

The **source** of **nectar** is **communicated** between worker bees by a **"wag dance"** on a vertical **honeycomb**. The **direction** and **distance** information about food is obtained from the dance. The **dance** is "8" shaped. The information is conveyed in two ways:

a) **Dance angle** from the **vertical** on the **honeycomb** (i.e tilting of the dance) shows the **angle** between the **line** drawn from the **hive** to the **food**, and the line drawn from the **hive** to the **point** that is **perpendicular** to the **sun**.

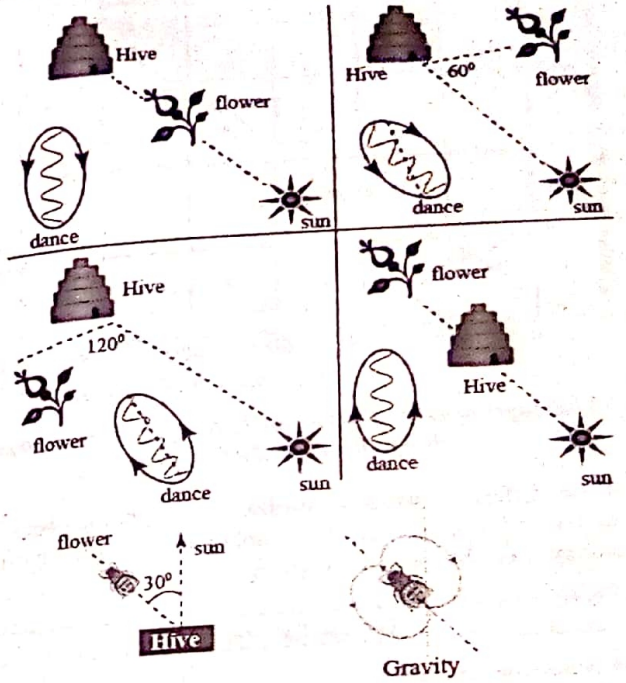

Figure 10.4. "Wag dance" in honeybees

b) The **length** of the **"straight-run"** or **"straight dance"** part in the center of the waggle dance is proportional to the **distance of food** from the **nest** (hive). **Shorter** "straight runs" show **nearer food sources**, and vice versa. According to recent findings, **sound** is also used to communicate **distance** of food between worker bees. It may also be used as **criteria** for **"recruiting"** other worker bees which may become **foragers** (nectar, water, pollen, etc collectors). In this case, the **duration** of **sound production** on the honeycomb after coming back from foraging, is **correlated** (proportional) to the **distance** of nectar (food) **source** form the hive (nest).

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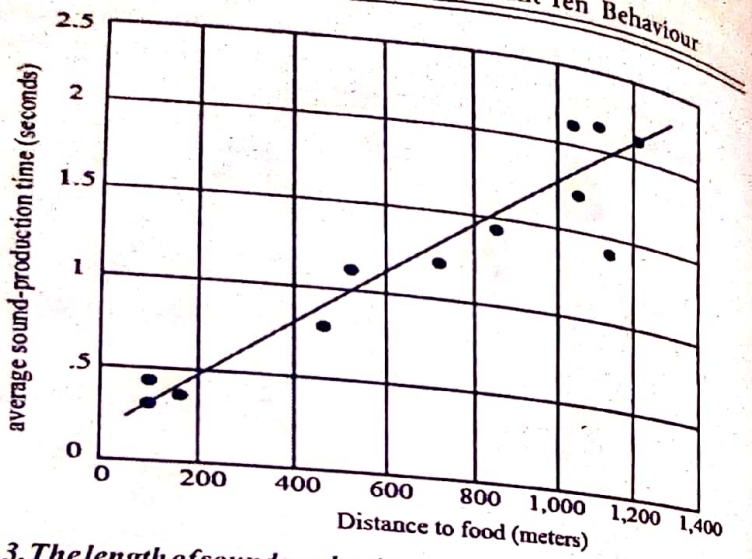

Figure 10.3. The length of sound production by foraging workers is proportional to the distance of a food source

The role of the different **castes** is **similar** in the bumblebees. The queen maintains "**order**" in the same way. But, bumblebee nests (colonies) are annual, and a new colony is established every spring.

Review Examples

31. Human behaviour has both_____and_____components.
    - **Innate** and **learned** components: it is **difficult** if not impossible to sort out the relative contributions of the **2** to a particular behaviour.
32. The study of the biological bases of social behaviour is called_____.
    - Sociobiology
33. Programmed behaviours that are less affected by experience are called_____.
    - Innate behaviours.
34. Which learning types are the simplest?
    - Habituation and sensitization
35. Which of the following types of movements in response to a stimulus has no specific direction? **(UEE, 2004 E.C)**
    (A) Taxis
    (B) Kinesis
    (C) Gravitropism
    (D) Phototropism

Explanation:
- **Kinesis** is a **non-directional movement** in response to a **stimulus** in **small animals**. It is an example of orientational behaviour. **Taxis, gravitropism** and **phototropism** are all **directional** responses.

Answer: B

36. Why is that woodlice are typically found under logs, stones, bark and amongst leaf litter? **(UEE, 2004 E.C)**
    (A) To be sheltered in a dry windy environment
    (B) To run away from the area where the air is humid.
    (C) To make sure that they are in the hottest place all the time.
    (D) To reduce the rate at which water is lost from their bodies.

Explanation:
- **Woodlice** are very **small** and they have **large surface area to volume ratio**. They have **no hard** outer **cover**. Both these characteristics **increase** their **rate** of **water loss**. So, taking shade under logs, stones and bark minimizes this water loss.

Answer: D

37. Which one of the following do bees use to inform other bees about the location and distance of a new source of nectar they discover? **(UEE, 2005 E.C)**
    (A) Pheromones
    (B) Waggle dance
    (C) Buzzing noise
    (D) Vibration of wings

Explanation:
- Though **buzzing** sound is also being investigated, **waggle dance** is a well known means of **communication** between worker bees about the **distance** and **direction** of **nectar**.

Answer: B

38. Which one of the following is not classified as a learned behaviour?
    (A) Insight
    (B) Innate
    (C) latent
    (D) Conditioned

Explanation:
- **Innate** behaviour is **inborn**, it is **not acquired** from **experience**.

Answer: B

39. To which kind of animal behaviour can the spinning of a web by a spider be classified **(UEE, 2005 E.C)**
    (A) Learned behaviour
    (B) Experiential behaviour
    (C) Instinctive behaviour
    (D) Accidental behaviour

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Explanation:

- **Web spinning** in **spider** is **innate behaviour** (an **instinct**) that shows a **fixed action pattern** (FAP). It is less affected by **experience.**

40. Which of the following honey bee castes are fertile? **Answer: C**
(A) Drones (C) Workers
(B) Queen (D) A & B

Explanation:

- Both the **drones** and **queen** are **reproductively active** (i.e. fertile). The worker bees are **sterile** females.

41. The activity that preceds and results in mating and reproduction is: **Answer: D**
(A) Courtship behaviour (C) Social behaviour
(B) Territorial behaviour (D) Learned behaviour

Explanation:

- In almost all animals, **courtship behaviour** is needed before mating and **reproduction**. It helps in species **recognition.**

Answer: A

42. What are the 4 types of pheromones?
- **Alarm** pheromones
- **Releaser** pheromones
- **Territorial** pheromones
- **Sex** pheromones

43. What are the methods of communication in animals?
- Use of **pheromones**
- **Touch**
- **Songs**
- **Visual** displays

44. What are **conspecific conflicts**?

Explanation:

- These are **contests** (disagreements) between **individuals of the same species.** **Example**: fighting between gorillas, disagreements between humans, etc.

45. Some migratory birds fly in a "V" shape. What is the survival advantage of this behaviour?

Explanation:

- This **reduces wind resistance** and saves energy, but the **lead position** should be **rotated** between individuals to **share** the **burden** at that point.

1. Discuss the adaptive significance of habituation, imprinting, classical conditioning, operant conditioning and insight learning.

Explanation:

- **Learning** - is a sophisticated process in which the responses of the organism are modified as a result of experience. There are various forms of learning.
- **Habituation** - is a form of learning in which an animal learns to ignore a stimulus upon repeated exposure to it. It helps an organism to survive danger.
- **Operant conditioning** - a form of learning in which a certain response to a stimulus is reinforced. It is also called **a trial and error learning.** It helps animals to repeat or avoid a behavior up on reward or punishment.
- **Imprinting** - helps an animal to form a social attachment to another organism soon after birth or hatching.

2. Discuss the hypothesis that optimal foraging behavior is adaptive.

Explanation:

Optimal foraging is the most efficient strategy for an animal to get food, often enhances reproductive success.

3. Analyze social behavior in terms of costs and benefits, and describe modes of animal communication.

Explanation:

- **Social behavior** is an adaptive interaction, usually among members of the same species. Animal societies can be characterized a means of **communication, division of labour,** and **tendency to stay together.**

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- **Animal communication** - involves the exchange of mutually recognizable signals, which can be **auditory, visual, electrical,** or **chemical.**

4. Relate the experiment of Ivan Pavlov's experiment of classical conditioning with learned behavior; explain to which type of learned behavior it is highly related.

Explanation:

Ivan Pavolv's experiment of ringing bell on a dog is an example of classical conditioning in which the food is rewarded on a dog in order to make it repeat the behavior (salvation).

5. Compare innate behavior with learned behavior.

Explanation:

Innate behaviour	Learned behaviour
• Genetically determined and common to all members of a species. • Behavior is fully functional at the first attempt. • There is generally, no modification of the behavior. • Adaptive behavior that has been refined as a result of natural selection.	• The behavior is changed by, or develops through, experience and may vary from individual to individual. • The animal develops the behavior through trial and error or by insight. • The behavior may be modified by new experiences. • Behavior is learned by each member of the species and may not be adaptive.

6. Describe and give examples of courtship, territorial and social patterns of behavior in animals.

Explanation:

i. **Courtship behaviour** - is an activity that precedes and results in mating and reproduction. It allows different members of the same species to recognize each other. This may involve chemical, visual or auditory stimulus.

Example: seasonal courtship behaviour between dogs.

ii. **Territorial behaviour** - gives the holder of the territory or area a chance of getting food, attaching mate and reduces vulnerability to predators.

Examples: Bees

iii. **Social behaviour** -is the set of interactions that occur between two or more individuals of the same species that modify the behaviour of individuals of the same species in a way that is usually beneficial to the group as a whole.

7. Describe the interactions of the heredity, environment, and maturation in animal behaviour.

Explanation:

- Behaviour results from the interaction of genes (innate behaviour) and environmental factors. The capacity for behaviour is inherited, but behaviour is modified in response to environmental experience, this is **learned behaviour.**
- An organism must be mature - physiologically ready to produce a given behaviour before it can perform that pattern of behaviour. Walking and many other behaviours that we view

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is as automatic are **motor programs,** coordinated sequence of muscle actions.

- **A fixed action patter (FAP)** - is an automatic behaviour that once activated by a simple sensory stimulus, continues to completion regardless of sensory feedback. It can be triggered by specific unlearned **sign stimulus**, or **releaser.**

POST TEST

1. The males of bird species do a "dance" and "sing" a song each spring. What is the main purpose of these behavious?

   A. To scare off young birds.
   B. To imprint off young birds.
   C. To lean their feathers.
   D. To attract female birds.

2. At night, moths travel toward light. Which type of behaviour does this describe?

   A. Habituation
   B. Imprinting
   C. Innate behaviour
   D. Learned behaviour

3. A sophisticated process in which the response of an organism are modified as a result of experience is called:

   A. Fixed action pattern
   B. Habituation
   C. Imprinting
   D. Classical conditioning
   E. Learning

4. The behaviour reduces an individuals reproductive fitness white increasing the fitness of the family.

   A. Altruism
   B. Agonistic behaviour
   C. Cooperation
   D. Imprinting

You want to train your puppy to wait at the crub until you tell him to cross the road. Your friend advices you to give your dog a treat every time he does as you ask. Your friend is advising that you train the dog using.

A. Operant conditioning
B. Classical conditioning
C. Imprinting
D. Fixed action pattern
E. Habituation

_____ is learning that occurs during a sensitive or critical period in early life and is irreversible for the length of period.

A. Operant conditioning
B. Habituation
C. Classical conditioning
D. Trial and error learning

"Marry had a little lamb; it's fleece was white as snow. And everywhere that Marry went, the lamb was sure to go." The behaviour of the lamb is best described as:

A. Habituation
B. Imprinting
C. Operant conditioning
D. Classical conditioning
E. Fixed action pattern

Fixed action patterns are imitated by external stimuli called:

A. Fixed action pattern.
B. Signal stimuli.
C. Agonistic behaviour.
D. Dominance hierarchies.
E. Sensitive periods.

Animals that help other animals are expected to be:

A. Stronger than other animals
B. Related to the animals they help
C. Male
D. Female
E. Disabled in some way

An animal that scarifies itself for its relatives is exhibiting:

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A. Operant conditioning
B. Kin selection
C. Classical conditioning
D. Imprinting
E. Habituation

11. The contemporary study of behaviour in natural environments from the point of view of adaptation is:

A. Ecology
B. Ethology
C. Behavioural ecology
D. Evolutionary ecology
E. Behavioural science

12. The responses of an organism to signals from its environment are its:

A. Behaviour
B. Culture
C. Ultimate behaviour
D. Releasers
E. Motor programs

13. A form of learning in which a young animal forms a strong attachment to a moving object (usually its parent) with in a few hours of birth is:

A. Classical conditioning
B. Operant conditioning
C. Imprinting
D. Insight learning
E. Parental investment

14. An animal learns to ignore a repeated, irrelevant stimulus. This is:

A. Classical condition
B. Operant conditioning
C. Imprinting
D. Insight learning
E. Habituation

15. Chemical signals that convey information between members of a species are:

A. Pheromones
B. Hormones
C. Neurotransmitters
D. Leks
E. Neuropeptides

1. Behavioural ecologists have demonstrated that young tiger salamanders can become cannibals and devour other salamanders. investigators observed that the cannibal salamanders preferred eating unrelated salamanders over their own cousins and preferred eating cousins over their sublings. Explain this behaviour based on what you have learned in this chapter.

2. How is the society of a social insect different from human society? What are some similarities between the transmission of information by heredity and by culture?

Review Examples

I. Choose the correct answer for the following questions

1. What do we call the behaviour if an animal learns not to respond to a stimulus that is repeated a number of times without any effect?
(A) Habituation
(B) Sensitization
(C) Latent learning
(D) Associative learning

2. Which of the following is **not** an example of innate behaviour?
(A) Reflex action
(B) Orientation behaviour
(C) Operant conditioning
(D) Instinctive behaviour

3. Which of the following is the function of a queen in the community of honeybee?
(A) Guarding the hive from intruders
(B) Laying eggs of the community
(C) Rearing the young of the community
(D) Building the hive of the community

4. Among the following, which one is a learned behaviour? **(UEE, 2005 E.C)**
(A) Reflex action
(B) Habituation
(C) Imprinting
(D) Fixed action pattern

5. During seasons of reproduction, males of some species of birds produce colorful features to attract females. What do ethologists call this method of communication in animals? **(UEE, 2004 E.C)**
(A) Visual
(B) Chemical
(C) Auditory
(D) Touch

6. What is the role of worker honeybee just after it emerges? **(UEE, 2004 E.C)**
(A) Forage for nectar, pollen & water
(B) Clean out dirty honeycomb
(C) Guard the hive
(D) Build honeycomb

7. In the classical conditioning experiments on dogs made by Ivan Pavlov, which of the following was the conditioned response?
(A) Ringing of bell by Pavlov
(B) Salivation of dog for meat
(C) Salivation of dog for bell ringing
(D) Presentation if meat by Pavlov

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8. Insight learning involves:
   (A) Trial & error
   (B) Mental reasoning
   (C) Operant conditioning
   (D) Innate behaviour

9. In the classical conditioning experiment performed by Pavlov on dogs, which of the following alternatives is the **unconditioned stimulus?** **(UEE, 2004 E.C)**
   (A) The sound of the bell
   (B) The smell of the food
   (C) Salivation at the sound of the bell
   (D) Salivation at the smell of the food

10. What do we call the learned behaviour if a mouse that had just escaped from the mouth of a cat jumped violently at a slight touch by a trivial object? **(UEE, 2005 E.C)**
   (A) Latent learning
   (B) Sensitization
   (C) Conditioning
   (D) Imprinting

11. The behaviour known as ritual fighting is mostly associated with:
   (A) Courtship behaviour
   (B) Territorial behaviour
   (C) Social behaviour
   (D) Orientation behaviour

12. Suppose a boy, after watching his father mending shoes on several occasions, finally made it himself. What type of learned behaviour is this?
   (A) Classical conditioning
   (B) Latent learning
   (C) Habituation
   (D) Sensitization

13. The group of the Ethiopian wolf which does not contribute to territory marking with their urine containing pheromones is_____. **(UEE, 2005 E.C)**
   (A) Adult males
   (B) Adult females
   (C) Sub-adult males
   (D) Sub-adult females

14. Which of the following is the correct route that connects a stimulus and a response? **(UEE, 2005 E.C)**
   (A) Receptor --- Coordinator --- Effector
   (B) Coordinator --- Receptor --- Effector
   (C) Receptor --- Effector --- Coordinator
   (D) Effector --- Receptor --- Coordinator

15. In a reflex action, to which of the following does the impulse reach last? **(EGSECE, 2003 E.C)**
   (A) Coordinator
   (B) Receptors
   (C) Efferent neuron
   (D) Effector

16. Which of the following is an example of **hydrotropism?** **(EGSECE, 2003 E.C)**
   (A) Shoots grow towards light source
   (B) Pollen tubes grow towards ovaries
   (C) Tendrils grow around stems
   (D) Roots grow towards a source of water

17. Which of the following is **true** about innate behaviour?
   (A) It is acquired through experience
   (B) It is an adaptive behaviour
   (C) It does not need a stimulus
   (D) It is environmentally determined

18. The larvae of most insects move away fro light. What kind of behaviour is this?
   (A) Taxis
   (B) Tropism
   (C) Kinesis
   (D) Imprinting

19. Which of the following components of behaviour is essential in triggering a response?
   (A) Receptor
   (B) Stimulus
   (C) Effector
   (D) All

20. Which of the following is the increased rate of a non-directional movement in response to a stimulus in simple animals?
   (A) Kinesis
   (B) Taxis
   (C) Phototropism
   (D) Geotropism

21. One f the following statements is **incorrect.**
   (A) Ethology is the study of animal behaviour
   (B) Ethology has no contribution to the study of human behaviour.
   (C) Ethology contributes to the animal welfare
   (D) Neuroethology deals with how behaviour is linked to the nervous system.

22. The type of innate behaviour which involves a fixed action pattern for a key stimulus is:
   (A) Reflex action
   (B) Orientation behaviour
   (C) Instinctive behaviour
   (D) All

23. Which of the following statements is **not true?**
   (A) Reflex actions are protective
   (B) "Knee jerk reflex" is a somatic reflex.

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(C) Heart rate is controlled by autonomic reflexes.
(D) Peripheral nervous system comprises of brain and spinal cord.

24. Which of the following statements about instinctive behaviour is not true? Instinctive actions:
(A) Can sometimes be modified by experience
(B) Are innate
(C) Requires some learning
(D) Are adaptive

25. In one of the following learned behaviours, a naturally occurring stimulus is associated with a conditioned stimulus.
(A) Habituation (C) Insight learning
(B) Sensitization (D) Classical conditioning

26. Responses from the environment that neither increase nor decrease the probability of a behaviour being repeated are
(A) Reinforcers (C) Neutral operants
(B) Punishers (D) Neutral stimuli

27. Worker bees are:
(A) Sterile females (C) Sterile males
(B) Fertile females (D) Fertile males

28. The coordinated response of an organism to internal and external stimuli is called
(A) Aggression (C) Behaviour
(B) Entropy (D) Fixed action

29. The biological clock of mammals is called:
(A) Hypothalamus (C) Prostate gland
(B) Pineal gland (D) Suprachiasmatic nucleus

30. The biological clock of mammals is located in the:
(A) Cerebrum (C) Cerebellum
(B) Hypothalamus (D) Spinal cord

31. Which of the following is **not** an example of a circannual rhythms?
(A) Sleeping and waking (C) Migration
(B) Hibernation (D) Reproduction

32. Which of the following behaviour is genetically determined and common to all members of a species?
(A) Sensitization (C) Habituation
(B) Imprinting (D) Conditioning

33. Which of the following statements is correct?
(A) Bumblebees and honeybees have the same number of queen per colony.
(B) Colony size of honeybee is larger than that of bumblebees.
(C) Colony life-span is longer in honeybees than that of bumblebees which forms every year.
(D) All are correct

34. Plant responses to uniformly distributed stimuli such as light intensity, are known as:
(A) Nastic movements (C) Aerotropism
(B) Photoperiodism (D) Thigmotropism

35. Which of the following statements about tropisms in plants is **incorrect**?
(A) Roots are positively geotropic
(B) Shoots are positively phototropic.
(C) Roots are positively phototropic.
(D) Most climbing plants are positively thigmotropic

36. Which term means "a behavioural adaptation of travelling long distances to avoid a season that the animal can not survive in"?
(A) Hibernation (C) Aestivation
(B) Migration (D) Mimicry

37. Which of the following is an **instinct**?
(A) A man teaching her child how to cross the street
(B) A dog barking to defend itself.
(C) A kid teaching his dog to sit.
(D) A driver training his son how to drive

38. Young ducklings first see a researcher's red boots and then begin to follow him a round. What kind of behaviour are the ducklings demonstrating?
(A) Imprinting (C) Classical conditioning
(B) Habituation (D) Trial & error

39. Which of the following types of chemicals released into the air by female gypsy moths and female silk moths attract males of their respective moth species?
(A) Ecdysones (C) Auxins
(B) Pheromones (D) Phytotoxins

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40. On average, social behaviour:
(A) Increases the number of deaths
(B) Increases individual fitness.
(C) Increases agonistic behaviour.
(D) Decreases group survival

41. Pill bugs are terrestrial isopods that use modified gills for respiration. A scientist prepares a "choice chamber" that consists of a dry area separated from but connected to a damp area. Then pill bugs were put in each area. The scientist observed them during the next half an hour. The pill bugs placed in the dry chamber migrated to the damp area. Those that were put in the damp chamber randomly moved within the damp area, slowing down their movement as time progressed. What kind of behaviour was demonstrated by the pill bugs placed in the dry chamber?
(A) Taxis (C) Imprinting
(B) Kinesis (D) Habituation

42. What kind of behaviour was demonstrated by the pill bugs placed in the damp chamber?
(A) Conditioning (C) Kinesis
(B) Imprinting (D) Taxis

43. A behaviour that an organism is born with is called:
(A) Innate behaviour (C) Imprinted behaviour
(B) Conditioned behaviour (D) Learned behaviour

44. A bird building a nest is an example of:
(A) Instinctive behaviour (C) Conditioned behaviour
(B) Learned behaviour (D) Imprinted behaviour

45. Sneezing and shivering are examples of:
(A) Instincts (C) Trial & error behaviour
(B) Learned behaviour (D) Reflexes

46. A chimpanzee reaches a hanging banana by:
(A) Conditioning (C) Imprinting
(B) Insight (D) Reflexes

47. According to Pavlov's experiment on dogs, the dogs learn to salivate when:
(A) Food is provided (C) The smell of food is present
(B) The animal is hungry (D) The bell rings

48. The type of behaviour in the above question is:
(A) Innate behaviour (C) Reflex
(B) Instinct (D) Learned behaviour

49. Pheromones are used by some animals in:
(A) Action communication (C) Sound communication
(B) Light communication (D) Chemical communication

50. Hibernation and migration are examples of_____behaviour:
(A) Cyclic (C) Courtship
(B) Aggressive (D) Submission

51. Movement directed towards a light source is called:
(A) Positive chemotaxis (C) Positive kinesis
(B) Positive phototaxis (D) Negative kinesis

52. A sea slug normally responds to being poked by curling up. After being poked repeatedly, the slug no longer curls up. This is an example of:
(A) Dishabituation (C) Sensitization
(B) Habituation (D) Operant conditioning

53. During courtship, a male *Drosophila melanogaster* (fruit fly) will vibrate his wings, tap the female with his forelegs, and extend one wing out at 90 degrees from his body. This behaviour is:
(A) The result of operant conditioning
(B) The outcome of classical conditioning.
(C) The result of instinctive behaviour
(D) The result of insight learning

II. Give short answers to the following questions

54. Into how many parts is our nervous system divided physically? And what are they?

55. What are the two components of the **central nervous system**?

56. What are the components of the peripheral nervous system?

57. Into how many categories (parts) is our nervous system divided functionally? What are they?

58. What are the components of the autonomic nervous system?

60. What are the 2 types of reflexes? And what are the 2 types of habituation?

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Answer To Review Questions of Unit Ten

I. Multiple choice

1.	A	12.	B	23.	D	34.	A	45.	D
2.	C	13.	D	24.	C	35.	C	46.	B
3.	B	14.	A	25.	D	36.	B	47.	D
4.	B	15.	D	26.	C	37.	B	48.	D
5.	A	16.	D	27.	A	38.	A	49.	D
6.	B	17.	B	28.	C	39.	B	50.	A
7.	C	18.	A	29.	D	40.	B	51.	B
8.	B	19.	D	30.	B	41.	A	52.	B
9.	B	20.	A	31.	A	42.	C	53.	C
10.	B	21.	B	32.	B	43.	A		
11.	B	22.	C	33.	D	44.	A		

II. Short answer

54. **Two**: **Central** and **Peripheral** nervous system
55. **Brain** and **spinal cord**
56. **Cranial nerves** (which branch from the brain) and **spinal nerves** (which branch from the spinal cord)
57. **Two**: **Somatic** nervous system and **Autonomic** nervous system
58. a) Sensory division
    b) Sympathetic division and
    c) Parasympathetic division and
59. Learning how to drive, how to solve mathematical calculations, etc are adaptive as have advantageous. But, learning to smoke, drink alcohol and the like are non-adaptive as they are mostly harmful.
60. **Reflexes**:
    a) Somatic reflexes
    b) Autonomic reflexes

    Habituation:
    a) Sensory habituation
    b) Brain habituation

A

Absorption spectrum: The spectrum of light absorbed by a particular pigment.

Abyssal zone: The deepest part of the ocean, with depth generally greater than 1000 meters, where light does not reach.

Accessory pigment: A pigment that captures light energy and transfers it to chlorophyll a; beta-carotene is an example of an accessory pigment.

Accuracy: How close the measured value is to the true value.

Acetyl: The 2-carbon remainder of the carbon skeleton of pyruvic acid that is able to enter the mitochondrion.

Acetyl CoA : A coenzyme with the formula $C_{23}H_{39}O_{17}N_7P_3S$ that is formed when acetic acid reacts with coenzyme A at the start of the citric acid cycle.

Acquired immunodeficiency syndrome (AIDS): An infectious disease caused by a retrovirus called the human immunodeficiency virus (HIV).

Actin: A protein that is the main component of the thin filaments in a muscle myofibril.

Actin filament: The smallest (4-7nm in diameter) of the three types of filaments that comprise the cytoskeleton.

Activation energy: Energy added to molecules so that they collide with enough energy to break chemical bonds. The initial energy required to begin a chemical reaction.

Active site: Site on an enzyme where the substrate bonds and the reaction is catalyzed.

Active transport: Movement of solutes across a membrane from low to high concentration; active transport requires energy from cellular metabolism.

Adaptive radiation: Evolution of many different groups from a common ancestor.

Adenine: A nitrogen base with the formula $C_5H_5N_5$ found in DNA and RNA.

Adenosine di phosphate (ADP): A coenzyme compound made of adenine, ribose and two molecules of phosphoric acid that can be converted into ATP

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Adenosine tri phosphate (ATP): A nucleotide consisting of adenine, ribose, and three phosphate groups; the major source of usable chemical energy in metabolism.

Adipose Albinism: A condition caused by the lack of melanin production in the iris, hair, and skin.

Aldose : Monosaccharide containing an aldehyde group. The prefix *ald-* indicates the presence of an aldehyde group, and the *–ose* suffix indicates a carbohydrate.

Alleles: Different forms of a gene that code for slightly different traits.

Allopathic speciation: Speciation induced by geographical or physical separation of the ancestral population.

Alpha helix: Secondary structure of a protein in which the polypeptide backbone is arranged in to a regular spiral shape.

Anabolism: Part of metabolism that uses energy to build larger, more complex molecules from simpler ones.

Anaerobic respiration: A biochemical pathway that does not require oxygen for the production of ATP and does not use as its hydrogen ion acceptor.

Aneuploidy: A condition in which the chromosome number is not an exact multiple of the haploid set.

Anhydrase: An enzyme involved in the removal of water from acompound.

Anticodon: Three nucleotide bases found on each transfer RNA that code for a specific amino acid and complement the codon on messenger RNA.

B

Bacilli: Rod-shaped bacteria.

Bacteriophage: A virus that infects bacteria.

Base deletion: A mutation in which a nucleotide base is lost from the DNA sequence.

Base insertion: A mutation in which an extra nucleotide base is added to the DNA sequence.

Base pair: A pair of complementary nucleotides. In DNA, A-T and G - C. In RNA, A-U and G-C.

Base substitution: A single base change in a DNA molecule that produces a mutation.

B-cell: A lymphocyte that produces antibodies.

Behavioral isolation: A genetic isolating mechanism that prevents interbreeding between species because of differences in behavior.

Beta-glucose: The form of glucose whose structure, when drawn in flat plane, has a hydroxyl group at the first carbon that points up.

Bilayer: In referring to phospholipids, a bilayer is a spontaneously formed double layer of lipid, with an interior of hydrophobic hydrocarbons and an exterior of hydrophilic phosphate groups.

Biochemical: Organic and inorganic chemicals that occur in living organisms and are involved in the processes of life.

Biological clock: An internal biological timing system that influences cyclic phenomena.

Biome: A large geographical area with characteristic life forms.

Biosphere: The life zone of earth that includes the lower atmosphere, land, and water to a depth of 2 kilometers.

Biotic potential: The highest rate of reproduction for a population under ideal conditions.

Bipedal: Walking on two limbs.

Birth rate: The rate at which new individuals are added to the population.

C

C3 plant: Plant in which the first fixation of carbon is via the Calvin cycle. The first stable product of photosynthesis is a three carbon compound.

C4 plant: Plant in which the first fixation of carbon produces a four-carbon acid. C4 plants can fix CO2 in places where the concentration is low.

Calvin cycle: Series of enzymatic reactions in which CO2 is reduced to 3phospho glyceraldehyde (a three-carbon compound) and the CO2 acceptor (ribulose, 1,5-bisphosphate) is regenerated.

Camouflage: Any way of blending with the environment.

Cancer: An abnormal, uncontrolled growth of cells that usually causes death if not treated.

Carbon fixation: The second stage of photosynthesis which occurs in the chloroplasts in which carbon dioxide is converted into sugar molecules.

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Carbonyl: Family of organic compounds composed of a carbon double-bonded to an oxygen.

Carbonyl group: Functional group characterized by a carbon oxygen double bond.

Carrying capacity: The limit at which the environment can support a certain population.

Catabolism: The chemical reactions that break down large molecules into smaller molecules with the release of energy.

Catalyst: Compound that increases the rate of a reaction.

Cell cycle: Sum of the phases of growth of an individual cell type; divided into G1 (gap 1), S (DNA synthesis), G2 (gap 2), and M(mitosis)

Cellular respiration: The process of converting food energy into a form usable by cells that takes place in the mitochondria.

Cellulase: An enzyme that breaks down cellulose into smaller units by cleaving the 1,4 linkages between molecules of beta- glucose.

Cellulose: Plant polysaccharide made up of b 1,4 linked glucose.

Cenozoic: The geological era from the Mesozoic to the present (about 65 million years).

Chemotropism: In plants, growth towards or away from a chemical stimulus.

Chitin: A hard carbohydrate material found in the exoskeletons of arthropods and also in the cell walls of some fungi.

Chloroplast: A plastid containing chlorophyll

Cholesterol: Most abundant steroid in animals. An important membrane lipid with significant health implications.

Chromatid: A single strand of the chromosome pair formed by splitting of a chromosome during mitosis.

Chromatin: A compound of nucleic acids and proteins that makes up chromosomes.

Chromosomal mutation: A change in the gene arrangement in a cell as a result of breaks in the DNA molecule.

Chromosome: In eukaryotes, a DNA molecule complexed with histone proteins to form a threadlike structure containing genetic information arranged in a linear sequence.

Circadian rhythm: Cyclic behavior patterns that are repeated every 24 hours.

Citrate (citric acid): A six-carbon organic acid that is converted to iso citric acid in the second step of the Krebs cycle.

Citric acid cycle: A series of reactions in the aerobic phase of cellular respiration that breaks down acetyl CoA to form carbon dioxide, water and releases energy.

Classical conditioning: A form of learning in which an organism associates a previously unrelated stimulus and response.

Cleavage: One of the first cell divisions in a zygote.

Climax community: The final stable stage of change in a community of plants and animals.

Clone: A group of identical cells or a complete individual produced from a single somatic cell.

Cloning: A method of asexual reproduction in which a single somatic cell develops into a complete identical organism.

Cocci: Eubacteria with spherical Shapes

Co dominance: A condition that occurs when both alleles of a heterozygous gene are expressed equally.

Codon : A sequence of three nucleotides in a gene or molecule of mRNA that corresponds to a specific amino acid or to a stop signal at the end of a gene.

Coenzyme: A non protein molecule that works with an enzyme to catalyze a reaction.

Cofactor: A non-protein substance required by enzymes for proper function.

Colour blindness: Inability to see any difference between some or all colours. The most common is red green colour blindness which is controlled by a recessive gene on the X chromosome.

Commensalism: A symbiotic relationship in which one organism benefits and the other is unaffected.

Competitive inhibition: The formation of a temporary enzyme-inhibitor complex that interferes with the normal formation of enzyme-substrate complexes, resulting in a decreased turnover.

Competitive inhibitor: Reversible inhibitor that binds to the active site of an enzyme and thus competes with the substrate.

Complementary base: A base that can form hydrogen bonds with another base of a specific nucleotide.

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Complete dominance: A condition that occurs when the phenotype of one allele completely masks the phenotype of another allele for a heterozygous gene.

Complex carbohydrates: Macromolecules composed of simple sugars (fructose, glucose) that have joined together by releasing molecules of water; or digestible polysaccharides of the diet, mostly starches.

Concentration gradient: The gradual change in the number of molecules per unit of volume over distance.

Conditioned response: The behavior displayed when the neutral stimulus is given after association has occurred.

Control group: The situation used as the basis for comparison in a controlled experiment.

Controlled experiment: An experiment in which only one factor is varied.

Convergent evolution: The process whereby similar characteristics are independently evolved by different groups of organisms in response to similar selective pressures.

Cori cycle: Cycle in which lactic acid produced in muscle is converted back to glucose in the liver.

Courtship behavior: A specialized behavior pattern that occurs before mating takes place.

Crassulacean acid metabolism (CAM) : A type of photosynthesis in which CO_2 is fixed at night into four-carbon acids; during the day, the stomata close and the carbon is fixed via the Calvin cycle; CAM helps plants conserve water and is often characteristic of xerophytic plants.

Cristae (singular, **crista) :** The tubular or vesicle shaped folds of the inner membrane of mitochondri.

Crossing-over: Exchange of parts between two homologous chromosomes

Cytochrome : Heme-containing proteins that carry electrons in respiration and photosynthesis.

Cytochrome complex: A cluster of cytochromes that carry electrons

Cytokinesis: The process of cytoplasmic division in a cell.

Cytosine: A nitrogen base with the formula $C_4H_5ON_3$ that is found in DNA and RNA.

D

Dark reaction: Part of photosynthesis that uses ATP, NADPH, and carbon dioxide to make glucose.

Daughter cells: Two cells formed by cell division.

Daughter chromosomes: Chromosomes produced by DNA replication that contain identical genetic information; formed after chromosome division in anaphase.

Daughter nuclei: Two nuclei formed by mitosis.

Death phase: The portion of some population growth curves in which the size of the population declines.

Decarboxylation: Loss of CO2. This reaction takes place very readily for beta keto acids.

Deciduous forest: A forest made of trees that annually shed all their leaves.

Degenerate code: Term used to describe the genetic code, in which a given amino acid may represented by more than one codon.

Dehydration: Formation of an alkene from an alcohol as a result of the loss of water.

Dehydration synthesis reaction: A reaction that results in the formation of a macromolecule when water is removed from between the two smaller component parts.

Deletion: A mutation in which a chromosomal segment is lost to the cytoplasm.

Demographic transition: A shift in birth and death rates that affects the size of a population.

Denaturation: Alteration of the structure of a protein so that some of its original properties are diminished or eliminated.

Denitrifying bacteria: Bacteria that convert nitrates to nitrogen gas.

Deoxy ribonuclease : A class of enzymes that breaks down DNA into oligonucleotide fragments

Deoxyribose: A 5-carbon sugar molecule that is a component of DNA.

Desert: A biome characterized by scant rainfall and widely spaced vegetation.

Development: The stages that an organism goes through from when it begins life until it reaches adulthood.

Dihybrid cross: A genetic cross involving two sets of trait.

Dipeptide: A molecule made of two amino acids.

Diploid: Having two of each type of chromosome; the somatic number of chromosomes

Disaccharide: A compound sugar composed of two monosaccharides.

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Discontinuous variation: Phenotypic data that fall into two or more distinct classes that do not overlap.

Disulfide bond or Disulfide Bridge: A type of covalent bond between the sulfur atoms of separate amino acids in the same protein; disulfide bonds strengthen the tertiary structure of proteins.

Divergent evolution: A basic evolutionary pattern in which individual speciation events cause many branches in the evolution of a group of organisms.

DNA (deoxyribonucleic acid): A very large nucleic acid macromolecule found in the cell nucleus, composed of repeating units of deoxyribose sugar, phosphate, and nitrogen bases in a double strand formation.

DNA code: A sequence of three nucleotides of a DNA molecule.

DNA ligase: A type of enzyme that joins adjacent nucleotides together by catalyzing the formation of sugar-phosphate bonds in a strand of DNA.

DNA polymerase: An enzyme that bonds new DNA nucleotides together when they base-pair with an existing DNA strand.

DNA replication: The process by which the genetic material (DNA) of the cell reproduces itself prior to its distribution to the next generation.

Dominant allele: An allele that expresses itself and masks the effects of other alleles for the trait.

E

Ecological isolation: A genetic isolation mechanism that prevents interbreeding between individuals of the same species because they live in different areas; also called habitat preference.

Electron-transport chain: Series of electron carriers that transport electrons from the reduced coenzymes NADH and FADH2 to oxygen.

Embryology: The study of an organism's development from the zygote or fertilized egg stage.

Endocytosis: The transport of materials into cells by vesicles.

Endosymbiont theory: The proposal that self-replication cellular organelles such as mitochondria and chloroplasts were originally free-living organisms that entered into a symbiotic relationship with nucleated cells

Enzymatic competition: Competition among several different available enzymes to combine with a given substrate material.

Enzyme: A protein substance produced by living cells capable of speeding up specific chemical changes such as hydrolysis, oxidation, or reduction, but is unaltered itself in the process; a biological catalyst

Ethologist: A person who studies the behavior of animals in natural settings.

Ethology: The scientific study of the nature of behavior and its ecological and evolutionary significance in its natural setting.

Evolution: The theories concerning the processes of biological and organic change in organisms such that descendents differ from their ancestors.

Exergonic (exothermic): A reaction that releases energy and occurs spontaneously.

Exocytosis: The transport of material out of a cell by means of vesicles

Experimental variable: The one factor in an experiment that is period of time during population growth when the population increases at an accelerating rate.

F

F1 generation: The first generation of offspring in a genetic cross.

F2 generation: The second generation of offspring in a genetic cross.

Facilitated diffusion: Transport of substances across the membrane of a cell in which proteins speed up the movement.

FAD (flavin adenine dinucleotide): A hydrogen carrier used in respiration

Fermentation: An anaerobic process whereby organic compounds such as carbohydrates are broken down to form products such as lactic acid or alcohol.

Fibrous proteins: Fiber-like proteins that is usually insoluble.

Fluid-mosaic model: The concept that the cell membrane is composed primarily of protein and phospholipid molecules that are able to shift and flow past one another

Fossil: An imprint or remains of an organism that lived long ago, preserved in rock

Fossil fuel: Combustible materials that are the remains of ancient, fossilized plants and animals

Fumarate (fumaric acid) : A four-carbon organic compound that takes on a molecule of water and becomes malic acid in the seventh step of the Krebs cycle.

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Functional group: A group of atoms in an organic molecule that gives the molecule distinctive properties.

G

G_1 Phase: During interphase, the part of the cell cycle that occurs between the end of mitosis and the beginning of DNA synthesis; G_1 refers to first gap.

G_2 Phase: During interphase, the portion of the cell cycle that begins at the end of the S phase and lasts until the beginning of mitosis; G_2 refers to the second gap.

Gene: A piece of chromosomal DNA that codes for a particular trait; the basic unit of heredity; the information in a DNA molecule that codes for one polypeptide chain.

Gene conversion: The change of one allele to another during crossing over

Gene frequency: The proportion of a particular allele in population.

Gene gun: An instrument that shoots tiny beads coated with DNA directly into cells; some cells treated this way integrate the foreign DNA that is shot into them, thereby becoming transgenic.

Gene pool: All the genes present in a population

Genetic code: The nucleotide triplets (codons) that code for the

Genetic counselor: A professional with specific training in human genetics who can advise on the likelihood of genetic defects being passed to children 20 amino acids or for chain initiation or termination

Genetic drift: A random variation in gene frequency in a small population

Genetic engineering: The use of recombinant DNA and cloning techniques to produce new genes, biological products, or organisms with permanently altered abilities.

Genetic recombination: The gene mixing that occurs during sexual reproduction.

Genotype: The set of genes present in the cells of an organism.

Geographic barriers: Geographic features that keep different portions of a species from exchanging genes

Geographic isolation: The division or separation of a population by a physical barrier such as a mountain range

Glycerol: A three-carbon alcohol with three –OH groups

Glycolysis: The anaerobic, stage of cellular respiration in which glucose or other carbohydrate is broken down to form pyruvic acid or lactic acid, ATP and NADH.

Glycoprotein: A type of protein that has sugars attached to it

Glycosidic bond: Bond between the anomeric carbon of a cyclic sugar and the –OH group of another sugar or an alcohol.

Gram stain: A crystal violet stain that is retained by gram-positive bacteria and not retained by gram- negative bacteria, after alcohol or a similar solvent is applied.

Grana: Areas of the chloroplast membrane where chlorophyll molecules are concentrated.

Grassland: A biome in which grasses are the dominant form of vegetation.

H

Haemolysis: Swelling and rupturing of a cell that occurs when blood cells are placed in a solute ion of less than isotonic concentration, and water moves in through the cell membrane.

Haemophilia: A sex-linked condition in humans in which the blood fails to clot

Haploid: Having half of the full complement of chromosomes. The reduced number of chromosomes typical of gametes, as opposed to the diploid number of somatic cells.

Helix: Anything of a spiral shape; in biology it refers to the shape of DNA molecules, which occur as double helices

Heterozygote: An organism in which the pair of alleles for a trait is composed of different genes (usually dominant and recessive).

Histones: Proteins bonded with DNA in the nucleus that comprises the protein component of chromatin

Hominid: Modern humans and closely-related primate ancestors.

Homologous: Having similar structure and origin but having different functions in various species

Homologous chromosomes: A pair of chromosomes in a diploid cell that contain similar genes at corresponding loci throughout their length.

Homozygote: An individual with identical alleles at one or more loci. Such individuals will produce identical gametes and will therefore breed true

Homozygous: A term used to describe an individual with two identical alleles for a trait.

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Human immunodeficiency virus (HIV): A human retrovirus associated with the onset and progression of acquired immunodeficiency syndrome (AIDS).

Hybrid: An individual produced by crossing two pure lines.

Hybrid vigor: The increased size and strength of hybrids resulting from the cross of two inbred lines.

Hybridization: Production of offspring from crossing different species or between genetically different populations

Hydrocarbon: Compound consisting only of the elements hydrogen and carbon.

Hydrolysis: A chemical reaction in which a large molecule is split into two smaller molecules by the addition of a molecule of water.

Hydrophilic: Readily absorbing or dissolving in water

Hydrophobic: Tending not to combine with, or incapable of dissolving in, water

Hypertonic: Having a higher concentration of dissolved substances than the solution to which it is compared. As applied to blood, a solution that has an ionic concentration greater than that of blood

Hypothesis: A possible explanation for a set of observations

Hypotonic: Having a lower concentration of dissolved substances than the solution to which it is compared. As applied to blood, a solution that has an ionic concentration less than that of blood.

I

Immigration: The movement of individuals into a new area

Immune system: A collection of cells and tissues that defends the body against pathogens.

Imprinting: A type of learning in which an animal forms a social attachment to another organism soon after birth or hatching.

Inbreeding: Crossing two closely related individuals

Incomplete dominance: A situation in which neither allele for a trait is dominant over the other; blending.

Inheritance: Characteristics passed from generation to generation through the genetic material.

Insight learning: Learning in which past experiences are reorganized to solve new problems

Instinctive behavior: Automatic, preprogrammed, or genetically determined behavior not clearly learned

Intraspecific competition: Competition between members of a population for the same resource

Irreversible inhibitor: Molecule that binds tightly to an enzyme and reduces or eliminates the activity of the enzyme

Isolating mechanism: Any barrier to the exchange of genes between different populations of a group of organisms..

K

Karyotype: The set of a cell's chromosomes classified by number, size, and shape.

Krebs cycle: The series of reactions in aerobic cellular respiration, resulting in the production of carbon dioxides, the release of pairs of hydrogen, and the formation of an ATP molecule.

L

Lamarkism: A false theory proposed by Jean Baptiste Lamark, who suggested that characteristics developed during the lifetime of an organism can be inherited by the next generation.

Learning: A change in behavior that results from experience. eg. trial and error learning.

Light microscope: An optical instrument that uses light to magnify images of specimens.

Light reactions: The first stage of photosynthesis in which energy from the sun is captured as chemical energy to make ATP and NADPH

Limiting factors: Environmental influences that limit population growth

Linkage group: Genes that occur together on the same chromosome.

Lock-and-key model: Model for substrate-enzyme interaction that states that the two molecules are complementary to each other before binding and fit together like a lock and key.

Locus (plural, loci): The place on a chromosome where an allele is located

Lysogenic virus: A type of virus that causes the total destruction of a cell

Lytic cycle: A reproductive cycle found in viruses where the host cell lyses after producing new virus particles.

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M

Macromolecule: An organic molecule made of a very large number of atoms.

Malate (malic acid): A four carbon acid that is oxidized by the reduction of NAD+ to NADH in the eighth step of the Kreb's cycle.

Meiosis: A process of cellular division in which the number of chromosomes is reduced by half from the diploid to the haploid number.

Membrane system: The interconnected membranes of a cell, including the plasma membrane and the various organelle membranes

Messenger RNA (mRNA): A type of RNA that carries the instructions for protein synthesis from the DNA to the ribosome where it acts as a template

Metabolism: A group of processes that includes nutrition, production of energy (respiration), and synthesis of more protoplasm; the sum of the constructive (anabolism) and destructive (catabolism) processes in organisms that collectively sustain life.

Microfilaments: Long, fiber-like structures made of protein and found in cells, often in close association with microtubules; provide structural support and enable movement

Microtubule: Long, cylindrical protein (Tubulin) structures in the cytoplasm that give support to the cell and inside cilia and flagella

Migration: Movement of individuals from one geographic area to another.

Mitosis: A type of nuclear division which produces two daughter cells with the same number of chromosomes as the parent cell.

Monomer: Smallest repeating unit from which polymers are made.

monosaccharide : A simple sugar that cannot be decomposed into smaller sugar molecules; contains five to six carbons such as glucose.

Mortality: The number of individuals leaving the population by death per thousand, individuals in the population.

Mutation: Damage to the genetic mechanism of a cell, causing it to reproduce in a different form

Mutualism (symbiosis): A symbiotic relationship that is beneficial to both organisms.

N

NAD (nicotinamide adenine dinucleotide): A coenzyme that acts as a hydrogen acceptor in cellular respiration beneficial to both organisms.

Natality: The number of individuals entering the population by reproduction per thousand individuals in the population.

Nitrification: The oxidation of ammonium ions or ammonia to nitrate, by certain free living bacteria in the soil.

Nitrogen fixation: A process in which certain bacteria and cyanophytes can convert atmospheric nitrogen into usable nitrogen compounds.

Nitrogenous base: A category of organic molecules found as components of the nucleic acids.

Noncompetitive inhibitor: Reversible inhibitor that binds to an enzyme at a location other than the active site and thus does not compete with the substrate.

Noncyclic photophosphorylation: The light driven flow of electrons from water to NADP+ in oxygen-evolving photosynthesis; requires both photosystems I and II.

Non-disjunction: The failure of homologous chromosomes to segregate during meiosis.

Nucleic acid: Any of several organic acids made of phosphoric acid, sugar, and nitrogen bases.

Nucleoside: A purine or pyrimidine base covalently linked to a ribose or deoxyribose sugar molecule.

Nucleotide: A molecule consisting of phosphate, 5-carbon sugar (ribose or deoxyribose), and a purine or a pyrimidine base; the purines are adenine and guanine, and the pyrimidines are cytosine, thymine, and uracil.

O

Operant conditioning: A form of learning in which a certain response to a stimulus is reinforced.

Organic compound: A chemical compound containing one or more carbon atoms in chain or ring form.

Outbreeding (outcrossing): Mating an inbred individual with an individual that is not closely related.

Oxaloacetate (oxaloacetic acid) : A four-carbon organic acid that is converted to citric acid by the addition of an acetyl group in the first step of the Krebs cycle.

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Oxidative phosphorylation: Process that synthesizes ATP from ADP and inorganic phosphate using energy stored in a proton gradient.

P

Pangaea (Pangea): Single supercontinent that existed during the mid-Paleozoic and consisted of all modern continents in direct physical contact.

Peptide bond: A chemical bond that forms between the organic acid group of one amino acid and the amino group of another amino acid.

Peptidoglycan : A large carbohydrate polymer found in the walls of true bacteria. It is composed of long chain molecules interconnected by short chains of peptide.

PGAL (phosphoglyceraldehyde): A three-carbon sugar that is an intermediate product of photosynthesis.

Phagocyte: An amoeba-like white blood cell that engulfs and destroys invading microorganisms.

Phagocytosis: The process through which food particles are surrounded by cell membrane and incorporated into the cell cytoplasm, forming a food vacuole.

Phenotype: The observable appearance of an individual as a result of its genetic makeup.

Phosphate: An ion composed of one phosphorus atom bonded to four oxygen atoms. Part of a nucleotide.

Phosphodiester bond: A bond between a sugar group and a phosphate group.

Phospholipid: A molecule composed of a lipid and a phosphate group.

Photochemical reactions: The "light" reactions of photosynthesis. **Photon:** Packet of light waves, sometimes called a "quantum." The fundamental particle of light.

Photorespiration: The light-dependent formation of glycolic acid in chloroplasts and its subsequent oxidation in peroxisomes.

Photosystem: A complex of chlorophyll and other pigments embedded in the thylakoids of chloroplasts and involved in the photochemical (i.e., "light") reactions of photosynthesis.

Phytoplankton: Tiny photosynthetic organisms that float on or near the surface of a body of water.

Pigment: Any substance that reflects or absorbs light to give color.

Pinocytosis: The process by which a cell engulfs some molecules dissolved in water.

Pioneer community: The first organisms to inhabit a specific environment.

Plasmid: A small circular piece of DNA found in bacteria.

Plasmodesma (pl. **plasmodesmata) :** A tiny, membrane-lined channel between adjacent cells

Plasmolysis: Loss of turgor pressure in plant cells due to loss of water.

Plastid: A type of organelle that is bounded by a double membrane and is associated with different pigments and storage products; chloroplasts are green, photosynthetic plastids; amyloplasts are storage plastids that contain starch.

Point mutation: A mutation in which one base replaces another in the DNA chain.

Pollution: Any unfavorable change in the environment caused partly or wholly by the actions of humans.

Polymer: A molecule consisting of many identical or similar monomers linked together by covalent bonds.

Polymorphism: The existence of two or more separate phenotypes in a population.

Polypeptide: A molecule consisting of many joined amino acids, not as complex as a protein

Polyploidy: A condition in which cells contain multiple sets of chromosomes.

Polyunsaturated: A term used to describe an organic molecule having more than two double bonds and having less than the maximum number of hydrogen atoms.

Population growth curve: A graph of the change in population size over time.

Population sampling: A method in which data obtained from part of a population is assumed to be true for the entire population.

Post zygotic isolation mechanism: Factors that prevent or reduce inbreeding by acting after fertilization to produce nonviable, sterile hybrids or hybrids of lowered fitness.

Pre zygotic isolation mechanism: Factors that reduce inbreeding by preventing courtship, mating, or fertilization.

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Primary protein structure: The sequence of amino acids in a polypeptide chain.

Primary succession: Ecological succession occurring in an area not previously colonized

Progeny: The offspring produced from a mating.

Prophage: A phage genome integrated into a bacterial chromosome. Bacterial cells carrying prophage are called lysogenic.

Prosthetic group: Group in a protein that is not an amino acid and that plays a major role in the activity of conjugated proteins.

Protease inhibitor: Any chemical that inhibits the activity of enzymes that digest proteins (i.e., proteases); protease inhibitors can also be proteins.

Protoplasm: The living portion of a cell as distinguished from the nonliving cell wall.

Purine: A two-ringed nitrogen-containing base that is part of a nucleotide; the most common purines are adenine and guanine.

Pyrimidine: A one-ringed nitrogen-containing base that is part of a nucleotide; the most common pyrimidines are thymine, cytosine, and uracil.

Pyruvic acid: A three-carbon compound that is the end product of glycolysis.

Q

Quaternary protein structure: Interaction between two or more polypeptide chains within a protein molecule. e.g. hemoglobin.

R

R group: A general term for the side group of a molecule, such as a methyl group, a hydroxyl group, or a monosaccharide.

Recessive: A term used to describe a gene or trait that is hidden by a dominant gene.

Recombinant DNA: DNA that has been constructed by inserting new pieces of DNA into the DNA of another organism, such as a bacterium.

Reproductive isolation: A barrier to interbreeding caused by varied breeding times or geographical factors.

Resolving power: The ability of an instrument to separate and distinguish two objects.

Retrovirus: Viruses with RNA as genetic material that use the enzyme reverse transcriptase for replication into double stranded DNA.

Reverse transcriptase: A type of enzyme from viruses that catalyzes the synthesis of DNA from an RNA template; in genetics, reverse transcriptase is used for making DNA of eukaryotic genes.

Reversible inhibitor: Molecule that binds temporarily to an enzyme and, while bound, reduces or eliminates the activity of the enzyme

Ribosomal RNA (rRNA): Ribonucleic acid molecules found in ribosomes.

RNA polymerase: An enzyme that attaches to the DNA at the promoter region of a gene when the genetic information is transcribed into RNA.

S

Scientific method: A way of analyzing the physical universe; observations are used to construct a hypothesis that is tested by observations or experiments; the hypothesis is accepted or rejected based on experimental evidence.

Seasonal isolation: A genetic isolating mechanism that prevents interbreeding between species because they reproduce at different times of the year.

Secondary protein structure: Helical or pleated structure of a protein that is due to hydrogen bonding between atoms of the polypeptide backbone.

Secondary succession: Ecological succession occurring in an area stripped of its previous community.

Segregation: The separation and movement of homologous chromosomes to the poles of the cell.

Sex-influenced traits: Characteristics that are dominant in one sex and recessive in the other. eg. Baldness in men.

Sex-limited trait: A phenotypic characteristic that appears only in individuals of one sex. eg. Lactation in female mammals, face hair in male humans.

Sex-linked genes: Genes located on the X chromosome.

Sex-linked traits: Characteristics determined by genes on the X chromosome. eg. Red-green colour blindness.

Somatic mutation: A mutational event occurring in a somatic cell.

Spectrum: The range of colours of light given off by an object .

Spontaneous mutation: Natural changes in the DNA caused by unidentified environmental factors.

Substitution: Mutation in which a single base has substituted for another in a DNA strand; a single codon is affected.

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Substrate: The molecule on which an enzyme acts. The enzyme catalyzes its conversion to product.

Substrate-level phosphorylation: The transfer of a phosphate group from a substrate, such as phosphoenol pyruvic acid, to ADP, thereby making ATP without oxygen and independent of thenelectron transport system.

Succession: A process where plant and animal species of an area progress from initial colonization to a climax population.

Succinate (succinic acid): A four-carbon organic acid that is oxidized by the reduction of ubiquinone to ubiquinol in the sixth step of the Krebs cycle.

T

T-cell: A type of lymphocyte activated by the thymus that attacks infected cells.

Temperate phage: A bacteriophage that can integrate its DNA into the bacterial chromosome and remain stable for many generations without causing lysis.

Tertiary protein structure: The three-dimensional structure of a polypeptide chain produced when it folds upon itself, that is maintained by disulfide bonds, ionic interactions, or hydrophobic attraction between amino acids.

Test cross: A genetic cross using a homozygous recessive type to determine whether an individual is homozygous or heterozygous dominant.

Thylakoid: A tiny membrane bound sac containing the chlorophyll in a chloroplast.

Thymine: A nitrogen base (pyrimidine) with the formula $C_5H_6N_2O_2$ found only in DNA, that always pairs with the purine base adenine.

Trait: Any phenotypic variation of a particular inherited character.

Transcription: The process of copying the DNA code to form a strand of RNA which can be transfer RNA, messenger RNA or ribosomal RNA.

Transduction: A process in which a bacteriophage injects a portion of one bacterial chromosome into another bacterium.

Transfer RNA (tRNA): Ribonucleic acid molecules that transfer the correct amino acids into the protein that is being synthesized by ribosomes.

Transgenic: Refers to cells or organisms that contain genes that were inserted into them from other organisms by genetic engineering.

Triplet: Refers to a sequence of three nucleotides that together make a codon.

True-breeding: Refers to purebred strains for a given trait, which means that the gene for that trait is homozygous.

Turnover rate: The number of molecules of substrate that a single molecule of enzyme can react with in a given time.

U

Ubiquinone: A lipid-soluble quinone whose function is to accept electrons from electron donors like NADH and from the oxidation of fatty acids; also called coenzyme Q.

Ultraviolet (U.V.): A type of electromagnetic radiation with wavelengths from 18,000 nm to 33,000 nm. It is a powerful mutagen and is harmful to the human retina and skin.

Unsaturated fatty acids: Fatty acid that has one or more C-C double bonds.

Uracil: A nitrogen base (pyrimidine) with the formula $C_4H_4N_2O_2$ found only in RNA. Always pairs with adenine.

V

Valid: A term used to describe meaningful data that fit into the framework of scientific knowledge.

Variable: Any factor that can be changed or varied in an experiment

Vestigial organ: A body part that is no longer fully developed or useful. e.g. wings in ostrich.

Virulent phage: A bacteriophage that infects and lyses the host bacterial cell.

W

Wavelength: Length of one wave; mathematically equal to the wave velocity divided by the frequency.

X

X chromosome: The chromosome in a human female egg (and in one-half of sperm cells) that is associated with the determination of sexual characteristics. Female = XX; male = XY.

Xanthophylls: A yellow carotenoid; one xanthophyll, zeaxanthin, is the photoreceptor for blue-light in shoot phototropism.

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X-Linked gene: A gene located on one of the sex-determining X chromosomes.

Y

Y chromosome: The sex – determining chromosome in one-half of the sperm cells of human males that carries the genes that determine maleness.

Z

Zooplankton: Tiny heterotrophic organisms that float on or near the surface of a body of water.

Zygote: The diploid cell formed by the union of a male and a female gamete; the fertilized egg.

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